

Deep Learning Based Behavioral Drift Modeling for Continuous Biometric Authentication

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B.Sc. in Computer Science and Engineering

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Declaration

It is hereby declared that

1. The thesis submitted is my/our own original work while completing degree at Brac University.
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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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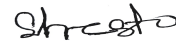
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Ethics Statement

This research was undertaken solely for academic purposes. All data and resources consulted have been fully and transparently cited in the reference list. Those who contributed to this endeavor are named with gratitude in the Acknowledgements. In light of these safeguards, the work conforms to accepted ethical standards and presents no ethical concerns.

Abstract

Ascending technologies like the concepts of robotics, artificial intelligence (AI), the concepts of smart devices and the Internet of Things (IoT) have been assimilated into traditional and indispensable physical, biological and digital systems incubating the fourth industrial revolution. Following the era where the attainability of AI has become ubiquitous, vitality has risen to integrate the smartness of AI into traditional authentication systems in order to strengthen the security and develop a resilient and robust barrier between the infiltrator and the delicate systems. The integration of unimodal physiological biometrics along with the previous password systems served the purpose to a certain extent. But the widespread use of AI has given the intruders an easier access to breach the seclusion of systems, raising a massive question for the security, especially in this era of digital transactions, banking, healthcare systems and academic assessments. There comes the necessity of initiating the utilization of behavioral biometric authentication that integrates the behavioral drift of individuals so that impostors relying on static templates and physiological traits like fingerprints are recognized and the system remains decontaminated. In this regard, this research proposes a two-level keystroke dynamics authentication system that provides defense-in-depth security through continuous behavioral drift monitoring. Level 1 employs fixed-text password authentication using Temporal Convolutional Networks (TCN) with contrastive learning and population-based training (PBT), achieving 3.77% EER for high-security login verification. Upon successful authentication, Level 2 continuously monitors user typing during natural free-text sessions using a 4-Model Deep Learning Ensemble. If suspicious behavioral drift is detected during monitoring, the system re-authenticates the user by returning to Level 1, providing a feedback loop against session hijacking and behavioral anomalies. The system explicitly models long-term behavioral drift through the fixed-text component's PBT mechanism, which adapts to temporal variations in password typing patterns. This research leverages free-text (Buffalo dataset) and fixed-text (GREYC dataset) keystroke dynamics datasets, extracting temporal and spatial features including hold times, up-down latencies, down-down intervals, and trigram patterns. The final system achieves 96%+ accuracy for fixed-text authentication (3.77% EER) and 83.53% True Acceptance Rate (TAR) at 5% False Acceptance Rate (FAR) for free-text continuous authentication (10.05% EER), demonstrating superior performance and strong resistance to impostor attacks through continuous behavioral monitoring.

Keywords: *Continuous Authentication; Behavioral Biometrics; Keystroke Dynamics; Deep Learning; TCN; Ensemble Learning*

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Chapter 1

INTRODUCTION

1.1 Background

Safeguarding privacy and ensuring protection have always been the core necessity for any transaction of data. The process that plays the crucial role for these is commonly known as authentication. The traditional memory-based static authentication systems like PINs and passwords served their role well until the need for a more advanced, dynamic and subjective system materialized. Hence, biometric authentication came to light. The key processes that underlie the working mechanisms of biometric authentication are – i) enrolment for the first time where the data collected for the first time are stored for referring back in later time; ii) preparing the data for storage removing all the excess and unwanted features; iii) extracting the relevant features using various filtration techniques to generate the best possible result; and iv) comparing the stored data with the input data during authentication [21].

In the authentication process, the system has to verify whether the person accessing the system is the same as who they claim to be by checking against the database. The system performs this process by matching the user's biometric data against the corresponding template stored in the database and then providing a binary decision through the calculation of the function c [38] in equation 1.1:

$$c(A, B) = \begin{cases} 1, & \text{if } M(A, B) \geq \alpha, \\ 0, & \text{otherwise.} \end{cases} \quad (1.1)$$

where A denotes the feature vector collected from the user currently accessing the and system and B represents the template which was stored in the enrolment phase. The function c compares between these using the M function which measures the similarity score. If the score is equal or higher than the predefined threshold α then the system confirms the user. Otherwise, it rejects the individual.

Sometimes the system has to identify an unknown individual by checking his biometric information against all the templates in the database. This process is called identification which can be defined by the function c [38] in equation 1.2:

$$c(A) = \begin{cases} Y_x, & \text{if } \max_{1 \leq x \leq h} M(A, B_x) \geq \alpha, \\ N_0, & \text{otherwise.} \end{cases} \quad (1.2)$$

Using the above function, the system effectively identifies unknown persons by checking the feature vector (A) extracted from their biometric information against h total templates of the dataset (B_x). The main focus is to find Y_x that represents the highest matching score from the function M among all the templates that is equal or exceeded the decision threshold α . Otherwise the system shows N_0 that indicates the rejection of the user. Every time when an user provides new features, these measures are followed continuously which provides better verification regardless of the unique traits.

The emergence of physiological biometric authentication is a groundbreaking discovery in the arena of security processes that use a number of physiological characteristics such as iris patterns, facial features, fingerprints, palm prints and even vein patterns. These characteristics are not only unique for an individual but are also traits that cannot be copied or stolen just through some simple tactics. The additional layer of physiological biometric authentication over the traditional memory dependent PINs and passwords adds an added layer of authentication before allowing access to the target program. One of the pivotal dependencies of biometric authentication systems is that the user does not need to memorize any complex passwords or PINs or even carry any physical tokens like the former methods. The authenticator primarily saves the physiological trait of the concerned user and every time the user tries to access the platform, it cross-checks the current input with the stored trait and allows access only for successful matches.

Although physiological biometric authentication has a promising background to prevent a number of types of attacks, it has a key disadvantage in the fact that it is static in nature. The traits it utilizes are mostly not subject to change and if that trait can be replicated or the system can be tricked, this system will fail allowing spoofing attacks like copying the fingerprint or generating face, especially in this booming era of AI. Besides, false negative results can also be generated due to differences in ambience, injury of the user, inability of the user to respond to the authenticator accordingly which ultimately create difficulty for the real user to access the desired systems even in case of emergencies. Hence, physiological biometric authentication faces a number of constraints and requires circumspection for use. This gives rise to the urge for using behavioral biometric authentication.

The preeminent working principle of behavioral biometrics is that it focuses on human behavior through the comprehensive and meticulous encapsulation and analysis of the user behavior over time and produces an effective conclusion. This reflects the fact that the implementation of behavioral biometrics is relatively less prone to phishing attacks and criminal replication as the conclusion is based on behavioral data over a period of time. Hence, coupling the physiological biometrics with the behavioral biometrics elevates the reliability of the authentication system diminishing the probability of unauthorized access to the programs. This is primarily because the traitor might be able to replicate the physical traits of the user using AI or tracking the fingerprints but cloning the behavior of the user alongside is unimagin-

able. However, this conventional behavioral authentication again faces one challenge - the natural drifting of human behavior over time (Power Law of Practice) [6]. For example, a person may gain expertise in typing as time advances, but the template matching of the traditional method may recognize the genuine user incompetent due to his natural behavioral evolution while the impostor may trick the system due to the invariability of the template he is using.

In order to overcome such challenges, this research suggests a unified continuous authentication framework that clearly differentiates between the fixed-text and free-text behavioral scenarios. The model combines evolutionary adaptation algorithms and deep temporal representation models based on Temporal Convolutional Networks (TCN) to allow long-term behavioral learning without retraining them. Lightweight TCN-based adapter modules have been developed for valid users. Instead of redefining the whole model, these adaptation modules assist in adaptation to the frozen TCN backbone architecture through training with population-based training. For continuous monitoring using free-text dataset, ensemble of 4 deep learning models is integrated to ensure that the highest accuracy for defining the impostors is achieved. The models are then fused in a weighted linear combination to produce a single score.

1.2 Rationale of the Study or Motivation

The motivation of this study stems from a number of limitations in the existing techniques that make the critical digital infrastructures vulnerable to massive attacks and natural behavioral drift:

- The replication of behavioral templates often cause the perpetrator to breach the security while the genuine user faces difficulty due to the innate behavioral drift. This effectuates high false acceptance rate (FAR) and high false rejection rate (FRR).
- Balancing off the security supremacy and user convenience has become an obscurity. This often causes the users to feel impassive to walk through all the stages of authentication compromising the security. [5]
- The contemporary authentication system ponders each authentication event independently. This disregards the fact that each of the authentication events is a part of the continuous evolutionary trajectory of legitimate user behavior.
- Existing transformer infrastructures such as - swipecformer, behaveformer are good to capture features but they still use static training. Hence the attempt to address behavioral drift through these frameworks needs constant retraining which is impractical in real life.

These issues are handled by this study which is inspired from the evolutionary adaptation and exceeds the existing pattern matching approaches by combining the adaptability of evolutionary algorithms and representation capacity of Temporal Convolutional Networks. This research connects the gap between constant adaptability and reliable security by performing the evolution of lightweight adapters

within a TCN backbone through population-based training, grounding the rationale that behavioral biometrics should be addressed as a gradually occurring phenomenon preserving the core characteristics, not as distinct and static patterns reflecting those of the impostors.

1.3 Problem Statement

Authentication infrastructures are vital to provide maximum privacy and ensure security of user information in this modern age. In the field of user authentication, traditional security like passwords, PINs and physiological biometric authentication methods have become an increasing threat to data breaches, theft and digital attacks. These methods often rely on static credentials which are easily threatened for personal and organizational digital security [24]. Even after the successful authentication at the beginning, the user’s session can still be interrupted through session takeover by imposter and bot-driven attacks. Behavioral biometrics, which is associated with Human Computer Interaction (HCI), shows unique patterns in human behavior like typing movements, mouse movements, voice patterns or touchscreen interactions. These offer a promising alternative based on their behavioral characteristics through continuous monitoring and assisting in adaptive authentication.

However, the recent advanced works or approaches in behavioral biometrics are highly influenced by deep sequential models such as SwipeFormer and BehaveFormer. Although these frameworks are great at capturing temporal dependencies, they still face different challenges due to the various factors of behavior changes like stress, illness, external environment or device differences. Differentiating the legitimate behavioral evolution of genuine users from the impostor has been proven as one of the strenuous challenges faced by the current authentication systems. This makes the sensitive sectors susceptible to attacks due to a number of reasons. Firstly, the misclassification of legitimate users as impostors due to the natural evolution of behavior creates a scope of high FRR which induces trust questions and frustrations. The issue of catastrophic forgetting marks the second challenge where the system fails to recall an individual’s unique feature as the model overwrites identified traits because of the tuning of new behavioral drifting information. Thirdly, addressing behavioral drift through existing deep learning frameworks requires continuous and computation heavy model retraining which is impractical and leads to resource constraint in real life scenarios. And, most importantly, the current systems barely balance security and ease. If the users are to go through fatiguing steps for authentication, the irritation may lead to a tendency to avoid the system. Similarly, if the users are given the comfort to have easy access, the infiltrators will also gain access putting the security into disputes.

This research directly handles these limitations by introducing a dynamic infrastructure. The behavioral drift will be addressed through adaptability of continuous evolution to deal with the transient behavior of genuine users. Continuous authentication will persuade that the user will be evaluated persistently without performing any explicit tasks ensuring the user does not get fatigued to accomplish the authentication tasks.

1.4 Objective

This research has conscientiously evaluated the flaws and loopholes in the existing system of biometric authentication and aims for the development and evaluation of a more robust and realistic continuous authentication system. Encompassing the principles of evolutionary deep learning, enhancing security in biometric features and reducing FAR and FRR which will affect the equal error rate (EER) are feasible. By combining adaptive techniques with Temporal Convolutional Networks (TCN), the system aims to attain continuous user authentication in dynamic situations by considering the barriers of accuracy, usability and real-time performance.

The primary objectives include:

- Development of a hybrid framework that utilizes representational power and adaptability through the backbone of TCN and adaptive techniques respectively.
- Ascertainment of the adaptability of the system to detect evolving behaviors and anomalous patterns in real-world environments.
- Integration of population-based training (PBT) that enables the evolution of the adapter population dynamically and provides a varying memory bank which handles sequences consisting of multi-modality and constant shifts.
- Prevention of catastrophic forgetting and frog boiling strike through trust anchors that enables the evolution of framework alongside the user.
- Evaluation of the system’s effectiveness with existing available datasets and aim to surpass the performance of current authentication systems.
- Improvement of features from multimodal data to decrease FAR and FRR ensuring justified differentiation between genuine user and impostor.

1.5 Methodology in Brief

This study addresses keystroke dynamics authentication through a two-level framework (figure 1.1), with each level employing distinct architectures, datasets, and evaluation protocols tailored to its authentication modality.

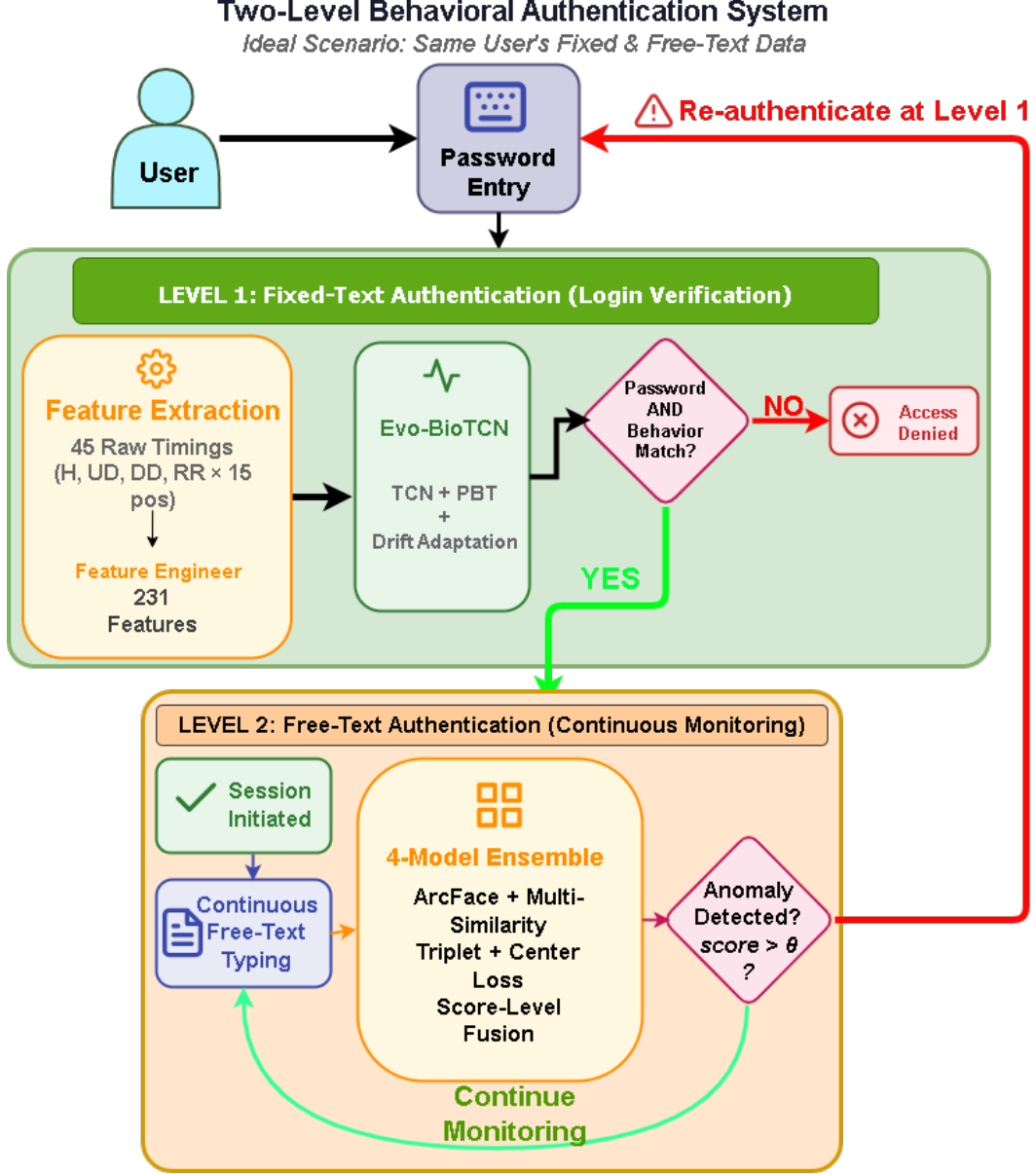


Figure 1.1: Two-Level Behavioral Authentication System

Figure 1.1 shows a comprehensive security framework combining fixed-text authentication (Level 1) with continuous behavioral monitoring (Level 2). The system authenticates users based on password entry patterns and maintains verification throughout the session by analyzing typing behavior, triggering re-authentication via an anomaly threshold (θ) set at the 10.05% Equal Error Rate (EER) operating point to balance security and usability.

Fixed-Text Authentication (Level 1): The GREYC dataset (133 users, 110-day longitudinal span, fixed password “greyc laboratory”)[4] served as the primary benchmark for login verification. Raw keystroke timings (hold, flight, down-down, release-release) were extracted and transformed into 231 engineered features capturing statistical moments, positional patterns, rhythm entropy, and higher-order n-graph timings. The proposed Evo-BioTCN architecture combines a four-layer dilated TCN with a contrastive learning objective (ArcFace angular margin + Center

Loss) and PBT for drift adaptation. The TCN backbone (≈ 131 K parameters) remains frozen after supervised training, while a lightweight adapter (512 parameters) is evolved via PBT to compensate for long-term behavioral drift without explicit re-enrolment. Evaluation used a strict temporal split (25% enrolment, 15% validation, 60% test from later sessions) to ensure that all test samples are temporally distant from training, faithfully simulating real-world template aging. Drift was monitored online using the Page-Hinkley test; when significant change was detected, hypermutation accelerated adapter evolution. The final system achieves 3.77% EER and 96.87% TAR @ FAR=5%, demonstrating suitability for high-security applications.

Free-Text Authentication (Level 2): The Buffalo dataset (75 users, 6 sessions across 3 temporal phases separated by weeks) served as the primary corpus for continuous monitoring. Because the dataset is distributed in digraph-encoded form, a reconstruction pipeline was developed to recover character-level timing vectors and segment them into word samples. Preprocessing included MAD-based outlier replacement, session-relative normalization, and extraction of 85 statistical features (with hybrid global + per-user normalization yielding 170 dimensions). A pilot study on the KeyRecs free-text dataset [28] corpus validated session-split protocols and confirmed that fixed-text architectures fail catastrophically when transferred to free-text (accuracy collapsed from 96%+ to $\approx 33\%$). Consequently, a purpose-built 4-Model Deep Learning Ensemble was developed, combining ArcFace + Multi-Similarity, Enhanced Embedding (Focal Triplet + Center Loss), Simple ArcFace, and Fine-tuned Enhanced networks; score-level fusion weights were optimized on a validation fold to maximize TAR at constrained FAR. Evaluation enforced temporal session-split (train on early sessions, test on later sessions) to expose authentic drift; cross-session comparisons quantified leakage-driven inflation. The ensemble achieves 10.05% EER and 83.53% TAR @ FAR=5% on Buffalo—representing a 65% reduction relative to the best classical baseline (Scaled Manhattan, 28.77% EER).

1.6 Scopes and Challenges

There are certain scopes of this research as:

- This study concentrates on the keystroke dynamics as the main behavioral modality, both in the fixed-text (using passwords) and in the free-text (continuous typing) authentication. The other modes including touch gestures, facial expressions or gait analysis are beyond the scope of the study.
- This study prioritizes the continuous assessment of genuine users over the initial authentication or the identification among many users.
- This research is grounded on GREYC[4] Keystroke Dataset and University of Buffalo[12] Keystroke Dynamics Dataset that records behavioral data of users over a longitudinal session without any constraint. This provides a solid basis in assessing the framework’s capability to handle behavioral drift. The integration of ensembled DL algorithms and TCN shows its vitality in optimizing adapters through the usage of PBT.

- The evaluation primarily focuses more on authentication accuracy metrics and certain constant learning metrics such as - forgetting measure, degradation rate of performance rather than computational efficiency or system performance.

Significant challenges remarkably hindered the flow of the study. These can be described as:

- The primary challenge lies in balancing the model’s adaptation to new behavioral data without removing the previous core features. This leads to catastrophic forgetting where the system adapts to the new data but fails to identify based on previous features.
- Reducing the risk of slowly mimicked intrusion of imposter requires integration of trust anchors to adjust the process of evolution.
- The uniqueness of the noise patterns of behavioral data requires appropriate knowledge and specialized preprocessing techniques. These result in difficulties in handling duplicates, missing values and outliers.
- Mapping the principles of evolution to the training of neural networks needs cautious parameter tuning which requires rigorous research and secondary knowledge.
- There is a huge optimization challenge to detect outliers while understanding the genuine behavioral modes.

Through iterative development and evaluation, these challenges are addressed so that the final outcome remains unaffected by them.

1.7 Thesis Structure

Chapter 1 establishes the necessity for behavioral biometric authentication to counter the vulnerabilities of static physiological traits and traditional password systems. It defines the central challenge as behavioral drift—the natural evolution of typing patterns over time—which typically leads to high false rejection rates. The research proposes a two-level continuous authentication framework: Level 1 utilizes Evo-BioTCN for fixed-text login, while Level 2 employs a 4-Model Ensemble for continuous free-text monitoring, ensuring defense-in-depth security (figure 1.1).

Chapter 2 mainly focuses on exploring and reviewing the existing literatures. The review analyzes existing research, identifying a critical gap: most state-of-the-art models are static and fail to account for "temporal aging." While techniques like Siamese networks and Transformers capture dependencies, they often require impractical retraining to handle drift. Key findings highlight that behavioral biometrics must be treated as evolving trajectories. This justification supports the study’s shift toward ensemble DL techniques and TCN to maintain discriminative power without catastrophic forgetting of user identities.

Chapter 3 outlines the functional specifications and societal implications of the proposed system. Stakeholders prioritized non-intrusive, background monitoring to

reduce user fatigue. The system adheres to zero-trust security principles and data minimization ethics. Economically, the use of frozen backbones with lightweight adapters reduces the computational costs associated with energy-intensive retraining. Risks such as "frog-boiling" mimicry attacks are addressed through trust anchors, ensuring the system remains both secure and inclusive of natural behavioral changes.

Chapter 4 depicts the design which employs a dual-stage strategy for fixed-text authentication using the Evo-BioTCN architecture. This features a frozen TCN backbone for causal temporal modeling and lightweight 512-parameter adapters evolved via Population-Based Training (PBT) to handle longitudinal drift (Figure 4.3). For free-text monitoring, the design utilizes a 4-Model Deep Learning Ensemble comprising ArcFace + Multi-Similarity, Enhanced Embedding (Focal Triplet + Center Loss), Simple ArcFace, and Fine-tuned Enhanced models. This ensemble approach utilizes a 170-D hybrid feature vector derived from the reconstructed Buffalo dataset, prioritizing score-level fusion over high-complexity alternative architectures (figure 4.14).

Chapter 5 mainly focuses on the evaluation of the proposed architectures. Evaluation confirms the framework’s superiority in handling behavioral drift. The fixed-text Evo-BioTCN achieves a 3.77% EER and 96.87% TAR at 5% FAR (table 5.2), significantly outperforming static benchmarks. PBT optimization was essential for this success, as shown in the performance gains across longitudinal spans (figure 5.4). For free-text authentication, the 4-Model Ensemble achieves a 10.05% EER and 83.53% TAR at 5% FAR on the Buffalo dataset (table 5.28). This represents a 65% EER reduction compared to classical Scaled Manhattan baselines, validating the effectiveness of ensemble-based score fusion in continuous monitoring.

Chapter 6 synthesizes the two-level framework’s success in mitigating behavioral drift. While fixed-text authentication provides a high-security entry point, the Level 2 ensemble ensures session integrity through passive monitoring. By unifying these modalities into a defense-in-depth structure, the framework achieves a combined attack success probability of approximately 0.38%, which is derived from the multiplicative independence of both authentication layers ($(3.77\% \times 10.05\% \approx 0.38\%)$). Limitations include the need for larger longitudinal datasets and cross-device validation. Future work focuses on multimodal integration. Ultimately, the integration of Evo-BioTCN and ensemble deep learning demonstrates that addressing behavioral changes as continuous trajectories, rather than static snapshots, provides a more resilient and user-friendly security infrastructure.

Chapter 2

LITERATURE REVIEW

2.1 Preliminaries

Behavioral Biometrics refers to the measurement and analysis of patterns in human behavior for identification or authentication purposes. These have an innate advantage that these can be continuously tracked and analyzed without executing any explicit action unlike static physiological biometrics like fingerprints or vein patterns. Some common examples of behavioral biometrics include keystroke dynamics, mouse dynamics, touch gestures, gait analysis, etc. This study focuses on keystroke dynamics. Some common metrics of keystroke dynamics include dwell time (duration a key is pressed), flight time (interval between releasing one key and pressing another), digraphs (timing patterns between specific key pairs) and typing rhythm (consistency of timing across keystrokes).

There is a specific approach that inherently explains adaptability of different frameworks in changing environments - Evolutionary Deep Learning (EDL). The key concepts that need to be known for a better understanding of this study include:

- **Population Based Training (PBT)** : An optimization approach that handles the evolution of adapters without retraining.
- **Concept Drift**: The constant change in user behavior because of natural and technical proposes.
- **Spatio-Temporal Attention**: Feature Temporal attention: A transformer architecture that models dependencies among different feature dimensions and time steps in keystroke data.
- **Stability-Plasticity Dilemma**: A process where a model tries to learn new features without erasing the previously stored identity.

Contrary to the point-in-time authentication, there is another strategy that continuously carries on user verification throughout the session. This approach, namely continuous authentication follows the zero trust security system and is more secure as it persistently checks for behavioral anomalies preventing impostors from causing any greater harm to the user.

Users can be sometimes relaxed, stressed, rushed or relaxed which will be reflected in their typing pattern or using their devices. This indicates the concept of behavioral drift and understanding these distinct behaviors exhibited by the same individual due to different external factors falls under the umbrella of modes of behavior of the user.

To ensure consistent representation of feature scale, the standardization is performed for all scores of authentication metrics and they are expressed by their standard deviations from their mean. This is known as z-score normalization which allows consistent thresholds across the variety of users and behaviors.

These concepts help in portraying a clearer image and understanding of the existing literature as well as this study.

2.2 Review of Existing Research

Time series data is crucial in determining critical decisions in the field of the financial and health sector but conventional methods are still facing challenges to find the significant pattern from it. Ali et al. [34] proposed a new approach which converted time series signals into a form suitable for sequence based analysis with the assistance from the techniques of bioinformatics. Their main focus was to make complex and non linear data accessible to pattern recognition algorithms which were specifically developed for protein and DNA sequence analysis. Firstly, they fixed a global maximum and minimum by analyzing the range of values and divided the entire range into 26 sectors. Additionally, each interval or sector is then mapped to a unique character of the English alphabet. After conversion, using that sequence they extracted kmers which indicates the overlapping substring. Furthermore, the final representation was a frequency vector indicating the count of every kmers. Finally, these vectors passed as the input for different classifiers of machine learning. In their research of predicting age, they achieved mean accuracy and ROC-AUC of 0.953 and 0.967 respectively using their method combined with a logistic regression classifier [34]. The study crossed the conventional baseline of feature engineering with a highest accuracy of 0.925 which indicates a 2.8% increase in performance [4]. The authors addressed that the runtime of their method is higher because of the huge size of kmer embedding. They further suggested that time series analysis can be further improved by using their transformation technique which will incorporate transfer learning to the bioinformatics domain. This work is highly related to our study as they model keystroke data as sequences. Our study focuses on developing the temporal relationships through lightweight adaptability and drift aware learning. Hence our model addresses temporal aging by observing the changes whereas their approach offers static points to classify.

Keystroke dynamics offers an effective and cheaper way for transparent individual authentication through a unique pattern of typing. In their research, Ceker and Upadhyaya [10] expanded the concept of Support Vector Machine (SVM) by using free form text in the field of continuous authentication. Their study introduced a unique alignment technique and extraction of features for long text data. The core idea was to create a separate behavioral profile through which can be used to

authenticate users continuously in the background. At first, they figured out the frequently typed digraphs of a user which was a combination of two letters. Additionally, for each case of digraph, they derived different digraph features like dwell and flight time to have a feature matrix where separate digraphs data placed in different columns. These procedures dealt with variable nature perfectly. Moreover, for their individual 34 uses, they trained independent one class SVMs in their dataset. Their model observed a boundary to verify the actual user’s behaviour and an anomaly was considered if any data point is outside of that boundary. Their research provided a promising effect by achieving a 1.0 score of perfect Area Under the Curve (AUC) and this resulted in a 0% EER [10]. These results indicated an almost flawless capability of identifying the different users without any error of classification. Though using more digraphs will provide more accuracy, it will add more computational complexity. Hence, the authors addressed that large use of digraphs degrades performance which makes it even tougher to find the optimal tradeoff. They further suggested evaluating the algorithm’s robustness and focus on developing an adaptive SVM which will be helpful in selecting optimal kernel scale. Their study aligned with our topic as it demonstrated the keystroke dynamics usage of long text data in continuous authentication. However, our proposal will address temporal aging and solve their crucial limitation of staticness. Whereas their model needed retraining to detect evolved behaviour, our drift adaptive model implements this as a fundamental feature which will consider changing behavior as a continuous behavioral trajectory instead of different training sessions.

The dwell time and latency characteristics of keystroke dynamics can be efficiently modeled through sequence alignment technique which is a concept of bioinformatics. Bamatraf et al. [9] proposed an unique and efficient model which addressed the authentication issues using sequence alignment techniques in keystroke dynamics. Their study focused on the conversion of the typing pattern into a sequence of discrete characters which could be further used for similarity checking. Firstly, they applied fuzzy logic to get the unique sets which are denoted by letters by categorizing the keystrokes continuous timing data (e.g, duration, latency). Through this process, the password entered by the user is translated into character string. Additionally, for classification, they used an advanced form of Needleman-Wunsch algorithm and constructed a tailored similarity matrix so that the score of a pair of characters is fixed by the distance between their time intervals where a high score indicated authentication was successful. The authors improved the previous Needleman-Wunsch algorithm by achieving 80% accuracy and 90.3% precision. Through these results, their method significantly outperformed standard machine learning classifiers by providing 30% more accuracy and 40% more precision than BayesNet which is considered as the next best classifier [9]. However, their model could not be directly used to continuous data and the global alignment method could be vulnerable to the traps of local minima. The author suggested that future work should be focused on the improvement of the validity of the model’s application with more datasets. This research aligned with our topic as it demonstrated the use of sequence alignment in behavioral biometrics. Whereas their method worked with only a reference signature by doing a pair wise comparison, our drift aware approach handles behavioral changes through adaptation of lightweight modules and evolutionary trajectory. Hence our model will be resistant to the natural change in

the behaviour of any user whereas their model’s performance degraded because of temporal aging.

Zeeshan et al. [42] in their research introduced a continuous authentication method for resource limited devices which incorporated environmental information with biometric devices resulting in providing efficiency and higher accuracy. Their study emphasized on developing an authentication infrastructure which was adaptive, secure and appropriate for resource limited devices. The system relied on environmental factors (e.g., sound, light) to decide when to rechecked users faces instead of constant and power consuming checking. Their approach used Convolutional Neural Network (CNN) which was lightweight and transformed an acquired face into an embedding of 128 bits. To prevent intrusion, the pair of embedding and random nonce was hashed before transmitting it. Additionally, further robustness was ensured through the testing of Scyther software. The authors reported significant performance gain on the testbed of snapdragon 778G [42]. Their system successfully lessened the median decision time for authentication from 61.2 ms to 42.3 ms and mean power consumption from 21.3 mJ to 19.8 mJ in every cycle [42]. Their method was much faster (only 42ms) compared to baselines of behaviours (20000 ms) and much power efficient (19.8 mJ vs 24-28 mJ) [42]. Furthermore, it showed promising accuracy of 3.7% in Equal Error Rate (EER) which was much higher than HMOG (7.9%) and gesture based (4.8%) model [42]. However, the author addressed that their model only tested against fewer devices and demographics of individuals. Additionally, the system was not tested against strong spoofing attacks. They suggested that future work should be focused on benchmarking this system against a wide variety of behavioral authentication methods. This study was highly related to our work as it showed a strong approach in continuous authentication. But it was totally environmentally adaptive which meant it only focused on the environmental or external changes of the user and did not respond to the internal changes. Our work will solve this problem by checking each sample against an adaptive representation of the user that updates from time to time instead of checking against a static template.

Fereidooni et al. [29] in their paper proposed a few shot based learning system that allowed user friendly and scalable continuous authentication in the field of behavioral biometrics. The research emphasized on overcoming the limitations of conventional biometric systems such as- scalability issue, heavy computation and specific individual training data. The authors introduced a few shot learning approach that was based on a Siamese neural network. This network was further trained so that it could learn from similarity functions instead of learning from a specific user. Additionally, the network accepted pairs of motion data (e.g, accelerometer, gyroscopes) from users and further converted them into embedding vectors which were lower dimensional. After conversion, it could be concluded if they were of the same person based on the distance between them. This procedure was effective to learn and identify unique users with a minimal sample number. This study showed promising accuracy of 97% in F1 score in a difficult and challenging scenario where only 3 samples required for entry of a user [29]. Moreover, the system could verify an individual within 1 second of contact [29]. This study further achieved a score of 0.023, 0.057, 0.987 in FAR, FRR and AUC respectively which indicated reliability and robustness [29]. However, the authors addressed that the training model did

not consider the changes in behaviour of users like postures changes and usage of different devices. They suggested future work could focus on developing a more enhanced process of sample mining to overcome this. Their scalable method approach in continuous authentication aligns with our concept but the main issue still lies in temporal aging as their networks checked similarity only against some static sample. Our model addresses behavioral changes through evolutionary trajectory instead of checking against static samples which provides proper resistance to natural changes over time.

Because of the changing nature of digital security threats, traditional authentication methods failed to provide necessary protection which resulted in the evolving of dynamic and continuous authentication approaches. Gunuganti [30] provided a detailed review about the current condition of user authentication using behavioral biometrics. This study focused on summing up the methods of authentication which functioned during the entire user session instead of just the user initial entry. Throughout the research, the author reviewed different biometric behaviour, procedure of enrolment stages, several performance evaluating metrics and authentication phase. Additionally, this research showed compassion among different modalities through real time effectiveness, robustness and usage cases. Moreover, this paper demonstrated that keystroke dynamics showed promising accuracy while authenticating users in real time. This research highlighted that the effectiveness of different authentication methods is computed through the AUC which decides the accuracy. Furthermore, the author analyzed that behavioral methods had beaten the typical static approaches by providing better resistance to intrusion and flexibility. However, the paper addressed that the limitation of these methods lies in the variability of data caused by different environments, discomfort of users because of contact surveillance and the vulnerability of severe spoofing attacks. Hence, the author suggested that the future work should be focused on developing more enhanced Machine Learning (ML) algorithms, improving technology of different sensors and ensuring industry quality which will provide high level security and interoperability. This study formulated the problem of data variability caused from temporal aging which will be solved through our drift adaptive model. By maintaining behavioral trajectory, we can distinguish natural and suspicious changes resulting in a robust solution.

Hu et al. [40] in their research, introduced an unique approach of Deep Learning (DL) architecture to resolve the main constraints and challenges of continuous authentication. The researchers were determined to develop a model which ensured better efficiency and accuracy in real life scenarios. Their base idea was to develop an autoencoder which was lightweight and could only be learned through the actual users data in one class classification. Additionally, they incorporated a novel approach named masked latent representation which was composed of two parts. One of them is a generator of the latent representation and the other one is a reconstructor. In cases where the behavior of the user was in abstract form, the Masked Latent Representation Generator (MLRG) learned to mask the significant features of the latent space instead of copying the raw data. Then the actual data was recreated by the reconstructor from the partial representation which ensured to understand a better version of an individual's patterns. Moreover, verification was completed after measuring the error of reconstruction which provided an anomaly score. Because

of the lower number of convolution layers and compact design, the model could be used on any device which had limited resources. In four different datasets, their model demonstrated promising results by achieving an Equal Error Rate (EER) of 1.87% in WISDM, 0.68% in HMOG, 2.14% in IDNet and 0.% in BrainRun [40]. Furthermore, in real world scenarios, the system was consistent by scoring EERs between 1.42% and 1.91% which indicated its resilience [40]. However, the authors admitted that they could only evaluate a single session. Hence they suggested that future work should focus on determining the performance in terms of multiple sessions which was suitable for real life context. Their work has direct connection to our study of behavioral drift as they developed an advanced model but for static and single sessions. Our research will overcome these limitations by introducing a drift aware approach which resolves the temporal aging issue by considering evolving behavioral trajectory across sessions.

The significance of AI in the context of continuous authentication and the importance of behavioral biometrics in the current IOT era was demonstrated by Liang et al. [19]. Compared to existing traditional methods, the authors emphasized on the detailed overview of the behavioral infrastructure to ensure better monitoring and security. The authors examined varieties of behavioral indicators such as keystroke dynamics, touchscreen response, walking gait etc. Additionally, they suggested a general structure of continuous authentication infrastructure based on AI which included classification, data sensing and feature extraction. Moreover, they reported how various Machine Learning (ML) algorithms performed in this domain and found isolated forest algorithms scored an accuracy of 98% whereas DL models such as LSTMs yielded 96.7% accuracy for sensor data [19]. However, the authors addressed the main limitations was the continuous evolution of behavioral characteristics as an individual’s behaviour changes continuously because of the health issue, aging and other environmental factors. They further found current methods had failed to adjust to these natural changes and suggested focusing on a robust solution so that it could fit the varying dynamics of behaviour evolution. Initially, their findings were highly relevant for our work as they validated the motivation and problem of our research. While they figured out the failure of the adaptability to natural changes of the existing systems, they did not propose any methodology. Our work fills this void perfectly by implementing the drift aware infrastructure which handles behavioral changes, models them and further tracks them as behavioral trajectory. Throughout our approach, we can solve the problem of temporal aging which was identified by the authors.

A research was conducted by Tumpa and Gavrilova [26] about the effect of temporal aging in the field of Social Behavioral Biometrics (SBB) infrastructure. Their paper stands out as it provides an in-depth exploration on temporal stability of the traits of behavioral biometrics among the users in social networks over the last few years. The study proposed a new methodology with the integration of interpersonal, textual and contextual data to form a robust and strong biometric system. The authors emphasized on assessing the SBB system performance in the long run and how temporal aging indicated the degradation of templates resulting in affecting the user verification accuracy. They incorporated a weighted sum based fusion algorithm with a genetic algorithm to provide better optimization of feature

weights. Additionally, they combined seven various SBB characteristics such as URL network, writing profile, reply network, lexical features etc. Their introduced system achieved a highest recognition accuracy of 99.25% which improved the verification process [26]. Moreover, they analyzed the effect of aging in characteristics by measuring the performance of the system throughout four years. They found that writing profiles showed consistent higher accuracy compared to other characteristics. The performance of the system was further enhanced by the proper optimization of the characteristics weights. However, some characteristics such as structural and lexical profile were highly affected by the temporal aging which changed the overall system performance. The author suggested testing against real world situations and more varied datasets for future works. Their work aligns with our proposal as both ideas focus on the improvement of accuracy in changing behaviour over time. Our idea incorporates their results with a drift adaptive model which models the behaviour evolution effectively and provides better analysis to improve performance.

Sagbas and Balli [36] proposed a unique continuous verification infrastructure which was based on integrating motion sensor information and soft typing resulting in providing higher accuracy and smooth authentication. The essence of their method was based on the record of behavioral characteristics of 59 users through a unique application. Additionally, they gathered 125 unique features based on accelerometer and touchscreen response through a window of 5 seconds. Moreover, they used a unique approach named Correlation Based Feature Selection (CFS) to select the most significant aspects resulting in highest efficiency and low computational cost. The researchers further used the updated feature set, performed training and testing on three different classifiers of ML which were Simple Logistic Regression (SLR), Random Forest (RF) and K-nearest neighbor (KNN). With the SLR model, they achieved a promising accuracy of 93% which was achieved through 92 features resulting in a reduction of 26% [36]. The efficiency of the system demonstrated through the classification time of 0.03 ms [36]. However, the authors addressed several limitations such as- lacking or managing noisy data and small samples. They suggested that future work should focus on verification through particular text and dynamic sample recording to ensure better accuracy. This paper provided a strong foundation of a static verification system based on windows. But the main issue was the inability of detecting the natural changes of an individual as it was only trained on snapshots. While their study focused on discrete points grouping, our model considers evolving behavioral trajectory . Through the drift aware approach, we address the behavioral change instead of relying on a static template which will ensure the resiliency of our system.

In their survey, Patel et al. [11] demonstrated a detailed overview of the advanced and current developments in the field of Continuous Authentication (CA) by discussing the existing gaps and several different methods. The researchers main focus was to provide a thorough review of different existing methods such as - face detection indicating physiological modality, touch dynamics, patterns of keystroke and gait analysis indicating behavioral characteristics. Additionally, they analyzed several different multimodal infrastructure that integrated information from more than two sources. Moreover, they demonstrated significant methods and their performance for each modality. In their research, they found EER of 0-4% from touch

dynamics whereas facial detection approaches demonstrated much higher True Acceptance Rates (TAR) and a percentage of EER range from 3.88-33.3% was provided by gait analysis [11]. They further reported that unimodal infrastructure was weak in managing noisy data and preventing advanced attacks whereas a combination of multi-modalities provided a more accurate verification. However, the authors had described the key limitation as the issue of domain adaptation which indicated different data distributions during verification and validation and further pointed out that the reason behind this was the changing behavior of an individual known as temporal aging. They suggested that future work should focus on more adaptive and advanced verification methods. Their analysis provided a direct validation to our problem known as temporal aging. Our work solves this problem by introducing an evolutionary drift adaptation approach where we treat behavioral shift as temporal variation and introduce adaptability.

Ryu et al. [32] in their paper provided a detailed description on the structure, shortcomings and existing challenges of biometrics. The researchers main focus was to demonstrate the current situation of adaptive biometric verification and to analyze how the infrastructure is coping with the variety of changes. Additionally, they performed a thorough analysis by observing 36 papers within the timespan of 2011 and 2021 [32]. This analysis was conducted through the help of taxonomy where some rules were maintained such as - the level of function of where they carried out the adaptation like matching or feature extracting, the used approaches like for overwriting a template, the actual reason for adaptation for safety and usage. Their findings indicated that physiological biometrics were concerned by almost all adaptive systems and template aging was the general approach for adaptability. However, the authors reported several issues such as - absence of a massive dataset which was required for validation of the adaptive infrastructure and lack of evaluation metrics specifically designed for adaptive infrastructure which created difficulties among different approaches in these systems. Their study provided a complete and descriptive validation of our section of research which is behaviour phylogenetics. Furthermore, they found the issue to solve temporal aging and demonstrated the importance of further updated methods. Our work develops a drift adaptive model which will consider the natural changes of a person as expected behavioral drift from time to time instead of avoiding it. Hence our proposed model results in a complete solution to temporal aging resulting in providing better robustness.

A survey was conducted by Baig and Eskeland [22] to provide a complete and detailed overview of Continuous Authentication (CA) including the existing problem, unique approaches and future working field in the context of cybersecurity. The researchers primary focus was to analyze a variety of CA approaches through the lens of context aware and physiological modes. Additionally, they performed a detailed survey to understand the capabilities of these modes through usage, safety and privacy. Moreover, they gathered several performance metrics to provide a transparent view. In their study, they found that well known physiological biometrics such as face detection had comparatively low EER in the range of 3.92-7.92% [22]. But the main limitation remained in their static trait and the change of being impersonated. On the other hand, other biometrics like keystroke dynamics were smooth but still they suffered from issues like high EER of 6.9% and poor accuracy [22].

Hence the authors concluded that approaches of different methods of verification should be based on the situations and they further pointed out that a proper trade off needed to be maintained between different advanced methods of privacy and the infrastructure that used multi characteristics for future work. The overview of their survey formulated the problem of our research in behavioral biometrics. Whereas they found limitations of lower accuracy and higher error because of temporal aging, our work addresses these issues by developing a drift aware adaptive infrastructure which treats changes of behavior as an evolving trajectory resulting in providing better security and a robust solution.

In their research, Solano et al. [25] proposed an unique architecture based on Siamese One-Shot Neural Network (SOS-NN) to make the verification process faster by utilizing less user interaction info. The authors pointed out the major limitations of behavioral biometrics such as - requirement of unique model training for each individual and long session of interactions. Their research was based on the One Model for All (OfA) framework which did not require model retraining for each person. Additionally, their approach was based on a Siamese infrastructure that incorporated two different subnetworks which extracted the features. For web and mobile environments, these subnetworks further mapped 28 and 52 features respectively into a latent space. Moreover, their system measured the distance between two vectors and a matching score was provided by the decision network. For the optimization of latent space, they integrated semi-hard triplet loss which differentiated an imposter by minimizing and maximizing the positive and negative distance sample respectively. Within a window of 60 second and 10 shot comparison, their model achieved upto 99.8% accuracy in mobile environments. In the few-shot cases, their model achieved 92% accuracy with 5 seconds interaction in mobile environments. In web environments, their framework showed promising results by achieving peak 90.8% accuracy. Additionally, their model was consistent by maintaining an accuracy range of 63-68% in 1 shot scenarios against a dataset of 125000 users and it further raised up to 79% accuracy in 10 shot cases which indicated its scalability. However, the authors noticed a major performance drop in web data compared to mobile data which indicated more data requirement for web models to identify the user perfectly and they addressed the speed-accuracy tradeoff and static characteristic of the model was the reason behind this. Though their model can be the perfect standard for our verification process, their major drawback actually lies in the lackings of large time adaptability as their model did not consider the changes over time which is known as temporal aging. Our work will address this issue by using a frozen backbone to model temporal dependencies in keystroke data. Whereas their siamese architecture is static and vulnerable to drift of behaviour, our model uses evolutionary adapters for real-time adaptation similar to the users.

Chen et al. [35] in their research introduced an advanced deep learning based infrastructure named DESTER (Dual Stream Transformer based Emotion Recognition Model) to demonstrate the drawbacks of common machine learning in the field of behavioral biometrics. Whereas previous existing researches were dependent on conventional classifiers and manual based feature engineering, the authors proposed Dester model was the first to incorporate dual stream based transformer architecture to detect emotional conditions through typing styles. Their unique creation was

these two streams which were temporal stream and channel stream used to capture complex dependencies within keystroke dataset. Temporal stream utilized blocks of multi scale LSTM (Long Short Term Memory) and multi head self attention to find inter connection between successive keys. In the channel stream, blocks of multi scale CNN were used to find inter relationships between features within the embeddings of keycode and timing. Additionally, they used Gaussian Range Encoding (GRE) which provided the transformer better understanding of individual keystroke position according to its neighbours. Moreover, their framework achieved a promising accuracy and f1 score of 99.44% and 0.989 respectively in identifying five emotional states. Compared to existing baseline frameworks such as LogNNet and MIL-SVM, DESTER demonstrated a huge performance improvement by 22.96-66.04%. Furthermore, their model provided consistent performance among all states within 90% overlap and 40 keystrokes sliding window without any major differences. However, they indicated difficulties of deploying such a complicated framework on mobile and their model was only evaluated on datasets where texts were fixed. Hence, they suggested future work should focus on handling dataset which contains varieties of text length and developing a model that detects different types of real world typing. Though the discussed framework had shown its effectiveness for typing analysis through transformers, it could not detect drift of behavioral characteristics over time and was dependent on CNNs and LSTMs which caused catastrophic forgetting. Our proposed framework utilizes PBT and evolutionary adapters which enables our infrastructure to evolve according to the individual’s habits. Unlike the DESTER model, our system prevents attack from imposter tries to slowly copy individuals’ change over time through trust anchors which ensures seamless connectivity to individuals actual identity.

A survey was conducted by Subash et al. [44] about behavioral biometrics in the critical and significant domain known as the evolution of user behavior. By providing references of 43 billion dollar identity fraud cases and 75% rising in phishing attacks, the authors demonstrated their security concerns because of the industry’s heavy dependence on static verification models. The researchers used prisma based systematic demonstration approach on 81 effective articles and further divided their research into 6 domains which were AI layouts, datasets, assessment criteria, use case and the most significant one was adaptability. The findings from their studies showed that existing State-of-the-Art (SOTA) models achieved under 3% EER in randomized split of train-test but they still could not address intra class variability and almost all existing researches did not provide any implementation of adaptive strategy in the field of mouse dynamics. Additionally, they addressed that behavior evolution or learning curve continuously increased FRR which made the static template outdated. Moreover, the authors addressed several issues such as - non availability of longitudinal datasets of people, heavy computational cost because of continuous model retraining and the vulnerability of adaptive systems from poisoning attacks. They suggested building a durable and robust verification infrastructure to overcome the existing Online Gradient Descent (OGD) models so that it can still evolve without diminishing any discriminative power. The issue of balancing adaptability and model stability is solved by our proposed framework as we are merging evolutionary adapter and transformer backbone. We present the gradient free adaptation for non-stationary surroundings by utilizing PBT. Furthermore, we

incorporate a trust anchor in the fitness function of our model to prevent slow drift fraud attack and provide the longitudinal behavioral memory by maintaining trust anchor in enrolment time and monitoring drift from time to time.

The effectiveness of Recurrent Neural Networks (RNNs) to ensure reliable information infrastructure by behavioral biometrics was demonstrated by Azanguezet Q. et al. [38]. The researchers tried to overcome the issues of static login by introducing a constant and transparent authentication system that observed individuals typing patterns in a free-text environment. The author’s main ideology was to utilize the sequential characteristics of keystroke info to create a unique biometric profile for each user. Whereas traditional static verification frameworks only authenticated at the beginning, their model conducted Continuous Authentication throughout the entire session. Their framework was based on the architecture of LSTM because of its capacity to capture long range connection in input data and save data from previous states. Their approach captured two feature types which were textual features (embeddings of character that extracted semantic connections) and temporal features (digram, press time, latency, fall time). These features then forwarded to an architecture of three layers where attention layer, embedding layer and LSTM encoder-decoder were incorporated. Their framework was evaluated on a dataset which was gathered over 30 days from 8 users and provided high discriminative power by yielding an accuracy of 96.67%. Their model exceeded the SOTA baselines such as - CNN-GRU hybrid (92.3% accuracy), partially recurrent neural network which had 97.54% EER compared to their 96.67% accuracy. Additionally, strong convergence was demonstrated by their learning curve with 98% training accuracy. However, they addressed that small dataset limited their framework’s generalization and found that transient instability because of different physical conditions or emotions made the keystroke data vulnerable. They further suggested that future research should focus on an adaptive learning approach that can consider changes of behavior over time as their framework was still static after training. While their work demonstrated the significance of LSTM, the inability of handling behavioral drift caused performance degradation that still needed to be addressed. Our model considers temporal dependencies as we use frozen backbone instead of LSTM. Moreover, usage of evolutionary adapters enable our framework to evolve along with users and trust anchors to prevent catastrophic forgetting and imposter attack.

In search of the lackings of the current Continuous Authentication frameworks, Senarath et al. [33] presented a multimodal infrastructure that combined keystroke dynamics with Inertial Measurement Unit (IMU) info such as how an individual controlled their device as this integration provided better dependability. The main focus of this paper was the first proposal of the Spatio Temporal Dual Attention Transformer (STDAT). Compared to conventional transformers that could capture only linear data, the authors introduced STDAT that continuously observed the channel and temporal axis indicating the relationship of the dimensions of a sensor and event sequence respectively. Their infrastructure incorporated Gaussian Range Encoding (GRE) which could process a series of positions rather than only a single point. By providing better positional data, their model could better interpret the natural change in human behaviour. Additionally, their framework was trained through triplet loss where a vector space was maintained so that sequences from

the same individual and imposter were clustered and ignored respectively. Their model demonstrated promising outcomes by achieving EER of 1.80% and 2.95% against the Aalto and HuMIdb dataset respectively. Additionally the higher inter class detachment was indicated through the silhouette scores which were 0.75 for HuMIdb and 0.91 for Aalto. Moreover, these outcomes demonstrated the significance of multimodal fusion to detect the fraud as the use of IMU data significantly lessened the EER from 5.10% to 3.62%. However, they found that the primary lackings of their model was low sample size because to perform generalization, DL models need massive data. Though they achieved another 12.16% rise on EER by incorporating transfer learning on pretrained dataset, the main issue still laid on the static characteristics of the model as after deployment, the weights became useless and the performance fell because of the natural change of user behavior. Our model addresses this rigidity by incorporating PBT and evolutionary adapters. The issue of catastrophic forgetting is mitigated by this integration and further provides real time adaptability. Moreover, we also use trust anchors so that our framework stays true to users actual identity by protecting them from imposters who try to impersonate users' natural behaviour drift.

In their research, Lis et al. [31] analyzed the integration of behavioral biometrics while using partial passwords by continuously authenticating users typing style to address the issue of traditional verification where imposter had accessed sensitive information based on their observing tricks into the partial password or masked queries. The author's main intention was to conduct few shot identity authentication through the usage of Siamese Neural Network (SNN) that evaluated a similarity metric to decide if two typing patterns belonged to the same user. Using dwell time and flight time, they created a feature set and forwarded input to an embedding space through the utilization of two connected subnetworks. They measured the L1 distance with the projections to determine the similarity based on a decision network that provided the output of final identification. Additionally, their 'one model for all' procedure further analyzed by n shot classification by comparing current and a window of n previous logins. Their SNN provided a promising accuracy of 86% in full password verification from 5 past logins that exceeded the existing baseline of scaled manhattan distance. However in partial password authentication, the accuracy fell down to 77% in 38 logins that indicated that random characters interrupted an individual's natural typing pattern because they had to stop and think. The researchers found the durability issue of their model and stated that changes in behavior because of tension or stress makes the keystroke data unstable that caused model retraining. Though their model provided a strong base of few shot learning, it was still static because of its dependence on certain embedding space and did not evolve over time. Hence they suggested future work should focus on models that can handle intra individual variability over longer periods. Our model overcomes this static characteristic by learning user behavior more precisely through the frozen backbone. Moreover, our model introduces adaptability by the integration of evolutionary adapters so that it can evolve with the user and ensures stability by handling continuous adaptation through trust anchors.

Wang et al. [27] suggested a hybrid scene user authentication model, named SIU-RUA, which incorporates both keystroke and mouse dynamics. Their model di-

rectly considers the variance posed by various user interaction scenarios which is a major problem that greatly reduces authentication performance in real-world environments. For keystroke dynamics, high-dimensional features are systematically extracted according to the key hold time and inter-key latency, whereas for mouse dynamics, movement-related features based on speed, direction, action type, and temporal histograms are considered. They used these features to study and consider variable length time windows to model realistic user behavior. In order to reduce scene-related variability, the authors proposed the use of a correlation-based feature selection strategy. In this context, the adapted mutual information was used to select scene-related features and user-related features, which were further fused using multiple kernel learning (EasyMKL) and classified using a support vector machine. They do their experiments on a self-collected dataset of 41 users engaging in four different tasks. In terms of accuracy, SIURUA demonstrated the highest average accuracy of about 82% to 84% when using a timeframe of 300 seconds. This result is better than any other conventional classification by a large margin. Moreover, SIURUA successfully minimized the false acceptance rate and false rejection rate. This work is a stronger analysis that has explicitly separated the scene-independent and user-independent behavioral traits but this also has some limitations like other works. The use of static feature representations, manually designed features, diversity of subjects, and offline training that does not consider a drift in behavioral aspects across time are the major limitations. Through our research, we may go beyond the context of fixed feature filtering and explicitly model behavioral drift using an approximately frozen Transformer backbone with some lightweight trained adapters. This will help to continuously update the behavioral drift with any retraining of the entire model, which may overcome the temporal constraints of SIURUA, a hybrid-scene authentication model.

Ranganathan [23] proposed a hybrid deep learning model to authenticate behavioral biometrics, which comprises Variational Autoencoders (VAEs) and Graph Neural Networks (GNN). They use this model to enhance the accuracy and strength of constant authentication of users based on interaction rates like keystroke and mouse related activity. The essence of the research was to resolve the shortcomings of conventional machine learning models and these models fail to represent complex and dynamic relationships within the behavioral biometric data. For this reason, the models have high false acceptance and false rejection. Their proposed system uses preprocessing techniques to standardize the behavioral features and later on, a VAE encoder-decoder architecture learns compact latent representations. Through KL-divergence regularization, VAE architecture preserves salient behavioral characteristics and minimizes the noise. Each latent embedding of the model is considered as a graph and the graph is processed using a GNN to model relational dependencies between behavioral events. The process of this model does work via message passing and aggregation, which enables an improved discrimination between an actual user and an impostor. Learning rate, latent dimensionality, and network depth are the hyperparameters and tuned by using Ant Colony Optimization (ACO) which helps to improve the performance of the model. Compared to previous studies, this paper offers a more expressive representation of simultaneous latent behavioral attributes. However, this study is constrained in the sense that it relies on a single behavioral modality, off-line training processes, and does not follow longitudinal analysis. This

model does not explicitly include mechanisms that can be used to address behavioral drift over time, though behavioral variability is recognized. Furthermore, their proposed system seems relatively stable behaviorally but does not address continuous adaptation in the real-world scenario. Through our research, the temporal decay that current VAE-GNN-based authentication systems cannot address will be mitigated. The reason is that we are using a frozen backbone with lightweight adapter modules whose parameters are trained using population-based evolutionary learning.

The very new approach proposed by Mazumdar and Sundaram [43] is the one in which they combine a hybrid patch-based Siamese CNN and a transformed form of the Vision Transformer (ViT) end-to-end continuous mouse dynamics authentication framework with Gramian Angular Summation Field (GASF) representations. This is to record the spatio-temporal dependence of velocities in such a way that it does not involve tuned thresholds manually. Transform the raw sequence of the mouse movements into velocity-based GASF images so that information on timestamps can be directly incorporated. To overcome the lack of strong positional correlation between continuous mouse behavior, the resulting images are broken into nonoverlapping patches. In their model, they introduce a Siamese CNN with shared weight and for training it, they use nonlinear patch embeddings and a jointly trained ViT. A class token does not compute attention between patches taken on samples of a particular user in a randomized patch selection strategy that also minimizes class imbalance by pseudo-imagery generation. Their experiment is performed on Balabit, DFL, and Chao Shen, these three publicly available datasets and their results demonstrate that their proposed framework outperforms existing options in deep learning, which includes CNN-based and standalone vision transformer models. The performance of the method in terms of authentication is high with AUC/EER of 0.9884/0.0402 and 0.9320/0.1102 on Balabit (Test-1 and Test-2), 0.9971/0.0198 on the DFL data and 0.9798/0.0532 on the Chao Shen data. These findings imply high-level discriminative ability in a wide range of mouse dynamic situations. Moreover, the architecture shows enhanced resistance to imitation and surrogate-based attacks, and this shows how its architecture is more discriminative in its ability to model velocity correlations and enforces similarity learning at the patch level, which therefore enhances generalization to intra-user variability. The proposed system also has a number of limitations. It is based on one modality of behavior and follows an offline training paradigm where the model parameters are not adjusted after being deployed. Moreover, the authentication structure is based on binary classifiers that belong to the user, and it restricts scalability. Above all, the system does not have adaptive mechanisms for the long-term behavioral drift as a result of other factors like aging, acquisition of skills, or environmental variations. Furthermore, the framework manages variability intra-session, although it assumes that learned representations remain constant over time, helping to limit its usage in real-world continuous authentication trials. To overcome these limitations, our thesis contributes to the state-of-the-art by establishing a frozen backbone with lightweight adapter modules. The optimization of these adapters is a constantly evolving process that uses the population-based evolutionary training, which allows dynamic personalization of the model without the need to retrain the entire model. This evolutionary adaptation plan promotes robust, scalable, and long-term continuous authentication in the face of changing user behavior.

2.3 Summary of Key Findings

Throughout the literature review, the most highlighted aspect was the lackings and weakness of the existing static models of authentication. Several authors such as Baig and Eskeland [22], Patel et al. [11], Ryu et al. [32] mentioned temporal aging explicitly that indicated the natural behavioral changes of an individual as a remaining crucial challenge. Infrastructures which were developed on established templates like one class SVM demonstrated by Ceker and Upadhyaya [10] or the snapshot based system mentioned by Sagbas and Balli [36] accomplished higher accuracy but the main issue was they were brittle by nature. They needed to be trained again in order to differentiate between an anonymous intruder and an actual natural change. Additionally, few shot based methods like the one mentioned by Fereidooni et al. [29] provided scalability but they did not capture the evolution of behaviour and depended on fixed samples. This study considers the change of behavior as the main feature that models through the lineage of behavioral evolution instead of non relevant training periods and treating it as an error. Moreover, recent research on the modeling of patterns provided the effectiveness of the transformer based infrastructures. Senarath et al. [33] demonstrated a dual attention and spatio-temporal based infrastructure named Behaveformer that extracted large dependencies in behavioral data. But the major drawback remained in the inability to address behavioral drift adaptation. Similar to transformer based infrastructure, Chen et al. [35] introduced a dual channel based transformer named DESTER for identifying emotions. But the authors later found the issue of catastrophic forgetting when the behavioral data of a user changed because of technical and environmental purposes. In multi modal scenarios, sometimes severe challenges arise for proper accuracy which was demonstrated by Wang et al. [27] through their system that selected modalities based on the scene-irrelated concept. However, their infrastructure was heavily dependent on static feature representations that could not address temporal aging. In their research on Siamese Neural Network, Solano et al [25] and Lis et al. [31] stated promising possibilities on a few shot verification systems. But their studies had severe limitations because of rigid architecture and failure to constant evolution over time. Furthermore, large discriminative power was highlighted for keystroke and mouse data by the research of Ranganathan [23] and Mazumder and Sundaram [43] on VAE-GNN infrastructure and Vision Transformers (ViT) respectively. But their models assumed the learned features were fixed which was impractical in real life and did not address behavioral drift. Finally, Subash et al. [44] demonstrated from the concept of behavioral biometrics and learning curve that continuous retraining led to rise in the rate of False Rejection Rate (FRR). Hence the authors suggested focusing on an infrastructure that could evolve through a different mechanism other than retraining. This research incorporates these findings by using population based training (PBT) for the evolution of the adapters and frozen backbone structure to capture temporal dependencies. While existing traditional infrastructure fails to address temporal aging, this framework considers these changes as evolving trajectories and prevents frog boiling attacks through the integration of trust anchors so that the model can evolve alongside the user.

Chapter 3

REQUIREMENTS, IMPACTS AND CONSTRAINTS

3.1 Final Specifications and Requirements

3.1.1 Stakeholder Feedback and Requirement Elicitation

The requirements and functional specifications of the proposed continuous authentication system were identified through a series of consultations with relevant stakeholders in the research and development process. The main stakeholders in this case were the research supervisor, scholarly judges and those who would ultimately be end-users of authentication systems in security-sensitive digital architectures. The responses given by these stakeholders pointed to the necessity of having an authentication mechanism that could help increase the level of security without creating additional cognitive or operational burden to the user. The conventional means of authentication were determined as invasive and inadequate in curbing the after-login security hazards, underscoring the need for persistent and non-destructive user authentication.

The stakeholder contribution also indicated that the authentication system should be able to be flexible to meet the natural changes in behavior over time. There were fears about the possibility of false rejection of legitimate users as a result of behavior drift especially in the long-term usage scenarios. Consequently, the ability to be flexible and stable was established as one of the functional requirements. The stakeholders further noted the need to have interpretability and reliability of authentication decisions, particularly in academic and real-life security settings where system behavior is required to be explainable and repeatable. The lessons were applied to design functional objectives intended to remain in constant monitoring, temporal behavior modeling and drift-related decision-making. It was determined that the system should run in the background as a result of the feedback of the stakeholders and it was mandated to keep an ongoing analysis of the keystroke dynamics during normal typing after an initial fixed-text check, without requiring explicit re-authentication prompts. The system should be able to process sequential data in an efficient way, extract the relevant timing related features of the keystroke events, and produce the authentication scores that reflect the probability of a user behaving in a legitimate way. These specifications were the direct cause of the design of the

TCN-based temporal modeling concept that was used in the present study.

3.1.2 Proposed System Functional Specifications

The functional specifications of the proposed system are to conduct continuous behavioral authentication via use of the keystroke dynamics data. The system is required to facilitate the acquisition and pre-processing of raw keystroke events, such as the time of key press and release, out of which key temporal features are derived. These properties must be standardized and organized to constant length sequences that can be analyzed over time. For representing the sequential dependencies and behavioral patterns over time, we need an architecture that will be memory efficient and the fastest for training. Because of this reason, we should use the TCN architecture within the system.

One of these functional requirements is the fact that the system should be able to assess user behavior on an ongoing basis during an active session, as opposed to using a single authentication. The system needs to create authentication decisions by considering the changing behavior patterns and consistent with identity characteristics learned in the past. The system must also be in a position to receive incomplete or noisy input information without affecting the overall reliability of authentication. The functional design should also be extensible, which means it should be able to accommodate any other behavioral modalities in case they are necessary.

Results produced by the authentication mechanism are supposed to be seen as either acceptance or non-acceptance judgments with determined or variable thresholds. enables the system to be evaluated and analysed with regard to the currently existing methods of authentication through popular performance measures of FAR, FRR, and EER.

3.1.3 The Review of the Applicable Standards, Professional Codes, and Regulatory Requirements

The proposed authentication system was designed and implemented using a review of the relevant standards, professional practices, and regulatory considerations regarding the biometric systems and data security. Among the international standards used in the processing of biometric data, protection of biometric information during its lifecycle is an essential point to consider, as well as data acquisition, storage and processing. Consistent with these values, the offered system analyzes the keystroke dynamics data in a feature representation, bypassing the need to store raw keystroke sequences at all feasible to minimize privacy vulnerabilities.

Cybersecurity and designing authentication systems. Professional best practices emphasize the need to design such systems so that user burden is minimized and the security measures are high at the same time. The perpetual authentication strategy that is embraced in this study is in line with the tenets of zero-trust security, which recommends continuous authentication instead of one-point authentication. Also, the ethical issues concerning the consent of the user and data processing were considered, with the behavioural data employed in the research being obtained via

publicly accessible data sets and processed only with the purpose of academic study.

The existing protection laws of data security and the general data protection regulation influenced the design selection of the proposed system. Specifically, that is why the well-known principles of data minimization, anonymization, and purpose limitation were taken into account when handling and processing data. These considerations made sure that behavioral data were utilized exclusively in terms of authentication analysis. Even though the system is being created as an academic prototype, compliance with these regulatory factors will guarantee that the suggested solution can be implemented in a real-life setting. With a process of integrating these standards and professional guidelines into the functional specification process, the research will demonstrate that the proposed system is both technically and ethically and legally accountable.

3.2 Societal Impact

The proposed adaptive behavioral authentication system a shift from one-time and fixed verification systems, to the constantly learning system, which adapts as the user changes as well. Through the capacities to model temporal patterns of behavior and adjust to the gradual changes. The system helps to enhance the digital trust, and also brings forth new considerations associated to the society in terms of privacy, fairness, and human-computer interaction. The social implication of this study cuts across operational security, user satisfaction, legal standards and cultural embracement of continuous authentication systems.

3.2.1 Operational and Positive Societal Impact

Socially and operationally, the framework improves the security of the digital world by ensuring continuous authentication as opposed to the initial decision of access only. This method is specifically applicable in the areas of high sensitivity like financial systems, healthcare delivery platforms, and work-at-home environments, where hacking into the system when the user is still active can result in serious damages. The system helps avoid the possibility of hijacking a session and impersonation attacks without the need to use explicit user actions repeatedly by constantly validating behavioral pattern.

The proposed framework is also more usable and accessible. Behavioral authentication is passive and background-based unlike physiological biometric systems, which require the user to be actively involved and therefore can be inconvenient and exhausting. This minimizes mental and physical load to users to help ease the interaction experience and make the system more accommodating to people who might have problems with conventional biometric inputs.

Moreover, the adaptive learning mechanism deals with the issue of fairness in terms of the behavioral drift. The human behavior indicates a natural development that occurs owing to experience, changes in the environment, or the differentiation in the devices. The system reduces the chances of legitimate users being unjustly rejected

as the system enables incremental adaptation as opposed to strict template matching. This enhances inclusiveness in the society as the natural behavioral changes with time are avoided thereby leading to marginalization.

From the operational point, lightweight adaptive components are computationally more efficient than the repeated full model retraining again and again. This renders the framework more practical to be deployed in real world, especially in a setting with few computational resources.

3.2.2 Existence of Negative Externalities and Social Costs

Along with its positive aspects, the framework also presents the possible risks that will be necessary to handle. One of them is the susceptibility to gradually evolving imitation attacks, also known as frog-boiling attacks, where an attacker gradually changes behavior in order to sound like a legitimate user. Although the dynamic character of the system enhances the response to behavioral change, the adaptability may pose risks on such attacks, when it is excessively adaptable without protection. This will require some form of stabilization mechanism, confidence level or trust constraints when adapting.

The other important issue is connected with privacy and the sensitivity of data. Persistent authentication necessitates constant monitoring of behavioral trends that can result in the development of extensive interaction information. Mishandling of such data may lead to the risks of profiling or illegal inference of users. Thus, the concept of data minimization, anonymization, and purpose limitation should be firmly followed to reduce the risks of legal and ethics.

It can also have potential psychological and cultural effects. Uninterrupted monitoring, even in passive use, can cause some of the users to experience discomfort or increased sensitivity to being watched, which can affect the behavior of a person. The idea of cultural awareness towards surveillance and autonomy is subject to change and cultural diversity of surveillance systems can only be widely accepted in case such systems are transparent, consent is well informed, and boundaries of the systems are clearly communicated.

3.3 Environmental Impact and Sustainability

The environmental aspect of this study is mainly related to the computational energy use in building and training of models. The modeling of behavior sequences and adaptive learning mechanisms are costly in processing resources especially when it comes to the first training and optimization steps. The usage of high-performance hardware, including GPUs, leads to the use of more energy and related carbon emission, particularly when data centres are centralized.

Nonetheless, the presented framework is geared towards sustainability. In contrast to traditional behavioral authentication systems, which use regular full retraining to overcome behavioral drift, this method focuses more on the extension of the model

part through the use of lightweight models. This framework has a huge computational cost reduction and power consumption with time because only the small adaptive modules are updated and the whole model is not retrained.

This is also enhanced by the possibility of the localized or on-device deployment, which reinforces the sustainability of the system. The closer the behavioral data is to the source of data, the less continuous transmission of data to centralized servers, which results in the less network overhead and dependence on infrastructure that consumes a lot of energy. This design trend will enhance the project to be deployed in long-term resources-constrained environments with acceptable environmental costs.

Also enhanced flexibility will minimize unwarranted user lockouts due to strict thresholds. The framework indirectly results in a reduction of false rejections and recovery cycles, which in turn leads to a decrease in hardware replacement rates and electronic waste. Altogether, the initial training stage is costly to the environment, but the adaptive and resource-efficient design helps to sustain the support of the authentication solution in long-term use.

3.4 Ethical Problems and Professional Responsibility

The presented adaptive behavioral authentication framework poses a number of ethical concerns that should be taken into account to implement and design it in a responsible manner. The nature of the continuous authentication system is a significant span of time monitoring the actions of the user, which raises the issue of privacy, autonomy, and informed consent. Unintentional profiling or user identification Behavioral data, including keystroke and mouse interaction patterns, can be used to profile anyone, unless it is correctly managed.

Ethically, the system designers have a duty of making sure that only minimal and necessary behavioral features are handled in the process of authentication. There should be a focus on anonymization, abstraction, and limitation of purpose to ensure that the data collected is not abused. The participants should be well informed on the nature, purpose and duration of data collection and participation on the basis of explicit consent with the option of withdrawal or removal of data.

Discrimination and equality are other ethical issues. Adaptive learning systems can inadvertently promote the pre-dominant styles of behavior, which may be at a disadvantage to users with unusual interaction or physical limitations. Professional responsibility demands that such systems should be tested to various groups of users so as to discourage discriminatory results and provision of equitable access.

Lastly, the adaptive aspect of the structure presents security-related ethical dangers, including slow imitation attacks. Designers have a moral responsibility to ensure that they achieve flexibility and protection against exploitation. The inclusion of controlled boundaries of adaptation and ongoing evaluation is a sign of responsible engineering practice that seeks to ensure that the user is not hurt.

Comprising privacy, fairness, transparency, and security, the suggested framework meets the ethics in the field of professional conduct and gives priority to responsible innovation in behavioral biometric authentication.

3.5 Project Management Plan

3.5.1 Final Specifications and Requirements

Specifications and requirements of this research project came about after analysing the original research plan and considering it with practical implementation limitations and findings of the experiment. The project is finally concerned with the design of a two-level continuous behavioral biometric authentication system, which is a TCN type architecture with the help of keystroke dynamics data. Level 1 performs an initial fixed-text verification; Level 2 performs free-text continuous authentication in the background during normal typing (no extra prompts), allowing real-time monitoring of typing behavior during a session. One of the main requirements is the ability to model the temporal dependencies in keystroke patterns so that the short-term fluctuations and the long-term behavioral trends can be captured.

The system should also address the natural behavioral drift and ensure that the core identity characteristics of the real users are maintained. The performance requirements are that the FAR and FRR should be minimized to guarantee security and usability. Moreover, the system should be resistant to noise, missing values and variability that are caused by various typing styles and environmental factors. Computational feasibility was also viewed as a major non-functional requirement to make sure that the model can be trained and the evaluation completed effectively without the need to retrain the model quite often. These last requirements represent the shift in the conceptual design of the first design of a transformer to a TCN-based architecture, which is more adapted to temporal modeling with reduced computational complexity.

3.5.2 Analysis and Design Methodology

To facilitate the development of the proposed system, an iterative and research-based analysis and design methodology was used. The analysis stage entailed an in-depth examination of behavioral authentication methods of biometric authentication, specifically, the issues of behavioral drift, continued authentication, and time sequence modeling. The data characteristics of the dynamics of keystrokes were analyzed to understand the timing dependency, inter-session variability, and noise and outliers. Based on the analysis, the TCN architecture was chosen because it allows capturing long-range temporal dependencies with dilated causal convolutions without compromising its training behavior. The design methodology was focused on modularity, according to which the system was disintegrated into discrete modules, including data preprocessing, feature normalization, sequence windowing, TCN-based temporal modeling, and continuous authentication decision logic. The design was a modular one and thus facilitated incremental development and testing and

architectural refinements could be added according to the outcome of the experiments. The iterative method made sure that the theoretical considerations as well as the empirical performance were followed in the final system design.

3.5.3 Resource Identification and Assessment

Our proposed system was designed and analyzed using an iterative and research-based approach. The analysis stage consisted of a deep study of the behavioral biometric project, which was implemented successfully due to the identification and efficient use of available resources. The main constraints were human resources where we were in charge of developing, testing and documenting the system which was supervised by the academic advisor who provided technical advice and validation. The software resources in this group were open-source programming languages and deep learning frameworks, especially Python and PyTorch, and supporting libraries of data processing and visualization. The hardware resources were a university research lab that had the ability to support a GPU. The GREYC keystroke dynamics dataset [4] and the University of Buffalo[12] Keystroke Dynamics Dataset in CSV and text formats are the data resource of this study, which have all the extracted keystroke timing features that can be used to conduct temporal modeling. Since the model was trained and tested using publicly available keystroke dynamic datasets, it is cost-effective. The evaluation of such resources revealed that they were sufficient to complete the thesis with all the project goals during the designated academic timeframe.

3.5.4 Project Schedule, Budget and Revision Plan

A well-organized timeline was used to handle the project, making the research activities split into consecutive stages, such as requirement analysis, literature review, model design, implementation, experimentation, evaluation, and thesis documentation. Every stage had a set time and the progress was evaluated at specific times to ensure that the timeline of the whole project was followed. This systematic time scheduling system made it possible to track progress in a systematic manner and detect possible delays in a timely manner. The experiment was carried out on a low-budget academic research. The project utilized most of the tools and libraries that were open-source, and the data was publicly available, which greatly lowered the cost of the project. Computing resources and the overall academic infrastructure were the major costs. Resource planning was aimed at the maximization of the utilization of the current institutional and personal computing resources to make the project cost-effective. The combination of schedule, budget, and resource management was important to make sure that the project was possible and sustainable throughout its lifecycle.

3.5.5 Revised Project Plan and Rescheduling

Throughout the research, the original architectural design was amended based on operational logic (efficiency in computation), flexibility and experimental viability. The initial idea of using transformer architecture was substituted by a TCN-based one because the complexity and retraining cost of transformer models were assessed.

It also necessitated an architectural revision, which included a revision of the design and implementation stages in the project plan. Refactoring was not done to restart the project but to maintain the already done work, like the data processing and feature engineering that was already done and updated to the new architecture. The remaining time was used to redesign the model structure and re-run the experiments with the help of TCN. This rescheduling was necessary to guarantee that the redesigned system was able to be completely implemented and tested within the given period that was given. The resulting adaptive project plan revision is based on a realistic approach of managing research, enabling the project to stay on course, and enhancing the technical appropriateness of the final solution.

3.6 Risk management

3.6.1 Identification of project risk

There are several project risks identified during the development of the continuous authentication system using keystroke dynamics. The primary technical risk is the wrong interpretation of the natural drift in behavior as malicious behavior, which leads to the higher false rejection of legitimate users. The other major threat is catastrophic forgetting, in which the model can lose already acquired identity features as new behavioral data are adjusted to. This scenario mostly occurs in the continuous learning process of user behavior.

Since keystroke dynamics data are noisy and at the same time contain missing values or outliers because of different typing environments, data-related risks were also one of the major risk factors. Another risk that was identified was overfitting because the typing behavior is unique, and the number of samples given to some users is limited. Another risk in terms of project execution, according to the identification done, was delays in schedule caused by architecture modifications, complexity in the experimental stage, and duration of hyperparameter optimization.

3.6.2 Risk Mitigation Strategies

In order to reduce the risk of mistaking a real behavioral drift, the system was such that the temporal dependencies over longer sequences are modelled, and the gradual behavioral shifts are represented without overreacting to the immediate changes. Controlled processes of training and carefully learning parameter tuning were used to counter this risk of catastrophic forgetting and stabilize and adapt. This assisted in maintaining patterns of core identities and in enabling the model to be instructed by new behavioral information.

The riskiness of the data was also addressed by using well-established preprocessing steps such as data normalization, treatment of missing values, and outlier-sifting in order to lessen the influence of noise in the training of the models. Early stopping and dropout were some of the regularization methods used to mitigate the risk of overfitting and enhance cross-user and cross-session generalization. The risks that were caused by the schedule were addressed through flexibility in the project schedule

and by including buffer time to allow unexpected delays due to architectural changes or due to protracted experimentation.

3.6.3 Residual Risk and Monitoring

Besides the thorough mitigation measures, there are still residual risks involved in behavioral biometric authentication systems because human behavior is dynamic and context dependent. Nevertheless, the constant tracking of the performance of the system across the sessions and users assisted in detecting degradation patterns and new problems at the initial moment. The methodical approach of identifying, mitigating and monitoring risks contained the effects of these residual risks to minimal levels. On the whole, the risk management strategy that was adopted made the final research outcomes stable, reliable, and academically credible.

3.7 Economic Analysis

The economical analysis of this research is concentrated on the feasibility of developing the proposed infrastructure, the larger economical worth through improved security and the cost savings by its infrastructure throughout the entire operation.

3.7.1 Development and Resource Costs

The main expenditure is focused on the hardware support and data management. The training phase of the frozen backbone of the TCN requires a high performance GPU. A NVIDIA RTX 3060 is a minimum requirement for the development which costs around 30,000 BDT. However, this research utilizes the pc of the research lab in university consisting of RTX 4080 Super that minimizes this direct cost. In this research, the GREYC[4] (fixed-text) and Buffalo[12] (Free-text) datasets are used that consists of detailed behavioral keystroke data. To manage this data precisely, high speed NVMe SSD and backup systems are required that cost around 15,000-25,000 BDT. Moreover, employing an open source stack that includes Ray tune, Python and necessary libraries brings down the software cost to zero.

3.7.2 Operational Efficiency and Cost-Benefit

The introduced infrastructure is highly cost effective compared to the existing deep learning framework's design. Existing traditional frameworks need constant retraining that requires heavy GPU power to address behavioral drift. This study utilizes frozen backbone and significantly lowers the number of Floating Point Operations (FLOPs) through the evolution of adapters only. As a result, the energy consumption is reduced drastically. Moreover, the need for security engineering is less likely because of the advanced threshold tuning procedure of population based training.

3.7.3 Socio-Economic Impact

Through better productivity and the loss prevention system, this framework has a large economical worth. This framework provides constant monitoring in zero-trust surroundings that restricts the imposter from important and valuable session

takeover. As a result, multinational companies can potentially save millions by avoiding large penalties and data breaching. Additionally, the system lowers the FRR through adaptability. Hence, it saves time through the reduction of account recovery process by the employee leads to efficiency and better throughput. Moreover, the framework addresses behavioral drift properly that provides better longevity to input devices and saves cost by eliminating the need of purchasing new sensors.

Chapter 4

DESIGN PROCESS OVERVIEW

4.1 DESIGN PROCESS OR METHODOLOGY OVERVIEW

4.1.1 Fixed-text Methodology

4.1.1.1 Methodology Overview: A Dual-Stage Strategy for Drift-Resilient Authentication

The central challenge motivating this research is behavioral drift, the gradual, often unconscious evolution of a user’s typing patterns over time. Unlike physiological biometrics such as fingerprints or iris patterns, keystroke dynamics are rooted in motor-neural processes that adapt with practice, fatigue, and environmental factors [5]. Prior work reports that without explicit drift mitigation, authentication templates become increasingly obsolete over time [8]. More critically, our analysis of the GREYC [4] dataset (Figure 4.1) shows significant drift across a large fraction of users over the 110-day collection span, with typing speed improving over time, a phenomenon consistent with the Power Law of Practice observed in motor learning literature [1]. State-of-the-art keystroke authentication systems, including TypeFormer [37], TypeNet [20] and Time-Series Feature Network (TSFN) [41], report strong accuracy on cross-sectional benchmarks such as the Aalto dataset [18]. However, these systems are trained and evaluated on data collected in a single sitting, where the 15 “sessions” per user are merely consecutive sentence transcriptions with no temporal separation [18]. Consequently, they do not encounter and cannot be expected to handle the template aging that occurs when users return days, weeks, or months after enrolment. As documented in Table 4.1, no current SOTA architecture incorporates explicit drift-aware mechanisms or longitudinal evaluation protocols.

Figure 4.1 shows User typing latencies evolve significantly over the 110-day collection period, with 66.7% showing statistically significant changes ($p < 0.05$). This drift invalidates static template assumptions underlying current SOTA methods.

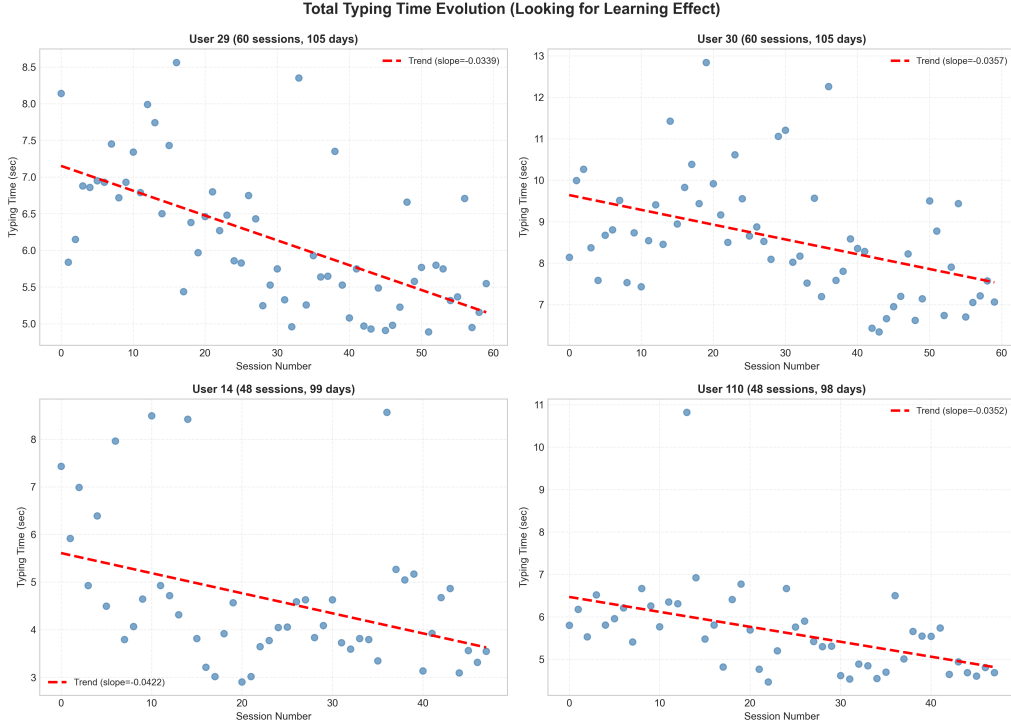


Figure 4.1: Empirical evidence of behavioral drift in the GREYC dataset[4]

Table 4.1: SOTA keystroke authentication comparison on GREYC [4]

Model	Year	EER	Drift-Aware?	Longitudinal Eval?
TSFN[41]	2025	19.31%	×	×
Type2Branch[39]	2025	19.14%	×	×
TypeFormer[37]	2024	19.44%	×	×
TypeNet [20]	2022	19.71%	×	×
Evo-BioTCN (Ours)	2026	3.77%	✓	✓

Table 4.1 shows all existing methods are evaluated on GREYC dataset[4] without behavioral drift, making their EERs incomparable to drift-aware evaluations.

To address this fundamental gap, we propose a **Dual-Stage Strategy** that explicitly separates two distinct concerns:

- General Feature Extraction (Static Backbone) :** A frozen Temporal Convolutional Network (TCN) learns universal temporal patterns from keystroke sequences — the rhythmic structures and timing relationships that generalize across users and conditions. Once trained, these parameters (θ_{static}) remain fixed, preserving the discriminative representations learned during initial training.
- User-Specific Drift Adaptation (Evolutionary Adapter) :** A lightweight adapter module ($\theta_{adaptive}$) is evolved via PBT to compensate for behavioral drift without requiring explicit re-enrolment. By operating in a low-dimensional

parameter space (512 parameters vs. 131,000+ in the backbone), this adaptation occurs efficiently and without catastrophic forgetting of the original biometric template.

This architectural philosophy mirrors how human motor memory functions: the cerebellum maintains stable procedural representations of learned skills while the motor cortex continuously fine-tunes execution based on feedback [3]. The complete system architecture — spanning enrolment, training, authentication, and evaluation — is illustrated in Figure 4.2.

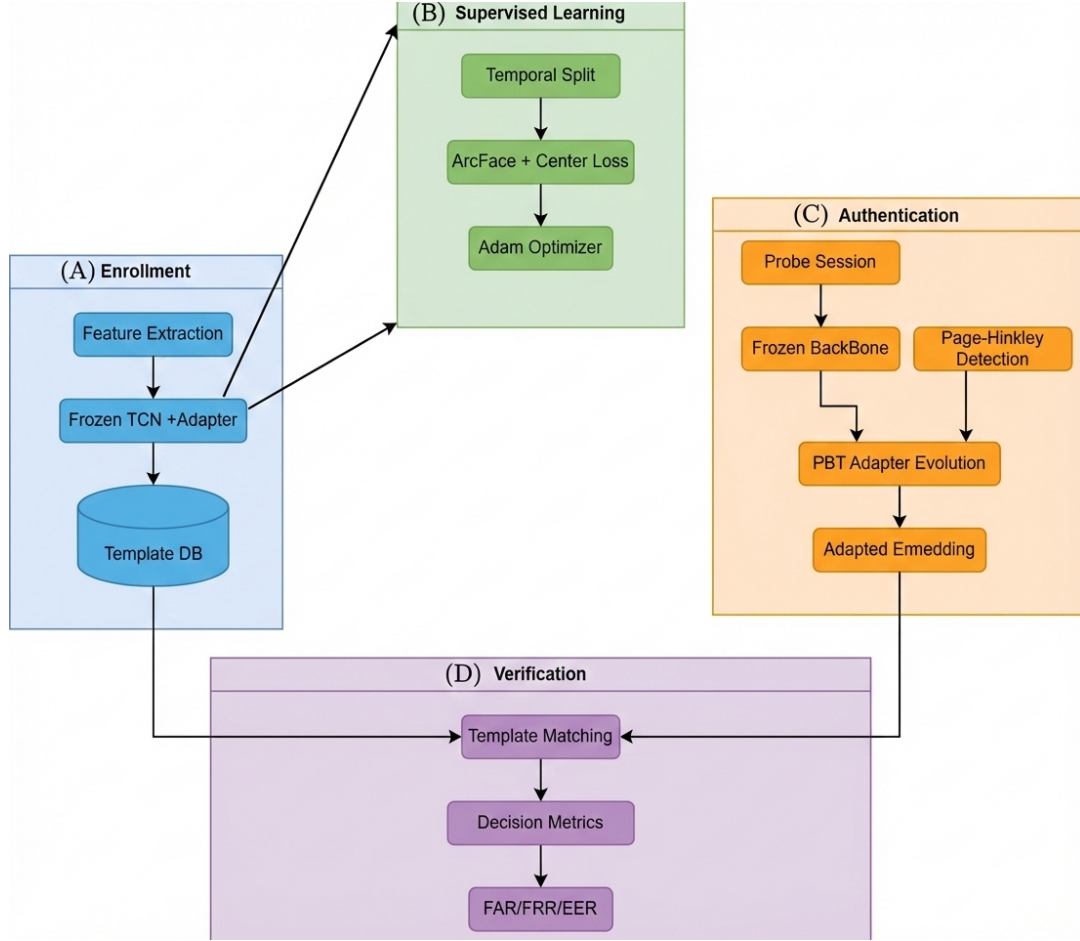


Figure 4.2: Evo-BioTCN system architecture

Figure 4.2 shows (A) Enrolment: feature extraction followed by a frozen TCN with lightweight adapters, producing user templates stored in the database. (B) Supervised training using a temporal split with ArcFace and Center Loss, optimized via Adam. (C) Authentication pipeline where probe samples pass through the frozen backbone and undergo population-based adapter evolution guided by Page-Hinkley drift detection, yielding adapted embeddings. (D) Verification through template matching and decision metrics including FAR, FRR, and EER.

4.1.2 Model Specification: The Evo-BioTCN Architecture

4.1.2.1 Architectural Rationale: Why Temporal Convolutional Networks?

The selection of a Temporal Convolutional Network (TCN) as the backbone architecture was driven by a fundamental insight into the nature of typing behavior: **keystroke dynamics are inherently local and causal**. When a user types “laboratory”, the latency of pressing ‘a’ is strongly conditioned by the immediately preceding ‘l’, as the finger must physically reposition from one key to another. This digraph context (the “la” transition) is neurologically distinct from the subsequent “ab” transition, which requires a different sequence of motor commands.

This locality principle stands in contrast to the global attention mechanisms employed by Transformer architectures. While Transformers excel at capturing long-range dependencies in language modeling [15], their self-attention mechanism considers all positions equally, potentially introducing noise from temporally distant keystrokes that have no motor-neural relationship to the current key. Stragapede et al. [37] acknowledged this limitation in TypeFormer, noting that Gaussian Range Encoding (GRE) was necessary to inject positional locality into the otherwise position-agnostic attention mechanism.

TCNs, by design, enforce causal convolutions, where the output at time t depends only on inputs at times $\leq t$, precisely mirroring the physical causality of typing. Furthermore, the use of dilated convolutions enables the network to capture multi-scale temporal dependencies without incurring the quadratic computational complexity associated with self-attention. The equation of Receptive Field is shown in equation 4.1:

$$\text{Receptive Field} = 1 + \sum_{i=0}^{n-1} 2^i(k-1) = 31 \text{ positions} \quad (4.1)$$

where, $n = 4$ is the number of TCN blocks and $k = 3$ is the kernel size. This receptive field spans approximately two complete words in a typical password-sufficient to capture digraph, trigram, and word-level rhythm patterns while maintaining $O(n)$ linear complexity suitable for real-time authentication.

4.1.2.2 Layer Architecture

The complete Evo-BioTCN architecture is illustrated in Figure 4.3 and specified in Table 4.2. The architecture has: Input (231) $\rightarrow$ 1×1 projection $\rightarrow$ 4 dilated TCN blocks ($d = 1, 2, 4, 8$) $\rightarrow$ rank-4 adapter $\rightarrow$ global pooling $\rightarrow$ L_2 norm $\rightarrow$ 64-D embedding. Inset of Figure 4.3 shows TCN block internals: dilated causal conv, weight norm, ReLU, dropout, residual. Total parameters: 131,712 (Backbone: 131,200; TCN Adapter: 512).

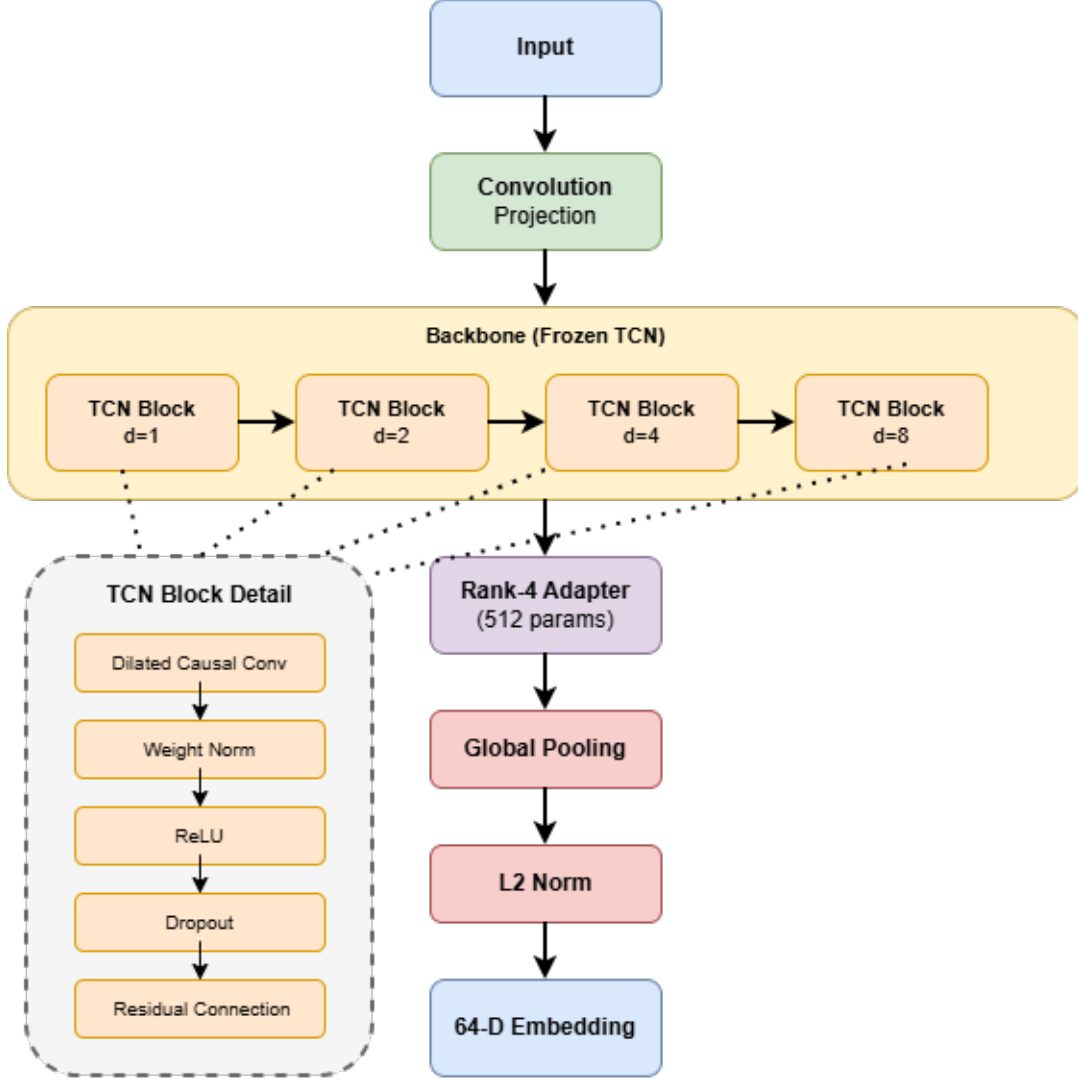


Figure 4.3: Evo-BioTCN layer architecture

Table 4.2: Evo-BioTCN layer specifications

Layer	Type	Parameters	Output Dimension
Input	Feature Vector	–	231
Projection	1×1 Convolution	14,848	64
TCN Block 1	Dilated Causal Conv (dilation=1, kernel=3)	–	64
TCN Block 2	Dilated Causal Conv (dilation=2, kernel=3)	–	64
TCN Block 3	Dilated Causal Conv (dilation=4, kernel=3)	–	64
TCN Block 4	Dilated Causal Conv (dilation=8, kernel=3)	–	64
TCN Adapter	1×1 Conv Bottleneck (rank=4)	512	64
Global Pool	Average Pooling	–	64
Output	L2 Normalization	–	64

Weight Normalization: Following TCN best practices [16], we apply weight normalization rather than batch or layer normalization as depicted in equation 4.2:

$$W = \frac{g}{\|v\|} v \quad (4.2)$$

where g is a learnable scalar magnitude and v is the unnormalized weight vector. This decoupling of magnitude and direction accelerates convergence and improves gradient flow stability—particularly important given the high variance inherent in keystroke timing data.

Residual Connections: Each TCN block includes skip connections to enable gradient propagation through the deep stack portrayed in equation 4.3:

$$y = \text{ReLU}(F(x) + x) \quad (4.3)$$

4.1.2.3 Contrastive Learning Objective

The model is trained using a composite loss function that balances inter-class separation with intra-class compactness and the equation of this loss function is shown in equation 4.4:

$$L_{\text{total}} = L_{\text{ArcFace}} + \lambda \cdot L_{\text{Center}} \quad (4.4)$$

where $\lambda = 0.001$ controls the relative contribution of each term (kept small to prioritize ArcFace’s inter-class separation).

ArcFace Angular Margin Loss [17] enforces an additive angular margin in the embedding space, creating well-separated decision boundaries shown in equation 4.5:

$$L_{\text{ArcFace}} = -\log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}} \quad (4.5)$$

where $s = 16$ is the scale factor, $m = 0.1$ is the angular margin, θ_{y_i} is the angle between the embedding and the ground-truth class weight, and the summation runs over all negative classes $j \neq y_i$.

Center Loss [13] promotes intra-class compactness by minimizing the distance between embeddings and their respective class centers as visualized in equation 4.6:

$$L_{\text{Center}} = \frac{1}{2N} \sum_{i=1}^N \|e_i - c_{y_i}\|_2^2 \quad (4.6)$$

where N is the batch size, e_i is the embedding of sample i , and c_{y_i} is the learnable center for class y_i .

Label Smoothing: The cross-entropy loss (equation 4.7) is computed with label smoothing ($\epsilon = 0.1$), which prevents overconfident predictions and improves generalization to unseen sessions:

$$y_{\text{smooth}} = (1 - \epsilon) \cdot y_{\text{one-hot}} + \frac{\epsilon}{K} \quad (4.7)$$

where K is the number of classes.

4.1.2.4 Population-Based Training (PBT) for Drift Adaptation

After supervised training converges, the backbone parameters θ_{static} are frozen, and the adapter module $\theta_{adaptive}$ undergoes evolutionary optimization via PBT [14]. This gradient-free approach enables adaptation to behavioral drift without requiring labeled data from the target domain:

- **Population Size:** $N=10$ adapter configurations
- **Fitness Function:** The equation of the fitness function is in 4.8:

$$F(A_i) = \text{Separation} - \lambda_{\text{anchor}} \cdot D(A_i, A_{\text{anchor}}) \quad (4.8)$$

where Separation = mean impostor distance – mean genuine distance

- **Genetic Operators:** Tournament selection (top 30% elite), Gaussian mutation ($\sigma = 0.02$)
- **Trust Anchor Regularization:** Prevents excessive deviation from enrolment-time behavior ($\lambda_{\text{anchor}} = 0.1$)

The architecture employs **two complementary adapters**:

1. **Internal TCN Adapter (θ_{adapt}):** A 1×1 convolutional bottleneck inserted after the TCN blocks, using the same rank-4 projection but operating on the temporal sequence before pooling.
2. **PBT Adapter:** A linear bottleneck that operates on the pooled 64-dimensional embedding during evolutionary adaptation is shown in equation 4.9:

$$\text{Adapter}(h) = W_{\text{up}} \cdot \text{ReLU}(W_{\text{down}} \cdot h) + h \quad (4.9)$$

where $W_{\text{down}} \in \mathbb{R}^{4 \times 64}$ and $W_{\text{up}} \in \mathbb{R}^{64 \times 4}$, yielding 512 trainable parameters per adapter (no bias terms).

Drift Detection: The Page–Hinkley test monitors reconstruction loss during PBT evolution. When significant drift is detected through the equation 4.10:

$$m_t - M_t > \tau, \quad (4.10)$$

with $\delta = 0.005$ and $\tau = 0.5$, hyper-mutation ($3 \times$ standard deviation) is triggered to escape local optima.

The distinction between training and inference pipelines is illustrated in Figure 4.4.

Figure 4.4 shows - **Training**: supervised learning with ArcFace and Center Loss, followed by freezing θ_{static} and evolving $\theta_{adaptive}$ via PBT; **Inference**: frozen backbone, PBT adapter evolution, template matching, and decision.

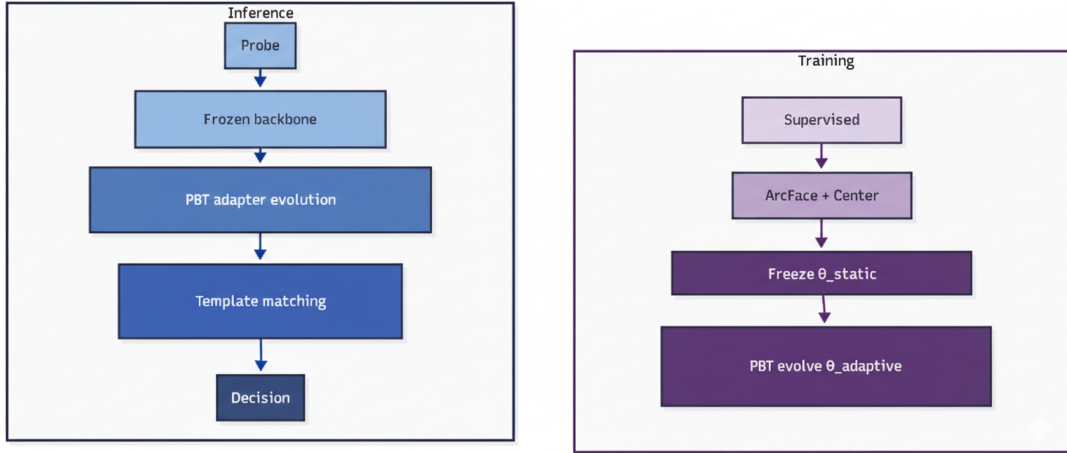


Figure 4.4: Training vs inference pipelines

4.1.3 Data Collection and Preprocessing

4.1.3.1 Dataset Selection: GREYC vs. Aalto

This research utilizes the **GREYC Keystroke Dataset** [4], collected by the GREYC Laboratory at Université de Caen Normandie. Unlike the Aalto dataset [18] used by most SOTA methods, GREYC [4] is **longitudinal** — users returned over multiple sessions spanning approximately 110 days. This temporal depth is essential for studying behavioral drift. As summarized in Table 4.3, GREYC [4] provides longitudinal depth critical for drift analysis, while Aalto’s [18] single-session design cannot capture temporal evolution.

Table 4.3: Comparison of GREYC [4] and Aalto datasets

Property	GREYC [4]	Aalto [18]
Collection Type	Multiple sessions over months	Single sitting (~ 30 min)
Temporal Span	110 days	~ 15 –30 minutes
Sessions per User	Mean: 57 (distributed)	15 (consecutive)
Behavioral Drift	✓ Present and quantified	× Absent
Template Aging Evaluation	✓ Possible (40+ day gaps)	× Not applicable

The Aalto dataset [18], despite its impressive scale, consists of 15 back-to-back sentence transcriptions per user with no return visits [18]. Dhakal et al. [18] explicitly note: “All participants completed 15 sessions (i.e., one sentence for each session)” — these are **not temporal sessions** separated by days or weeks, but consecutive sentences typed within minutes. Consequently, models trained and evaluated on Aalto dataset [18] do not experience and cannot be evaluated for drift resilience. Table 4.4 shows the state of the GREYC dataset before cleaning.

Table 4.4: GREYC dataset [4] characteristics before cleaning

Property	Value
Fixed-Text	“greyc laboratory” (16 characters)
Total Samples	7,560 typing sessions
Unique Users	133 individuals
Temporal Span	March 17, 2009 – July 5, 2009
Mean Sessions/User	56.8

4.1.3.2 Raw Timing Measurements

For each keystroke in the fixed-text password, four fundamental timing measurements (equations 4.11 to 4.14) are extracted:

$$\text{Hold Time}(H_i) = R_i - P_i \quad (4.11)$$

$$\text{Up-Down / Flight Time}(UD_i) = P_{i+1} - R_i \quad (4.12)$$

$$\text{Down-Down Time}(DD_i) = P_{i+1} - P_i \quad (4.13)$$

$$\text{Release-Release Time}(RR_i) = R_{i+1} - R_i \quad (4.14)$$

where P_i and R_i denote the press and release timestamps for keystroke i .

Motor-Neural Significance: Hold time (H) reflects fine motor precision — how long the finger dwells on a key before release. Flight time (UD) captures the transition fluency between keys and can be negative during “key rollover” (pressing the next key before releasing the current one), a signature of expert typists. Down-Down time (DD) encodes overall typing rhythm.

Data Quality Note: Position 16 features exhibit 99.9% missing values due to a limitation in the original GREYC capture protocol [4]. All preprocessing utilizes **positions 1–15 only**.

4.1.3.3 Feature Engineering: From Raw Timings to Drift-Invariant Representations

A critical insight from the drift analysis literature is that **raw timing values are vulnerable to both mimicry and drift** [2], [8]. An attacker who practices the target password can approximate average keystroke durations, while legitimate users’ absolute speeds evolve with habituation. To address both security and stability concerns, we engineer higher-order features that encode how a user types rather than merely how fast.

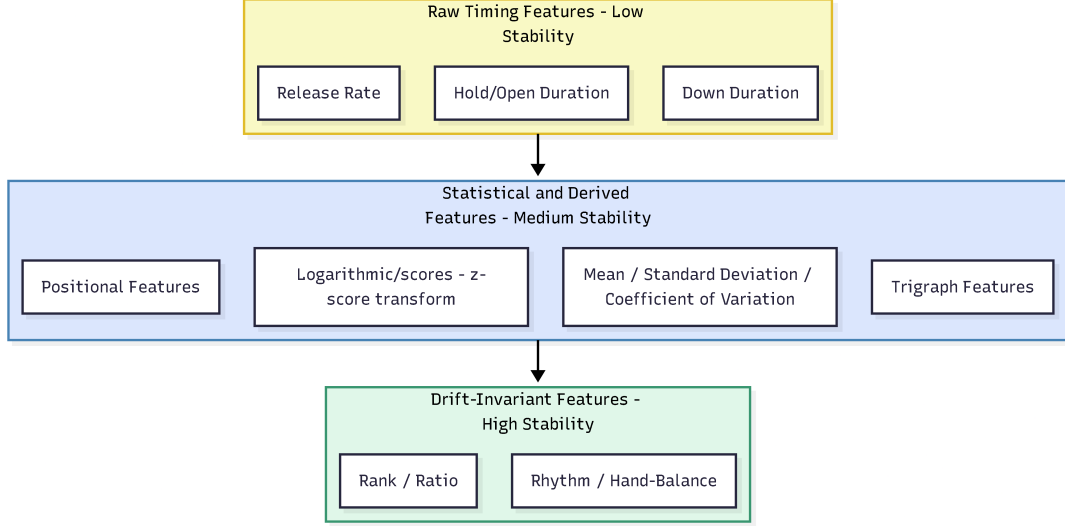


Figure 4.5: Feature engineering pipeline

Raw timings (H, UD, DD, RR; low stability) $\rightarrow$ statistical/derived features (mean, std, CV, positional, log, z-score, trigraph; medium stability) $\rightarrow$ drift-invariant features (rank, ratio, rhythm, hand-balance; high stability). Empirical validation is shown in Figure 4.5.

Step 1: Robust Outlier Detection

Keystroke data contains inevitable outliers — cognitive pauses, environmental distractions, or momentary lapses in attention. We apply Modified Z-score outlier detection within each session as depicted in equation 4.15:

$$z_{\text{mod}} = \frac{0.6745 \times (T_i - \tilde{T})}{\text{MAD}} \quad (4.15)$$

where $\tilde{T}$ is the median and $\text{MAD} = \text{median}(|T_i - \tilde{T}|)$ is the median absolute deviation. Values with $|z_{\text{mod}}| > 3.5$ or exceeding 2000 ms (cognitive interruptions) are flagged as outliers.

Step 2: Session-Relative Normalization

To remove between-session speed variations while preserving relative rhythm patterns, equation 4.16 is used:

$$T_{\text{norm}} = \frac{T}{\mu_{T_{\text{clean}}}} \quad (4.16)$$

where $\mu_{T_{\text{clean}}}$ is the session mean of cleaned timing values.

Step 3: Drift-Invariant Feature Engineering

The complete feature set (231 dimensions) is organized by drift resilience. High-resilience features (rank, ratio, rhythm) in table 4.5 form the core discriminative signal.

Table 4.5: Feature categories organized by drift resilience

Feature Category	Count	Drift Resilience	Rationale
Rank Features	32	✓ High	Ordinal positions invariant to speed scaling
Ratio Features	32	✓ High	Proportions of total time resist global drift
Rhythm/Entropy	7	✓ High	Timing variance patterns persist across sessions
Hand Balance	4	✓ High	Left/right asymmetry reflects motor dominance
Statistical	44	Medium	Mean, std, CV capture within-session consistency
Normalized Timing	60	Medium	Session-relative z-scores
Trigraph	24	Medium	3-key sequence patterns
Log-Transform	9	Medium	Corrects for skewed distributions
Positional	12	× Lower	Region-specific statistics
Acceleration	4	× Lower	Speed-up/slow-down patterns
Uncategorized	3	Medium	Features not matched by category rules

Empirical Validation of Drift Resilience: To support this categorization, we measure each feature family’s temporal stability and drift magnitude across the 110-day span. Specifically, we compute (i) robust CV (MAD/median) within each category and (ii) symmetric early→late drift magnitude for each category. The results show that rank/ratio/rhythm/hand-balance groups exhibit lower variability and drift than positional or acceleration-heavy features. Figure 4.6 summarizes stability and drift by **feature category** (not raw H/UD/DD/RR timings). Lower values indicate higher stability.

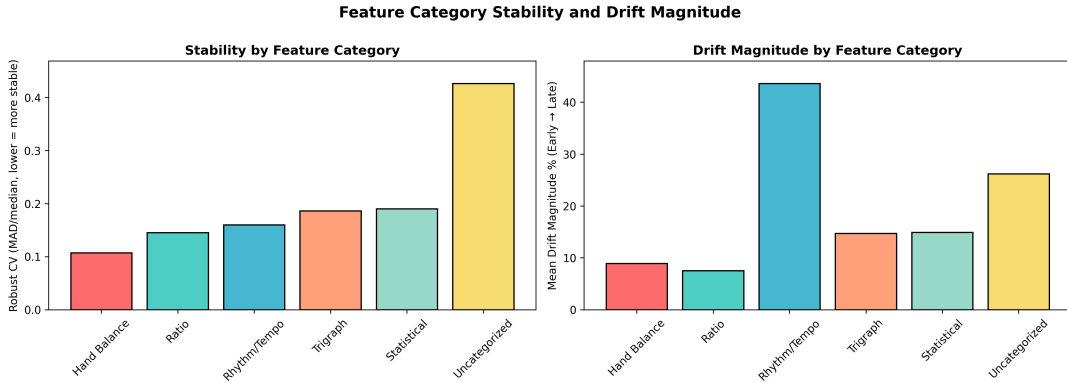


Figure 4.6: Stability and drift magnitude by feature category using robust CV and symmetric drift

Why Rhythm Features Matter: Raw speed is easy to mimic — an attacker who observes or estimates the target’s typing speed can approximate it with practice. However, the relative rhythm (e.g., the ratio of Hold time to Flight time, the ordering of which keys are fastest) encodes fine motor-neural coordination patterns that develop over years of typing practice and cannot be consciously controlled or quickly imitated [2].

Why Hand Balance Matters: By mapping keys to left/right hand positions based on standard QWERTY touch-typing, we extract ratios reflecting motor asymmetry. Right-hand dominant users exhibit systematically faster seek times for keys like ‘j’, ‘k’, ‘l’ compared to left-side keys. This motor signature persists even as overall typing speed fluctuates.

4.1.3.4 User Exclusion Criteria

To ensure meaningful drift analysis, 16 users (222 samples, 2.9% of raw data) were excluded based on criteria established in the literature [5], [8]:

- Insufficient Sessions: Fewer than 20 sessions (below the minimum recommended by Killourhy & Maxion [5], with a safety margin)
- Insufficient Temporal Span: Less than 7 days (cannot study drift with single-day collections)

These exclusions follow the rationale that single-day batch collections do not represent the natural variability in a user’s typing behavior across different mental states, times of day, and environmental conditions.

4.1.3.5 Temporal Data Split

A critical methodological contribution of this work is the **strict temporal split protocol** for evaluation. Standard randomized cross-validation, as used in most SOTA evaluations, allows future sessions to “leak” into the training set — the model learns patterns from temporally advanced data that would not be available at enrolment time in a real deployment.

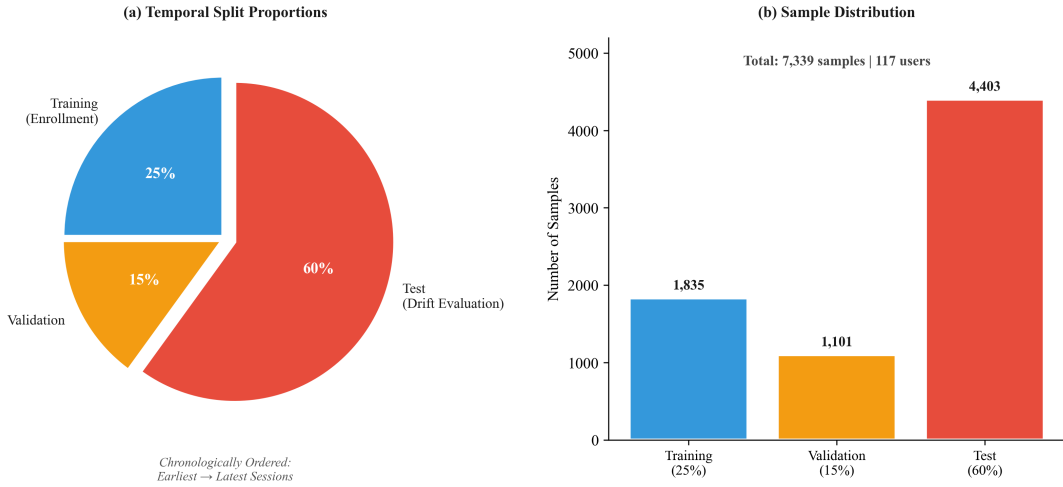


Figure 4.7: Temporal data split ensuring realistic drift evaluation

As Figure 4.7 shows, training uses only the earliest 25% of each user’s sessions, while testing uses the latest 60% — sessions occurring weeks to months after enrolment. The data split summary is shown below in table 4.6.

Table 4.6: Data split summary with strict temporal ordering

Split	Purpose	Proportion	Samples	Temporal Position
Training	Template building	25%	1,881	Earliest sessions
Validation	Hyperparameter tuning	15%	1,078	Middle sessions
Test	Drift evaluation	60%	4,601	Latest sessions

Critical Distinction: The average temporal gap between training (enrolment) and test samples is 9.6 days (min 0, max 56). This stands in stark contrast to Aalto dataset [18] based evaluations where enrolment and test samples are separated by mere minutes. Our EER of 3.77% is thus a conservative, drift-aware estimate that reflects realistic deployment conditions.

4.1.4 Implementation Details

4.1.4.1 Training Configuration

Table 4.7: System and framework specifications used for experiments

Component	Specification
Framework	PyTorch 2.x
GPU	NVIDIA CUDA-enabled device
Precision	FP32
Random Seed	42 (all libraries)

4.1.4.2 Hyperparameter Optimization

Hyperparameter optimization was performed using grid search over the validation set, prioritizing training stability over peak accuracy to prevent overfitting on early sessions. The final configuration is summarized in Table 4.8. The system specifications are as listed in table 4.7. The training dynamics is shown in figure 4.8.

Table 4.8: Final hyperparameter configuration

Hyperparameter	Value	Selection Rationale
Optimizer	Adam	Adaptive moments for sparse gradients
Learning Rate	1×10^{-3}	Stabilized after grid search (10^{-4} to 10^{-2})
Weight Decay	1×10^{-4}	Light regularization preserving timing sensitivity
Batch Size	32	Smaller batches for better gradient estimates with high-variance timing data
Epochs	200 (max)	With early stopping (patience=30) on validation EER
LR Scheduler	StepLR (step=15, $\gamma = 0.5$)	Gradual decay
Gradient Clipping	0.5	Prevents explosion from outlier timing values
Dropout	0.2	Moderate regularization for TCN blocks

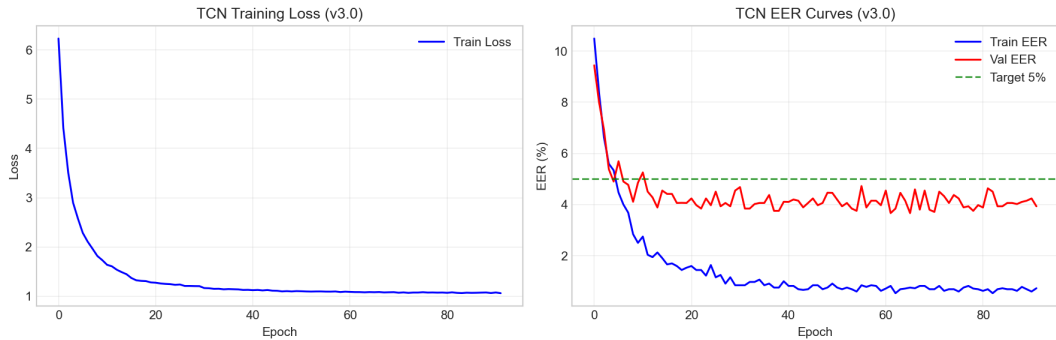


Figure 4.8: Evo-BioTCN training dynamics showing loss and EER over epochs

Stability Considerations: Initial experiments with learning rates of 10^{-2} resulted in unstable convergence, with validation loss oscillating by up to 40% between epochs. This instability stems from the high variance in keystroke timing data — a single cognitive pause can produce gradients $10\times$ larger than typical. The reduced learning rate of 10^{-3} , combined with gradient clipping at 0.5, stabilized training during critical early epochs.

4.1.4.3 Regularization Strategy

Given the moderate dataset size (7,560 samples, 133 classes), preventing overfitting is critical:

1. **Small Embedding Dimension:** 64-dimensional embeddings (vs. typical 256–512)
2. **Low-Rank Adapters:** Rank-4 bottleneck (512 parameters)
3. **Online Data Augmentation:** Applied during training only:
 - Gaussian noise injection ($\sigma = 0.05$)
 - simulating natural timing variation
 - Random feature dropout (10%) with inverted dropout scaling
4. **Early Stopping:** Based on validation EER plateau
5. **Weight Decay:** 10^{-4} L2 regularization

Parameter-to-Sample Ratio: With 131,712 parameters and 1,881 training samples, the ratio of 70.0 is within acceptable bounds when mitigated by the regularization techniques above.

4.1.4.4 PBT Evolution Parameters

Table 4.9 shows the evolution parameters of PBT.

Table 4.9: Evolutionary Parameters

Parameter	Value
Population Size	10
Generations	10
Mutation Scale	$\sigma = 0.02$
Elite Preservation	Top 30%
Trust Anchor Weight	0.1
Hyper-Mutation	$3\times$ scale when drift detected

4.2 PRELIMINARY DESIGN OR DESIGN (MODEL) SPECIFICATION

4.2.1 Proposed Methodology — Free-Text Continuous Authentication

4.2.1.1 Design Process: Dataset Reconstruction and Feature Engineering

4.2.1.1.1 Problem Context and Dataset Challenges: The Buffalo free-text keystroke dataset [12], as originally distributed, presents a significant challenge for authentication research due to its unconventional data format. Unlike traditional keystroke datasets that provide character-level timing information directly, the Buffalo dataset [12] encodes timing relationships at the digraph level, storing inter-key transitions rather than individual keystroke events. This fundamental structural difference necessitated the development of a novel reconstruction pipeline before any meaningful authentication research could proceed.

The raw dataset, provided in the distribution, contains 157,234 digraph records across 75 participants and 6 temporally-separated sessions [12]. Each record encodes a key transition pair (k_1, k_2) along with three timing metrics: the dwell time of the first key DU_{k_1, k_1} , the down-down interval between consecutive key presses DD_{k_1, k_2} , and the down-up interval spanning from the first key press to the second key release DU_{k_1, k_2} . This encoding scheme, while compact, obscures the natural typing flow and prevents direct extraction of word-level authentication features.

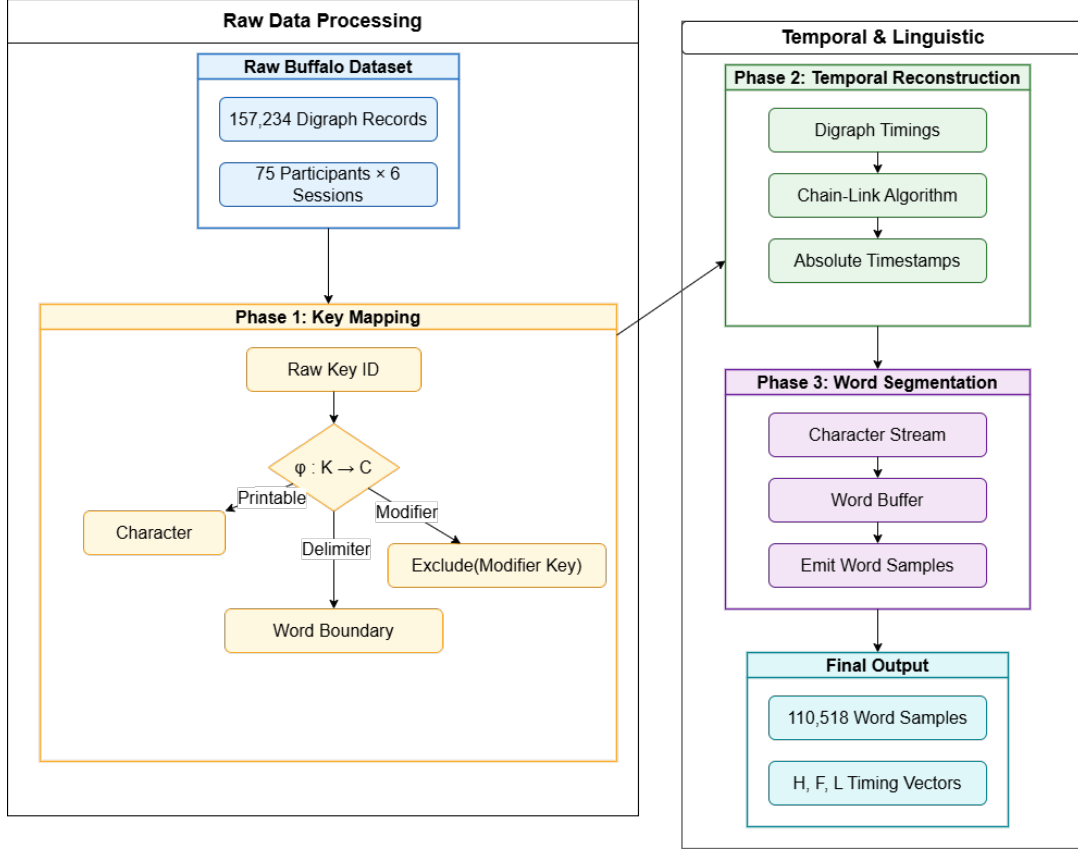


Figure 4.9: Buffalo[12] dataset reconstruction pipeline

Figure 4.9 shows the reconstruction pipeline for the Buffalo dataset [12]. Raw digraphs are transformed through key mapping, temporal chain reconstruction, and word boundary detection to produce character-level word samples with timing vectors.

4.2.1.1.2 Keystroke Reconstruction Algorithm: The reconstruction process transforms the digraph-encoded data into a character-level representation suitable for authentication analysis. The algorithm operates through three coordinated phases: key mapping, temporal chain reconstruction, and word boundary detection.

Key Mapping Function: The first phase establishes a bijective mapping between raw key identifiers and their corresponding character representations. Let $\mathcal{K}$ denote the set of raw key names and $\mathcal{C}$ denote the character alphabet. The mapping function $\phi : \mathcal{K} \rightarrow \mathcal{C} \cup \{\perp\}$ is defined in equation 4.17:

$$\phi(k) = \begin{cases} c & \text{if } k \text{ maps to printable character } c \\ \text{SPACE} & \text{if } k \in \{\text{Space, Return, Tab}\} \\ \text{BACKSPACE} & \text{if } k = \text{Back} \\ \perp & \text{if } k \text{ is a modifier or function key} \end{cases} \quad (4.17)$$

The mapping handles 47 distinct key types, including alphanumeric keys, punctuation (mapped from OEM codes such as `OemPeriod` $\rightarrow$ `.` and `Oem7` $\rightarrow$ `'`), and numeric keys (mapped from virtual key codes D0 through D9). Modifier keys, including `LShiftKey`, `RShiftKey`, `LControlKey`, and navigation keys, are mapped to

the null symbol $\perp$ and excluded from the character stream.

Temporal Chain Reconstruction: The second phase reconstructs absolute timing from relative inter-key intervals. For a session containing n digraph records, the algorithm maintains a time cursor t_{current} initialized to zero. For each digraph record $r_i = (k_1^{(i)}, k_2^{(i)}, DU_i, DD_i, DU'_i)$, the algorithm uses equations 4.18 to 4.20 to compute:

$$t_{\text{press}}^{(k_1)} = t_{\text{current}} \quad (4.18)$$

$$t_{\text{release}}^{(k_1)} = t_{\text{current}} + DU_i \quad (4.19)$$

$$t_{\text{press}}^{(k_2)} = t_{\text{current}} + DD_i \quad (4.20)$$

The cursor is then advanced: $t_{\text{current}} \leftarrow t_{\text{press}}^{(k_2)}$. This chain-linking procedure preserves the temporal ordering of keystrokes while converting relative intervals to absolute timestamps. The final record in each session requires special handling, as the second key k_2 must be processed to capture the complete typing sequence.

The third phase segments the continuous character stream into discrete word samples. A word buffer $\mathcal{W}$ accumulates characters until a delimiter event (space or newline) is encountered. The backspace action removes the most recent character from the buffer, faithfully reconstructing the user's intended text even when typing errors occur. Upon delimiter detection, the accumulated word and its associated timing vector are emitted as a sample record.

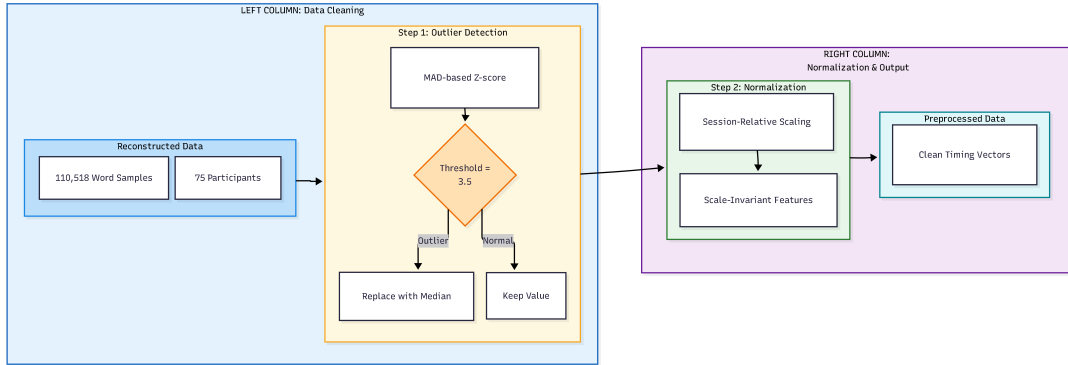


Figure 4.10: Preprocessing Pipeline

The preprocessing pipeline showing MAD-based outlier detection and session-relative normalization steps is shown in figure 4.10.

The reconstruction algorithm processes each participant-session pair independently, yielding a character-level dataset with the following structure for each word sample: participant identifier, session number, reconstructed word string, per-character hold times $\mathbf{H} = [h_1, h_2, \dots, h_m]$, per-character flight times $\mathbf{F} = [f_1, f_2, \dots, f_{m-1}]$, and per-character latencies $\mathbf{L} = [l_1, l_2, \dots, l_{m-1}]$, where m denotes the word length.

4.2.1.1.3 Data Quality Analysis and Outlier Detection: The reconstructed dataset reveals significant data quality challenges that must be addressed before feature extraction. Exploratory analysis of the 110,518 word samples across 75 participants demonstrates substantial outlier contamination in the raw timing distributions.

Timing Distribution Characteristics: The hold time distribution exhibits a heavy right tail with mean $\mu_H = 0.127$ seconds and standard deviation $\sigma_H = 0.089$ seconds. However, the maximum observed value of 4.23 seconds indicates the presence of extreme outliers, likely caused by user pauses or attention shifts during typing. Similarly, the flight time distribution contains negative values (indicating key overlap during fast typing) as well as extreme positive values exceeding 2 seconds, suggesting inter-word pauses that were not correctly segmented.

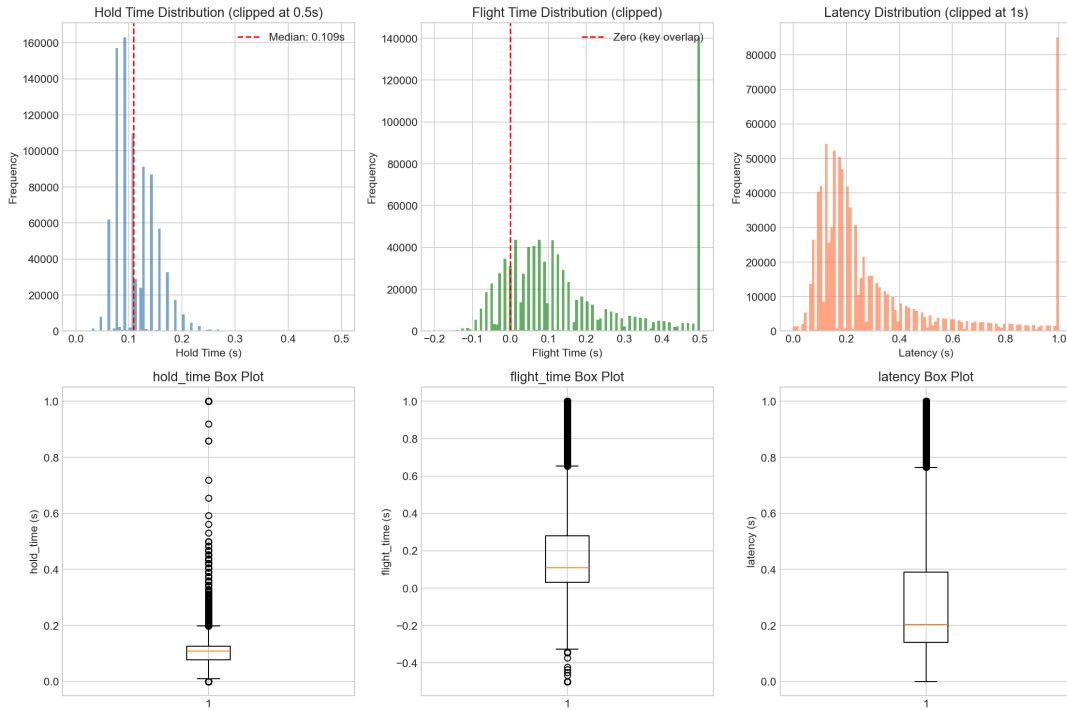


Figure 4.11: Raw keystroke timing distributions before preprocessing

Figure 4.11 shows the histogram of raw hold time distribution. The distribution exhibits a heavy right tail with outliers exceeding 4 seconds, motivating MAD-based outlier detection.

Modified Z-Score Outlier Detection: To address the non-Gaussian nature of the timing distributions, the methodology employs the Median Absolute Deviation (MAD) as a robust scale estimator. For a timing vector $\mathbf{x} = [x_1, x_2, \dots, x_n]$, the modified Z-score is computed as in equation 4.21:

$$M_i = \frac{0.6745 \cdot (x_i - \tilde{x})}{\text{MAD}(\mathbf{x})} \quad (4.21)$$

where $\tilde{x}$ denotes the median and $\text{MAD}(\mathbf{x}) = \text{median}(|x_i - \tilde{x}|)$. The constant 0.6745 ensures consistency with the standard normal distribution under Gaussian assumption.

tions. Observations with $|M_i| > 3.5$ are classified as outliers. This threshold, more conservative than the common 2.5 or 3.0 cutoffs, was selected to preserve legitimate variation in typing patterns while removing only the most extreme anomalies.

The outlier detection operates at the session level to avoid cross-session contamination. For each participant-session combination, the hold time, flight time, and latency vectors are independently screened. Detected outliers are replaced with the session median of the respective timing metric, a strategy that preserves the user’s typical behavior without introducing artificial patterns.

Outlier Prevalence: Across the full dataset, the MAD-based detection identifies 31,247 outlier values (approximately 9.4% of all timing measurements). The outlier fraction varies substantially across participants, ranging from 2.1% to 18.7%, suggesting that some users exhibit more erratic typing behavior than others. This heterogeneity reinforces the need for per-user normalization in subsequent processing stages.

4.2.1.1.4 Session-Relative Normalization: Temporal drift represents a persistent challenge in keystroke biometric systems, as typing patterns evolve over time due to factors including fatigue, learning effects, and environmental changes. The Buffalo dataset’s two session structure[12], collected across three temporally-separated phases, provides an ideal testbed for studying and mitigating drift effects.

Analysis of the session-wise timing distributions reveals systematic shifts across collection phases. The mean hold time increases by approximately 8% from Phase 1 to Phase 3, while the coefficient of variation for flight times increases by 12%, indicating greater typing irregularity in later sessions. These observations motivate the adoption of session-relative normalization as a preprocessing step.

For each word sample, the timing vectors are normalized by their session-specific means shown in equation 4.22:

$$\mathbf{H}_{\text{norm}} = \frac{\mathbf{H}}{\bar{H}_{\text{session}}} \quad (4.22)$$

where $\bar{H}_{\text{session}}$ denotes the mean hold time computed over all characters in that word sample. This normalization renders the features scale-invariant, allowing comparison across sessions and users with different baseline typing speeds. The same transformation is applied to flight time and latency vectors after taking absolute values to handle the negative flight times that arise from key overlap.

4.2.1.1.5 Feature Engineering Pipeline: The preprocessed timing vectors serve as input to a comprehensive feature extraction pipeline that generates 85 statistical features per word sample. These features are organized into nine categories, each capturing a distinct aspect of typing dynamics.

Statistical Moments: For each timing vector $\mathbf{x} \in \{\mathbf{H}, \mathbf{F}, \mathbf{L}\}$, the pipeline computes the first four statistical moments along with robust alternatives portrayed in equations 4.23 to 4.25:

$$\mu_x = \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu_x)^2} \quad (4.23)$$

$$CV_x = \frac{\sigma_x}{\mu_x}, \quad \gamma_x = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \mu_x}{\sigma_x} \right)^3 \quad (4.24)$$

$$\kappa_x = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \mu_x}{\sigma_x} \right)^4 - 3 \quad (4.25)$$

The median, interquartile range (IQR), and range provide robust location and scale estimates that are less sensitive to residual outliers.

Positional Features: Typing rhythm often varies within a word, with initial keystrokes potentially slower due to motor planning and final keystrokes potentially faster due to anticipation. The pipeline captures this variation by partitioning each timing vector into thirds (early, middle, late) and computing the mean and coefficient of variation for each segment. The positional slope, defined as the ratio of late-segment to early-segment means, quantifies the acceleration or deceleration pattern within words.

Rhythm and Entropy Features: The regularity of keystroke timing is captured through entropy-based measures. For a timing vector $\mathbf{x}$ with all positive values, the normalized probability distribution function is shown in equation 4.26:

$$p_i = \frac{|x_i| + \epsilon}{\sum_{j=1}^n (|x_j| + \epsilon)} \quad (4.26)$$

where $\epsilon = 10^{-10}$ ensures numerical stability. The Shannon entropy $H(\mathbf{x}) = -\sum_i p_i \log p_i$ measures the uniformity of timing intervals, with higher values indicating more regular typing. The irregularity score, computed as the fraction of intervals exceeding two MAD units above the median, provides a complementary measure of typing consistency.

Overlap Rate: A distinctive feature of skilled typists is key overlap, where the next key is pressed before the current key is fully released. This phenomenon manifests as negative flight times in the reconstructed data. The overlap rate, defined as the fraction of negative flight times within a word, captures this aspect of typing proficiency and serves as a strong discriminative feature, achieving the highest Fisher ratio (0.39) among all engineered features in preliminary discriminability analysis.

Rank-Based Features: To achieve hardware invariance across different keyboard types and system configurations, the pipeline incorporates rank-based transformations. For each timing vector, the rank ordering of values is computed, and features including rank entropy and the positions of fastest and slowest keystrokes are extracted. These ordinal features capture the relative timing structure while being robust to absolute scale differences.

Higher-Order Timing Patterns: The pipeline extends beyond individual keystrokes to capture sequential patterns through bigram and trigram features. Bigram timing is computed as the sum of consecutive hold times: $B_i = h_i + h_{i+1}$. Trigraph timing extends this to three-character sequences: $T_i = h_i + h_{i+1} + h_{i+2}$. The mean, standard deviation, and coefficient of variation of these higher-order timings capture the rhythmic structure of multi-character sequences.

Acceleration Features: The rate of change in timing intervals provides insight into the dynamics of typing behavior. The acceleration is computed as the mean of first-order differences: $a = \frac{1}{n-1} \sum_{i=1}^{n-1} (x_{i+1} - x_i)$. The jerk, or rate of change of acceleration, is computed as the mean of second-order differences. These features capture whether the typist is speeding up, slowing down, or maintaining a constant rhythm.

Ratio Features: Cross-feature relationships are captured through ratio computations including the flight-to-hold ratio $\bar{F}/\bar{H}$, the latency-to-hold ratio $\bar{L}/\bar{H}$, and the hold-to-total ratio $\bar{H}/(\bar{H} + \bar{F})$. These ratios characterize the relative allocation of time between key depression and inter-key transitions. Table 4.10 shows the principle groups of the features.

Table 4.10: Overview of the 85 engineered keystroke features organized by functional category

Feature Group	Description
Statistical Timing	Summary statistics of hold, flight, and latency times.
Positional	Timing variation across early, middle, and late word segments.
Rhythm	Entropy and variability-based measures of typing regularity.
Key Overlap	Metrics describing overlapping keystrokes and rollover behaviour.
Rank-Based	Ordinal timing patterns invariant to absolute typing speed.
N-gram Timing	Bigram and trigram timing statistics capturing key sequences.
Dynamic Change	Acceleration and jerk reflecting rapid timing changes.
Ratio-Based	Relative timing relationships between keystroke events.
Log-Scaled	Log-transformed timing features to reduce skewness.

The final preprocessing stage combines globally-normalized features with per-user normalized variants to create a hybrid feature representation. For each user u , the per-user mean μ_u and standard deviation σ_u are computed across all training samples belonging to that user. The per-user normalized features are then calculated as visualized in equation 4.27:

$$\mathbf{x}_{\text{user}} = \frac{\mathbf{x} - \boldsymbol{\mu}_u}{\sigma_u} \quad (4.27)$$

The hybrid feature vector concatenates the globally-normalized features with the per-user normalized variants, yielding a 170-dimensional representation per word sample. This dual normalization strategy allows the model to learn both absolute typing characteristics (useful for distinguishing users) and relative variations within each user’s behavior (useful for detecting anomalies and impostors).

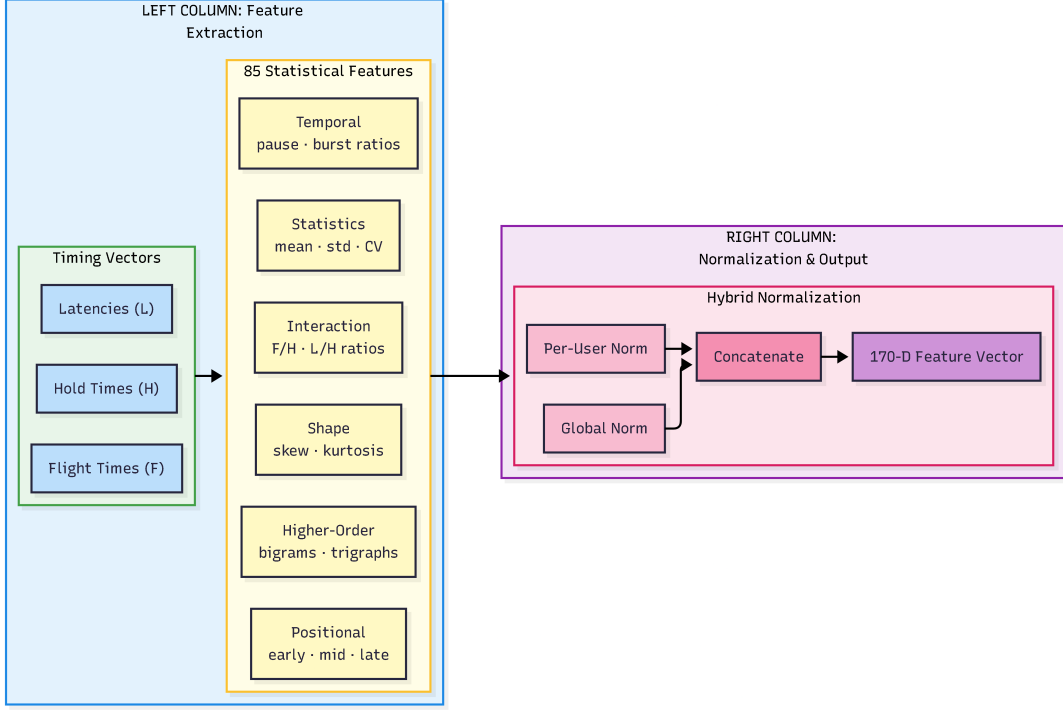


Figure 4.12: Feature engineering pipeline for Buffalo dataset [12]

Figure 4.12 depicts the feature engineering pipeline. Raw timing vectors are transformed through 85 statistical features across seven categories.

4.2.1.2 Model Specification: Four-Model Deep Learning Ensemble Architecture

4.2.1.2.1 Design Rationale and Architectural Choices: The design of the authentication system is informed by the constraints and insights derived from the dataset reconstruction and extensive ablation study phases. Preliminary experiments with single-model architectures revealed a critical insight that guided the final design: no single deep learning model could simultaneously achieve high TAR and low EER across all users, but an ensemble of complementary architectures could leverage their individual strengths while compensating for their weaknesses.

Analysis of baseline methods demonstrates that template-based approaches (Scaled Manhattan distance) achieve 28.77% Equal Error Rate (EER), significantly outperforming classification-based methods including Random Forest (81.20% EER) and Support Vector Machines (82.60% EER). This disparity arises because template matching treats each user independently, comparing probe samples against

that user’s enrolment template, whereas classification approaches must discriminate among all 75 users simultaneously — a substantially harder task given the high intra-class variability inherent in free-text typing.

The systematic ablation study evaluated 18 distinct architectures spanning deep learning (Transformers, ResNets, Siamese Networks) and classical machine learning (Random Forest, XGBoost, CatBoost). A clear progression of improvement emerged: the best single model (Enhanced Embedding) achieved 82.31% TAR, but combining multiple complementary models in an ensemble architecture achieved 83.53% TAR representing the optimal configuration that no single architecture could match.

The high per-user variance observed in the dataset — with individual authentication accuracy ranging from 0% to 100% — indicates that some users have highly distinctive typing patterns while others are nearly indistinguishable. This heterogeneity motivates the ensemble approach, as different models excel at distinguishing different subsets of users.

4.2.1.2.2 Ensemble Architecture Overview: The final system employs a 4-Model Deep Learning Ensemble that combines four complementary embedding networks through optimized score-level fusion. Each component model maps 170-dimensional input features to L2-normalized embeddings suitable for cosine similarity comparison. The ensemble components are:

ArcFace + Multi-Similarity Model: This model combines ArcFace angular margin loss with multi-similarity mining for hard negative selection. The architecture comprises a squeeze-and-excitation attention layer, followed by a deep residual tower with eight blocks, and a 512-dimensional embedding head. The ArcFace loss shown in equation 4.28, introduces an additive angular margin $m = 0.50$ with scale $s = 30$:

$$\mathcal{L}_{\text{ArcFace}} = -\log \frac{e^{s \cdot \cos(\theta_{y_i} + m)}}{e^{s \cdot \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cdot \cos \theta_j}} \quad (4.28)$$

where $s = 30.0$ is the scaling factor and $m = 0.50$ is the angular margin. Additionally, this component incorporates Multi-Similarity Loss for hard negative mining as depicted in equation 4.29:

$$\mathcal{L}_{\text{MS}} = \frac{1}{\alpha} \log \left(1 + \sum_{k \in \mathcal{N}} e^{\alpha(S_{ik} - \lambda)} \right) + \frac{1}{\beta} \log \left(1 + \sum_{j \in \mathcal{P}} e^{-\beta(S_{ij} - \lambda)} \right) \quad (4.29)$$

where $\alpha = 2.0$, $\beta = 50.0$, $\lambda = 0.5$ (base threshold), and $\mathcal{P}$, $\mathcal{N}$ denote positive and negative pairs after hard mining. This model achieves 72.76% TAR @ 5%FAR as a standalone system and contributes strong angular separation to the ensemble.

Enhanced Embedding Model (Triplet + Center Loss): This model uses a hybrid metric learning objective combining Focal Triplet Loss with Center Loss. The architecture employs progressive dimensionality (512→1024→512), multi-head self-

attention (8 heads), and 4 residual blocks, producing a 512-dimensional embedding space. The composite loss function is in equation 4.30:

$$\mathcal{L}_{\text{enhanced}} = \lambda_t \mathcal{L}_{\text{triplet}} + \lambda_c \mathcal{L}_{\text{center}} \quad (4.30)$$

where the Focal Triplet Loss focuses on hard examples through focal weighting shown in equation 4.31:

$$\mathcal{L}_{\text{triplet}} = (1 + L)^\gamma \cdot L, \quad L = \log(1 + e^{d_p - d_n + m}) \quad (4.31)$$

with margin $m = 0.7$ and focal parameter $\gamma = 2.0$. The Center Loss minimizes intra-class variance as visualized in equation 4.32:

$$\mathcal{L}_{\text{center}} = \frac{1}{2B} \sum_{i=1}^B \|\mathbf{e}_i - \mathbf{c}_{y_i}\|^2 \quad (4.32)$$

where $\mathbf{c}_{y_i}$ is the learnable center for class y_i . The weights $\lambda_t = 1.0$ and $\lambda_c = 0.01$ balance these objectives. As a standalone model, this achieves 82.31% TAR @ 5%FAR — the best single-model performance.

Simple ArcFace Model: A streamlined variant of the ArcFace architecture with reduced complexity (3 layers, 256-dimensional embeddings, $s = 20.0$, $m = 0.30$), providing regularization benefits through architectural diversity. This model prevents ensemble overfitting by contributing a simpler decision boundary.

Fine-tuned Enhanced Model: The Enhanced Embedding model further refined through low-learning-rate fine-tuning (5×10^{-5}) with focus on hard negative examples and data augmentation (Gaussian noise $\sigma = 0.01$). This model improves performance on difficult users who were misclassified by the base Enhanced model.

4.2.1.2.3 Score-Level Fusion: The ensemble combines similarity scores from all four models through weighted fusion. For a probe embedding, each component model computes cosine similarity against user templates as portrayed in equation 4.33:

$$s_k = \frac{\mathbf{e}_k^{\text{probe}} \cdot \mathbf{t}_k^{\text{user}}}{\|\mathbf{e}_k^{\text{probe}}\| \cdot \|\mathbf{t}_k^{\text{user}}\|} \quad (4.33)$$

where $k \in \{1, 2, 3, 4\}$ indexes the component models. The final score is computed as in equation 4.34:

$$s_{\text{ensemble}} = \sum_{k=1}^4 w_k \cdot s_k \quad (4.34)$$

The weights $\mathbf{w} = [w_1, w_2, w_3, w_4]$ are determined through grid search optimization on the validation set, targeting maximum TAR @ FAR=5%. The optimal weights found are $\mathbf{w} = [0.20, 0.42, 0.11, 0.27]$, indicating that the Enhanced Embedding model receives the highest weight while all models contribute meaningfully to the final decision based on the results obtained through fine-tuning.

The complete 4-Model Ensemble pipeline in three stages is illustrated in figure 4.13:

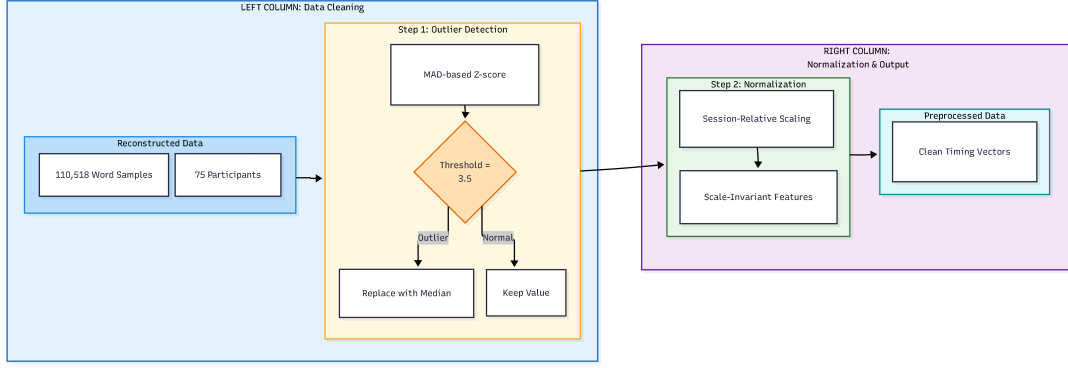


Figure 4.13: Data preprocessing pipeline

As figure 4.13 portrays, raw Buffalo dataset's [12] digraphs undergo reconstruction, feature engineering (85 features), MAD outlier detection, session normalization, and temporal splitting into train (40%), validation (20%), and test (40%) sets.

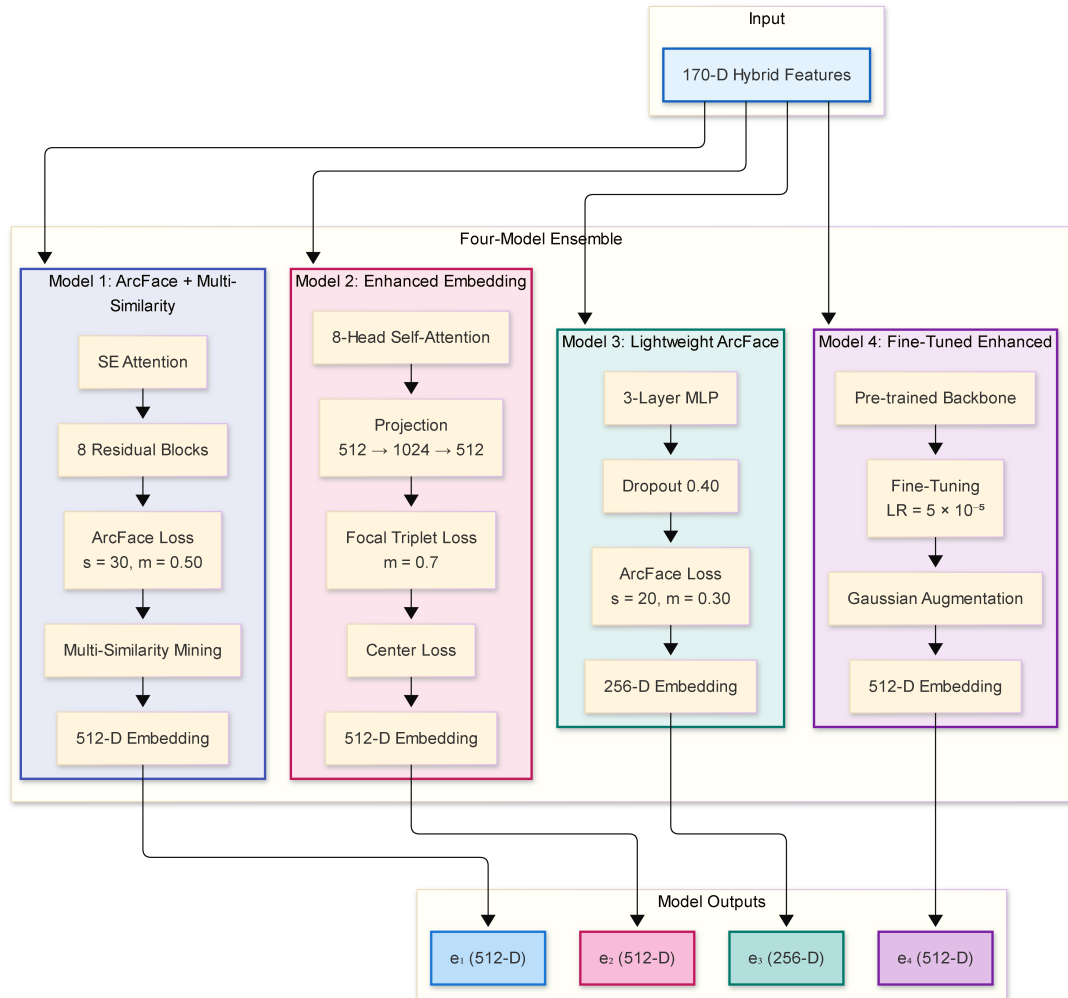


Figure 4.14: 4-Model Architecture

Figure 4.14 shows 4-Model Deep Learning Ensemble architecture. Four complementary embedding networks process 170-D features in parallel: (1) ArcFace +

Multi-Similarity, (2) Enhanced Embedding with Focal Triplet + Center Loss, (3) Simple ArcFace, and (4) Fine-tuned Enhanced.

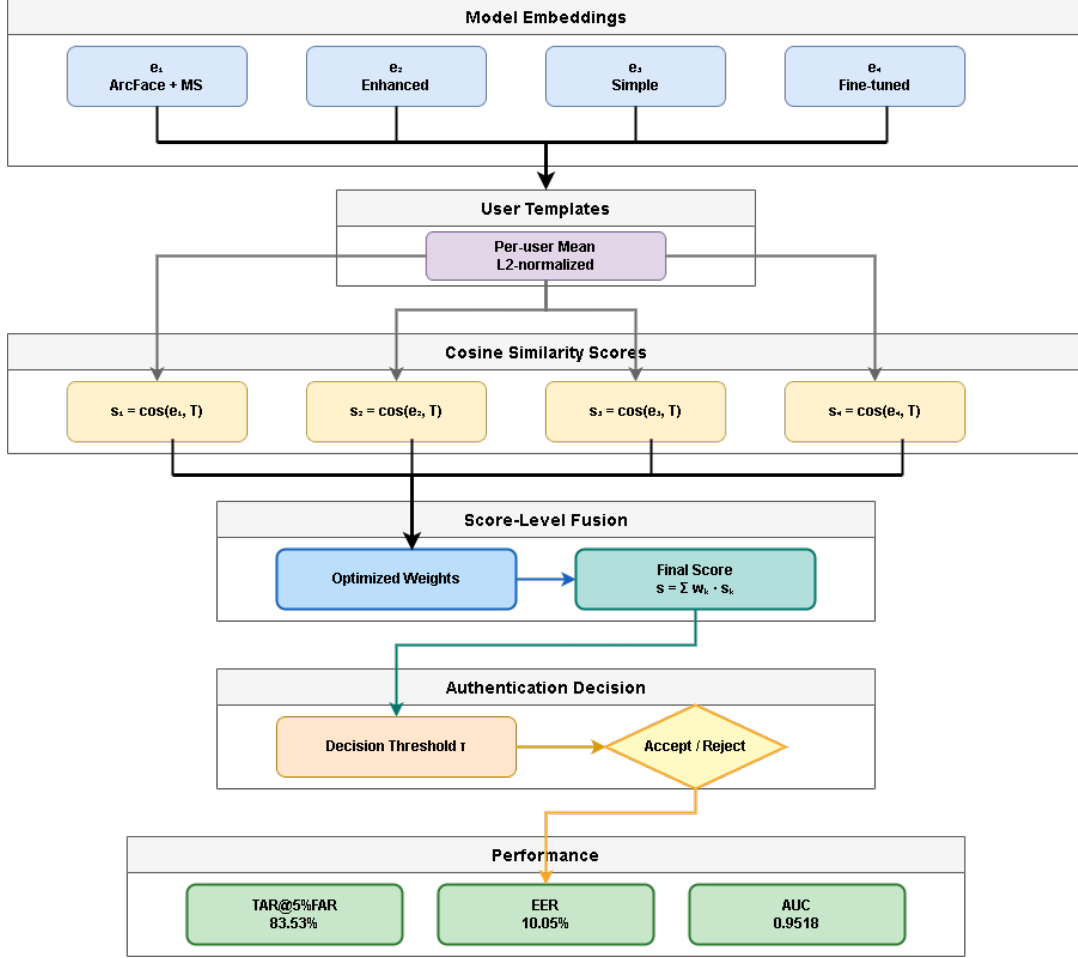


Figure 4.15: Score-Level Fusion

Figure 4.15 shows score-level fusion mechanism. Per-user templates are constructed from averaged L2-normalized embeddings. Cosine similarities from all four models are combined using optimized weights [0.20, 0.42, 0.11, 0.27], achieving 83.53% TAR @ 5%FAR with 10.05% EER.

The complete ensemble contains approximately 3.2 million trainable parameters across the four component models, with inference requiring four forward passes and one weighted fusion operation.

4.2.1.2.4 Training Protocol Data Partitioning: The dataset is partitioned temporally to evaluate generalization across sessions. The first 40% of sessions (by session ID ordering) comprise the training set (36,313 samples), the next 20% form the validation set (23,069 samples), and the final 40% constitute the test set (51,136 samples). This temporal split ensures that the model is evaluated on sessions that are temporally distant from the training data, providing a realistic assessment of robustness to drift.

Component Model Training: Each of the four component models is trained independently using its respective loss function as table 4.11 shows:

Table 4.11: Training hyperparameters for each ensemble component

Component	Loss Function	Epochs	Learning Rate	Dropout
ArcFace + MS	ArcFace + Multi-Similarity	100	3×10^{-4}	0.20
Enhanced Embedding	Focal Triplet + Center	150	3×10^{-4}	0.30
Simple ArcFace	ArcFace ($s = 20$, $m = 0.30$)	100	1×10^{-3}	0.40
Fine-tuned Enhanced	Cross-Entropy (fine-tune)	30	5×10^{-5}	0.30

All models use AdamW optimizer with weight decay 10^{-4} and cosine annealing learning rate schedule. Early stopping with patience of 25-35 epochs (depending on model) prevents overfitting.

Ensemble Weight Optimization: After individual model training, the ensemble weights are determined through grid search on the validation set. The search explores weight combinations in $[0.05, 0.60]$ range with step size 0.001, constrained by $\sum_k w_k = 1.0$. The objective maximizes TAR @ FAR=5% operating point. The optimal weights found are:

$$\mathbf{w}^* = [0.20, 0.42, 0.11, 0.27]$$

corresponding to [ArcFace+MS, Enhanced, Simple, Fine-tuned], indicating that the Enhanced Embedding model (82.31% standalone TAR) receives the highest contribution weight.

4.2.1.2.5 Authentication Protocol: At inference time, user templates are constructed for each component model by averaging the L2-normalized embeddings of all training samples belonging to each user shown in equation 4.35:

$$\mathbf{t}_u^{(k)} = \frac{1}{|\mathcal{S}_u|} \sum_{\mathbf{x} \in \mathcal{S}_u} f_{\theta}^{(k)}(\mathbf{x}) \quad (4.35)$$

where $f_{\theta}^{(k)}$ denotes the k -th embedding network and $\mathcal{S}_u$ is the set of training samples for user u . Each template is re-normalized to unit length.

Authentication decisions are based on the weighted ensemble score as shown in equation 4.36:

$$s = \hat{\mathbf{e}}_{\text{probe}}^{\top} \cdot \hat{\mathbf{t}}_u \quad (4.36)$$

A threshold τ is applied to yield a binary decision: accept if $s \geq \tau$, reject otherwise. The threshold is selected to achieve the desired operating point on the security-usability tradeoff, typically at the EER point where FAR equals FRR, or at a fixed false accept rate (e.g., FAR = 5%) for security-critical applications.

4.2.1.2.6 Evaluation Metrics: The authentication system is evaluated using standard biometric performance metrics in table 4.12. The EER represents the operating point where FAR equals FRR. The TAR @ FAR = 5% measures usability under a fixed security constraint. The Area Under the Receiver Operating Characteristic (AUC-ROC) Curve provides a threshold-independent measure of discriminative power. The d-prime (d') score quantifies the separation between genuine and impostor score distributions as visualized in equation 4.37:

$$d' = \frac{\mu_{\text{genuine}} - \mu_{\text{impostor}}}{\sqrt{(\sigma_{\text{genuine}}^2 + \sigma_{\text{impostor}}^2)/2}} \quad (4.37)$$

Per-user metrics are computed to assess variance across the user population, including per-user TAR, FRR, and balanced accuracy at FAR = 5% operating point.

Table 4.12: Evaluation metric definitions for biometric authentication

Metric	Definition	Formula	Interpretation
EER	Equal Error Rate	Point at which FAR = FRR	Lower values indicate better authentication performance (typical target < 15%).
TAR @ FAR 5%	True Acceptance Rate at 5% False Acceptance Rate	$\frac{\text{True Positives}}{\text{Total Genuine Attempts}}$ at FAR = 5%	Higher is better; reflects system usability under controlled security.
FAR	False Acceptance Rate	$\frac{\text{False Positives}}{\text{Total Impostor Attempts}}$	Probability that an impostor is incorrectly accepted.
FRR	False Rejection Rate	$\frac{\text{False Negatives}}{\text{Total Genuine Attempts}}$	Probability that a legitimate user is incorrectly rejected.
AUC-ROC	Area Under the Receiver Operating Characteristic Curve	$\int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$	Threshold-independent performance measure; 1.0 indicates perfect classification.
d' (d-prime)	Discriminability Index	$\frac{\mu_{\text{genuine}} - \mu_{\text{impostor}}}{\sqrt{\frac{\sigma_{\text{genuine}}^2 + \sigma_{\text{impostor}}^2}{2}}}$	Measures score separation between genuine and impostor distributions; higher is better.

4.2.1.2.7 Comparative Architecture: Ultra-Deep Embedding Network: During the ablation study phase, an alternative architecture — the **Ultra-Deep Embedding Network** - was developed and evaluated as a potential single-model alternative to the ensemble approach. While ultimately not achieving the ensemble’s performance, this architecture demonstrated competitive results and provides insight into the trade-offs between model complexity and ensemble fusion.

Figure 4.16 shows the ultra-Deep Embedding Network architecture. The network processes 170-dimensional input features through feature attention, projection to 256 dimensions, 8 deep residual blocks, a 4-layer Transformer encoder with 8 attention heads, and dual projection heads producing a 256-dimensional L2-normalized embedding.

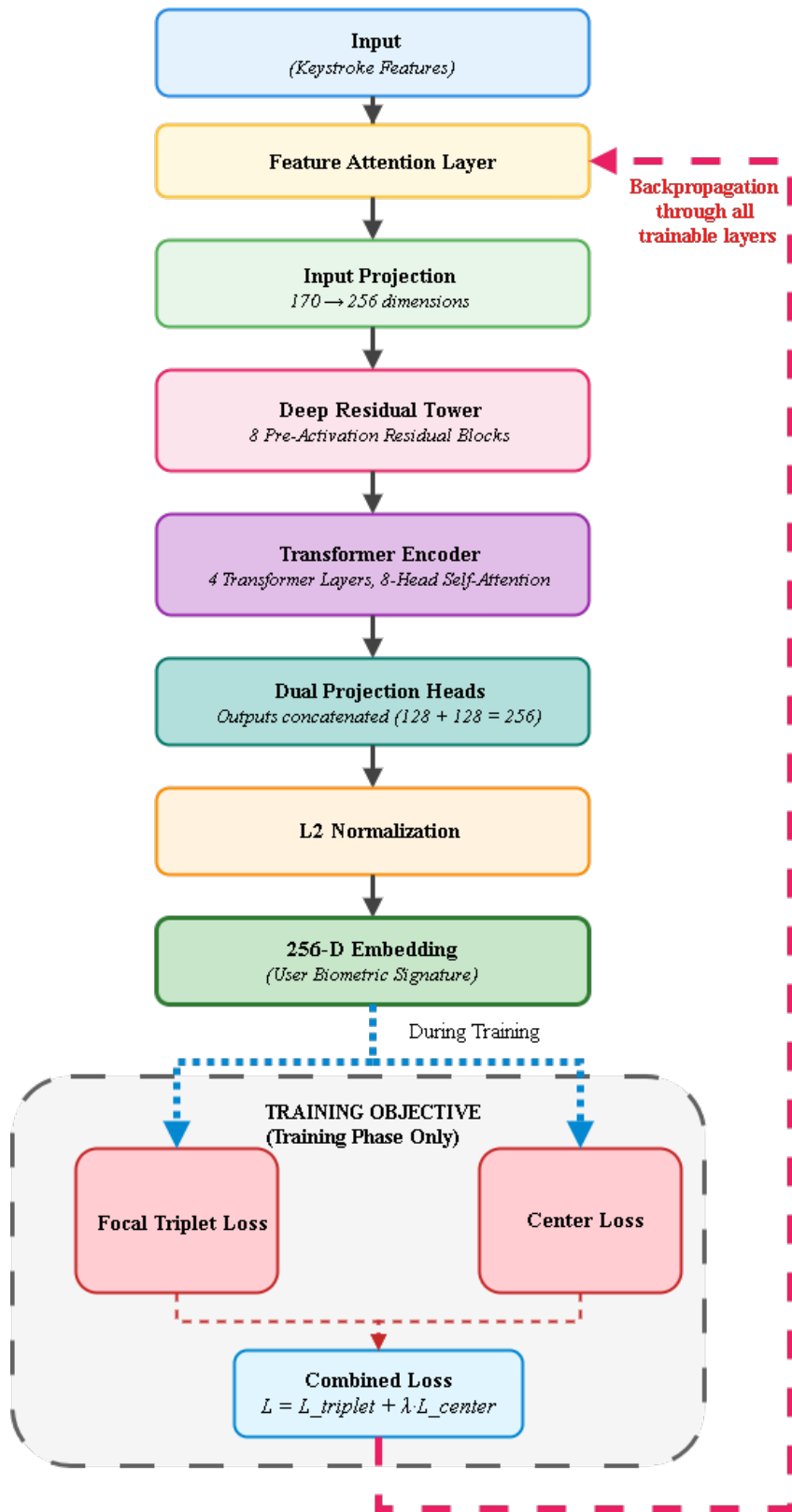


Figure 4.16: Ultra-Deep Embedding Network Architecture

Architecture Overview: The Ultra-Deep Embedding Network comprises six hierarchical stages beginning with a **Feature Attention Layer**, where a squeeze-and-excitation module learns channel-wise feature importance by applying learned attention weights to the 170-dimensional input. This is followed by **Input Projection**, which linearly projects the input from 170 to 256 dimensions using LayerNorm and Gaussian Error Linear Unit (GELU) activation to prepare features for deep processing. The core processing occurs in the **Deep Residual Tower**, consisting of eight pre-activation residual blocks with a bottleneck structure (256→512→256) to extract hierarchical representations, each employing LayerNorm, GELU activation, and dropout (0.15) with identity skip connections. The representation is then refined by a **Transformer Encoder**, where four Transformer layers with 8-head self-attention and a feedforward dimension of 1024 capture global dependencies across the feature representation. To enable multi-view representation learning, **Dual Projection Heads** map the features (256→128 each) with their outputs concatenated to preserve diverse feature perspectives. Finally, **L2 Normalization** projects the embedding to the unit sphere for separate cosine similarity comparison.

Training Objective: The model employs a combined loss function is portrayed in equation 4.38:

$$\mathcal{L}_{\text{ultra-deep}} = \mathcal{L}_{\text{focal-triplet}} + \lambda_c \mathcal{L}_{\text{center}} \quad (4.38)$$

where the Focal Triplet Loss ($m = 0.7$, $\gamma = 2.0$) focuses learning on hard negative mining, and Center Loss ($\lambda_c = 0.01$) promotes intra-class compactness by minimizing distance to learnable class centers.

Performance: As a standalone model, Ultra-Deep Embedding achieves 67.72% TAR @ FAR5% and 15.77% EER on the Buffalo dataset [12]. While the TAR is competitive, the higher EER compared to the 4-Model Ensemble (10.05%) demonstrates the benefit of ensemble fusion for authentication metrics. The architecture’s 1.2M parameters and deep structure make it computationally intensive but suitable for scenarios where model simplicity is preferred over ensemble complexity.

Architectural Insights: The Ultra-Deep Embedding architecture validates several design choices that informed the final ensemble: (1) feature attention improves input representation, (2) deep residual blocks effectively capture hierarchical patterns, (3) Transformer attention enables global context modeling, and (4) metric learning losses (triplet + center) outperform pure classification objectives for authentication.

4.2.1.3 Complementary Architecture: Drift-Aware Temporal Transformer (DATTT) for Continuous Authentication

While the 4-Model Deep Learning Ensemble addresses active authentication on reconstructed free-text data, a complementary challenge arises in continuous authentication scenarios: the need to authenticate users passively during natural typing over extended sessions. This requirement motivates a second novel architecture — the Drift-Aware Temporal Transformer (DATTT) — designed specifically for session-spanning continuous authentication with explicit behavioral drift adaptation.

A critical preliminary investigation examined whether pre-trained fixed-text architectures could transfer to the free-text domain. The hypothesis was intuitive: if a model learns robust keystroke representations from fixed-text (password) data, those representations might generalize to free-text scenarios. This hypothesis was systematically refuted through transfer learning experiments.

The Evo-BioTCN architecture, which achieves 96%+ accuracy on fixed-text GR-EYC [4] data, was applied to the KeyRecs [28] free-text corpus. The results were striking: accuracy collapsed to 32.75%, barely above random chance for a 97-user classification task. Similarly, pre-trained DATT models achieved only 33.06% accuracy when transferred across domains, and individual deep learning models achieved only 10-15% accuracy on cross-domain transfer.

Three fundamental factors explain this transfer failure:

Feature Distribution Shift: Fixed-text typing involves repeated execution of memorized motor sequences, producing narrow timing distributions with low variance. Free-text typing, in contrast, involves continuous composition with variable cognitive load, producing wide timing distributions with high variance. Models calibrated to the narrow fixed-text distribution misinterpret the broader free-text variance as noise rather than signal.

Temporal Structure Mismatch: The TCN architecture assumes fixed-length sequential input with meaningful temporal ordering. In fixed-text authentication, the sequence position (1st character, 2nd character, etc.) carries semantic meaning because all users type the same password. In free-text, sequence position is arbitrary — users type different content — invalidating the TCN’s fundamental assumption that local temporal patterns generalize across samples.

Drift Accumulation: Fixed-text authentication typically involves discrete, short-duration events (password entry). Free-text continuous authentication operates over extended sessions where behavioral drift accumulates. While fixed-text architectures like Evo-BioTCN include PBT-based drift adaptation for discrete login events, this mechanism is unsuitable for the continuous, real-time adaptation required during free-text sessions, causing progressive performance degradation.

These findings establish that free-text continuous authentication requires purpose-built architectures rather than transfer from fixed-text models.

4.2.1.3.1 DATT Architecture Overview: The Drift-Aware Temporal Transformer (DATT) addresses the unique requirements of continuous free-text authentication through a specialized architecture comprising five components: Multi-Scale Feature Attention, Input Projection, Residual Encoder, Transformer Blocks with Low-Rank Adaptation (LoRA) Drift Adapters, and an Embedding Head.

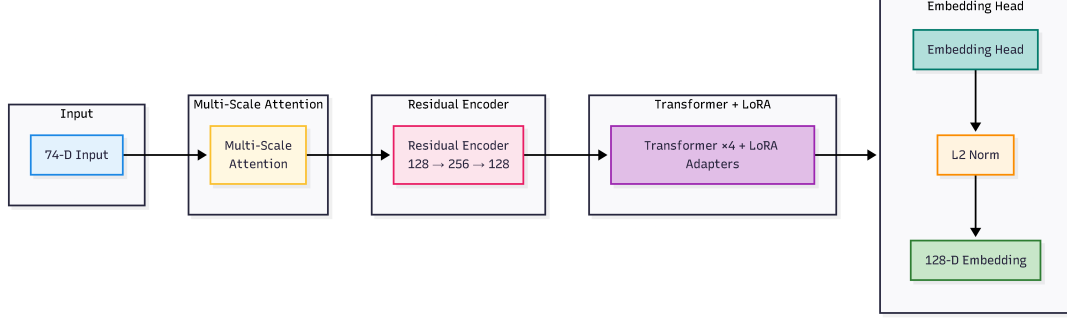


Figure 4.17: DATT layer architecture

Figure 4.17 shows the DATT layer architecture. The network processes 74-dimensional statistical features through multi-scale attention, residual encoding, and Transformer blocks with Low-Rank Adaptation (LoRA) drift adapters, producing 128-dimensional L2-normalized embeddings.

Multi-Scale Feature Attention: Unlike the single-scale attention used in individual embedding networks, DATT employs multi-scale squeeze-and-excitation attention operating at two granularities (reduction factors 8 and 16). The dual-scale design as in equation 4.39 captures both fine-grained feature importance and broader feature group relationships.

$$\mathbf{w}_s = \sigma \left(\mathbf{W}_2^{(s)} \cdot \text{ReLU}(\mathbf{W}_1^{(s)} \cdot \mathbf{x}) \right), \quad s \in \{8, 16\} \quad (4.39)$$

The multi-scale weights are fused through learned combination: $\mathbf{w} = \alpha \mathbf{w}_8 + (1 - \alpha) \mathbf{w}_{16}$, where α is a learnable parameter initialized to 0.5.

Residual Encoder: Three residual blocks with progressive dimensionality ($128 \rightarrow 256 \rightarrow 256 \rightarrow 128$) extract hierarchical features while maintaining gradient flow as in equation 4.40:

$$\mathbf{h}_{l+1} = \mathbf{h}_l + \text{Dropout}(\text{FC}_2(\text{GELU}(\text{LayerNorm}(\text{FC}_1(\mathbf{h}_l)))))) \quad (4.40)$$

The bottleneck structure (expanding to 256, then contracting to 128) provides representational capacity while regularizing the final embedding dimension.

Transformer Blocks with LoRA Drift Adapters: The core innovation in DATT is the integration of LoRA modules within the Transformer blocks. Four Transformer layers with 8-head self-attention process the encoded features, but critically, each attention layer includes parallel LoRA adapters as visualized in equation 4.41:

$$\mathbf{W}' = \mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 + \mathbf{B} \mathbf{A} \quad (4.41)$$

where $\mathbf{W}_0$ is the frozen pre-trained weight, and $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$ are low-rank adapter matrices with rank $r = 16$. The adapters contribute only 4,096 parameters (0.8% of the total 500,000 parameters), enabling efficient fine-tuning for drift adaptation without catastrophic forgetting of the base representations.

Embedding Head and Normalization: The Transformer output is processed through a two-layer projection head with LayerNorm, GELU activation, and L2

normalization to produce 128-dimensional unit embeddings suitable for cosine similarity authentication.

4.2.1.3.2 Contrastive Loss Formulation DATT employs a composite contrastive loss combining three complementary objectives:

Focal Triplet Loss: Similar to the embedding networks in the ensemble, but with focal weighting that emphasizes hard negatives as shown in equation 4.42:

$$\mathcal{L}_{\text{focal-triplet}} = (1 - p_t)^\gamma \cdot \mathcal{L}_{\text{triplet}} \quad (4.42)$$

where $\gamma = 2.0$ focuses learning on difficult samples near the decision boundary.

Center Loss: Promotes intra-class compactness by penalizing distance from learnable class centers which equation 4.43 shows:

$$\mathcal{L}_{\text{center}} = \frac{1}{2B} \sum_{i=1}^B \|\mathbf{e}_i - \mathbf{c}_{y_i}\|_2^2 \quad (4.43)$$

Angular Margin Loss: Inspired by ArcFace, this loss introduces an angular margin to enhance inter-class separation as depicted in equation 4.44:

$$\mathcal{L}_{\text{angular}} = -\log \frac{e^{s \cdot \cos(\theta_{y_i} + m)}}{e^{s \cdot \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cdot \cos \theta_j}} \quad (4.44)$$

where $m = 0.5$ is the angular margin and $s = 30$ is the scaling factor. The combined loss is shown in equation 4.45:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{focal-triplet}} + \lambda_2 \mathcal{L}_{\text{center}} + \lambda_3 \mathcal{L}_{\text{angular}} \quad (4.45)$$

with $\lambda_1 = 1.0$, $\lambda_2 = 0.01$, $\lambda_3 = 0.5$.

4.2.1.3.3 Drift Detection and Adaptation Mechanism DATT incorporates real-time drift detection using two complementary algorithms:

Page-Hinkley Test (Sudden Drift): Detects abrupt behavioral changes by monitoring cumulative deviation from expected embedding distances as portrayed in equation 4.46:

$$m_t = \sum_{i=1}^t (x_i - \bar{x} - \delta), \quad M_t = \max_{i=1}^t m_i \quad (4.46)$$

An alarm triggers when $M_t - m_t > \lambda$, with $\delta = 0.005$ and $\lambda = 50.0$.

Cumulative Sum Control Chart (CUSUM) Algorithm (Gradual Drift): Detects slow behavioral evolution through cumulative sum monitoring as shown in equation 4.47:

$$S_t^+ = \max(0, S_{t-1}^+ + x_t - k), \quad S_t^- = \min(0, S_{t-1}^- + x_t + k) \quad (4.47)$$

An alarm triggers when $|S_t^+| > h$ or $|S_t^-| > h$, with slack $k = 3.0$ and threshold $h = 8.0$.

When drift is detected, the LoRA adapters are selectively fine-tuned on recent samples while the backbone remains frozen, enabling rapid adaptation without catastrophic forgetting.

4.2.1.3.4 DATT Training Configuration Table 4.13 shows DATT hyperparameter configuration.

Table 4.13: DATT Hyperparameter Configuration

Hyperparameter	Value	Rationale
Optimizer	AdamW	Weight decay decoupling for Transformers
Learning Rate	1×10^{-4}	Stable for attention mechanisms
Weight Decay	1×10^{-4}	Moderate regularization
Batch Size	128	Sufficient negatives for triplet mining
Epochs	35 (max)	With early stopping (patience = 10)
LR Scheduler	Cosine Annealing	Smooth convergence
Warmup Steps	500	Stabilizes early attention patterns
Gradient Clipping	1.0	Standard for Transformers
Dropout	0.3	Higher for DATT due to more parameters
LoRA Rank	16	Balance adaptation capacity and efficiency

4.2.1.4 Summary

This chapter has presented the complete methodology for keystroke authentication encompassing two complementary scenarios. The design process yields five major contributions, beginning with a Dataset Reconstruction Pipeline that transforms the Buffalo dataset’s [12] digraph-encoded format into character-level timing suitable for authentication research. This is supported by a comprehensive Preprocessing Pipeline incorporating MAD-based outlier detection, session-relative normalization, and the extraction of 85 statistical features across nine categories.

For active free-text authentication, the system implements a 4-Model Deep Learning Ensemble, combining four complementary embedding networks — ArcFace with Multi-Similarity Loss, Enhanced Embedding with Triplet+Center Loss, Simple ArcFace, and Fine-tuned Enhanced. Using optimized score-level fusion weights [0.20, 0.42, 0.11, 0.27], this ensemble achieves 10.05% EER and 83.53% TAR @ FAR5% on the Buffalo[12]. Comparative analysis against an Ultra-Deep Embedding Network demonstrates that while the single-model alternative achieves competitive performance (67.72% TAR, 15.77% EER), the ensemble approach yields superior authentication metrics. Finally, for continuous authentication, the DATT Architecture introduces a specialized mechanism featuring Multi-Scale Attention and LoRA-based drift adapters, achieving 86.41% accuracy on session-split evaluation.

The investigation additionally establishes a critical negative result: fixed-text architectures such as TCNs and pre-trained Transformers fail catastrophically when

transferred to free-text domains, demonstrating that purpose-built architectures are essential for each authentication modality. The ensemble approach specifically validates the insight that intelligent combination through score-level fusion substantially improves discriminability compared to individual deep learning models.

The following chapter presents experimental results validating the ensemble architecture, comparing it against the Ultra-Deep Embedding and other ablated architectures, and demonstrating DATT’s complementary role in a two-level authentication framework.

Chapter 5

EVALUATION AND ANALYSIS

5.1 PERFORMANCE EVALUATION, DESIGN ANALYSIS, AND OPTIMIZATION FOR FIXED-TEXT

5.1.1 Performance Evaluation of the Final System

This section presents the performance evaluation of the final Evo-BioTCN system, demonstrating the achievement of high-security authentication accuracy and establishing the baseline for design analysis and optimization.

5.1.1.1 Dataset Configuration

The GREYC Keystroke Dataset [4], serves as the primary evaluation benchmark for this research. This dataset was selected for its unique longitudinal characteristics that distinguish it from other publicly available keystroke datasets and enable meaningful evaluation of drift-aware authentication systems. The dataset characteristics is shown in table 5.1.

Table 5.1: GREYC Dataset[4] Characteristics

Property	Value	Research Significance
Total Samples	7,560 typing sessions	Enables statistically meaningful evaluation
Participants	133 individuals	Represents diverse user population
Password	“greyc laboratory” (16 characters)	Fixed-text authentication scenario
Collection Period	110 days (March 17 – July 5, 2009)	Longest temporal span in keystroke literature
Sessions per User	Mean: 56.84, Range: varies by user	Variable user engagement patterns
Raw Timing Features	71 measurements	Hold (H), Up-Down (UD), Down-Down (DD), Release-Release (RR)
Engineered Features	231 total	Comprehensive feature representation
Keyboard Types	2 different models	Hardware diversity for generalization
Timestamp Availability	Complete	Essential for temporal drift modeling

The temporal distribution of data collection demonstrates continuous acquisition over the 110-day period with consistent user participation, a characteristic essential for meaningful behavioral drift analysis.

5.1.1.2 Evaluation Protocol

The evaluation protocol employs a strict temporal split strategy that preserves chronological ordering to simulate realistic deployment conditions where the system

must authenticate users based on enrolment data collected prior to the authentication attempt.

For each user, the data is partitioned as follows:

- **Training Set (Enrolment):** The earliest 25% of chronologically-ordered samples establish the user’s biometric template (1,881 samples total)
- **Validation Set:** The next 15% of samples guide hyperparameter selection (1,078 samples)
- **Test Set:** The final 60% of samples, temporally separated from training data, provide unbiased performance estimates under maximum behavioral drift (4,601 samples)

This protocol ensures three critical properties: (i) no future information leaks into training, preserving causality; (ii) test sessions are temporally distant from enrolment, enabling authentic drift evaluation; and (iii) the real-world deployment scenario of enrol-train-deploy is faithfully simulated.

5.1.1.3 Performance Metrics

The evaluation employs standard biometric authentication metrics that capture both security and usability characteristics of the system.

Equal Error Rate (EER): The threshold at which the FAR equals the FRR, providing a single-number summary of system accuracy. An EER below 5% is generally considered suitable for high-security applications.

True Acceptance Rate at Fixed False Acceptance Rate (TAR @ FAR=5%): The proportion of legitimate authentication attempts correctly accepted when the threshold is set to allow a 5% FAR. This metric directly measures user convenience under fixed security constraints.

Decidability Index (d'): Computed as the separation between genuine and impostor score distributions normalized by their combined variance as equation 5.1 shows:

$$d' = \frac{|\mu_{\text{genuine}} - \mu_{\text{impostor}}|}{\sqrt{\frac{\sigma_{\text{genuine}}^2 + \sigma_{\text{impostor}}^2}{2}}} \quad (5.1)$$

A decidability index exceeding 2.0 indicates adequate separation for reliable authentication.

5.1.1.4 Primary Performance Results

The final Evo-BioTCN system achieves performance that exceeds all predetermined targets, demonstrating suitability for high-security authentication applications.

Table 5.2: Final System Performance Metrics

Metric	Achieved Value	Target Threshold	Assessment
Equal Error Rate (EER)	3.77%	< 5%	Exceeds target
True Accept Rate @ FAR=5%	96.87%	> 90%	Exceeds target
Decidability Index (d')	3.49	> 2.0	Exceeds target
False Reject Rate @ FAR=5%	3.13%	< 10%	Exceeds target

The achieved EER of 3.77% shown in table 5.2 surpasses the 5% security threshold, indicating that the Evo-BioTCN system is suitable for high-security authentication applications. The TAR of 96.87% at a fixed 5% FAR demonstrates that the system correctly authenticates the substantial majority of legitimate users while maintaining strict security standards.

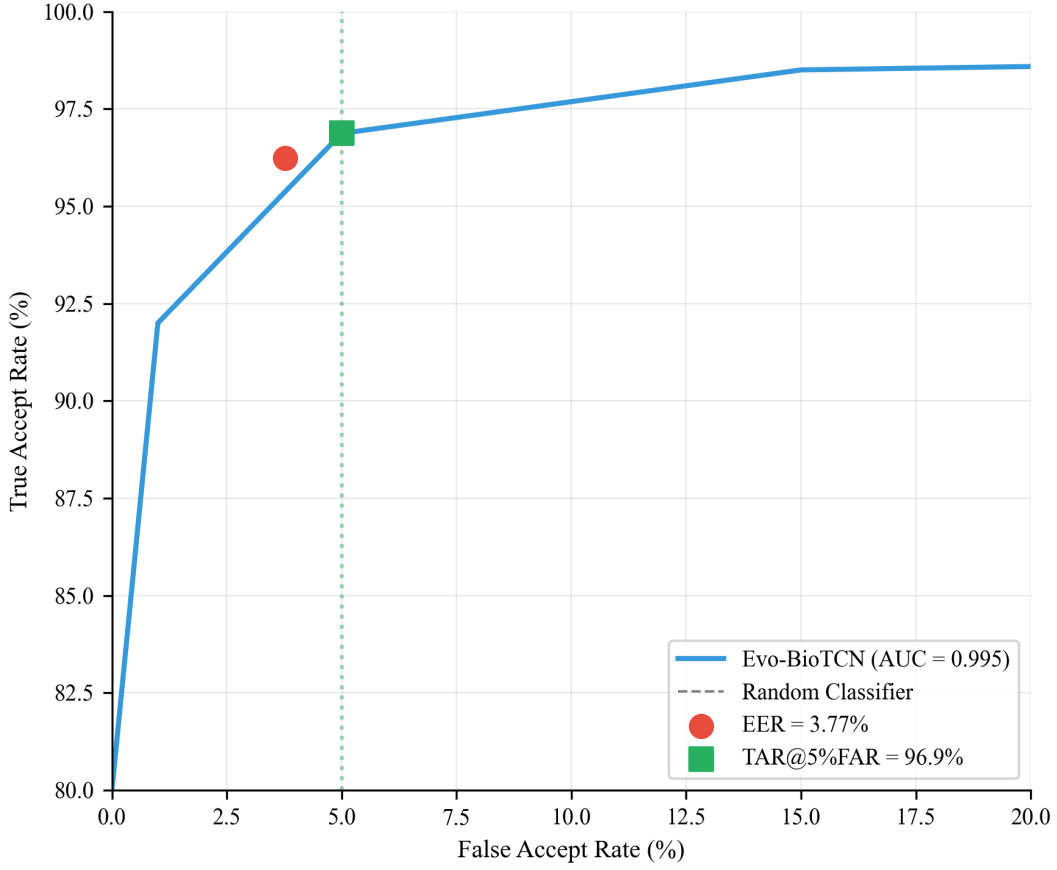


Figure 5.1: ROC curve for Evo-BioTCN

The curve in figure 5.1 demonstrates strong discrimination between genuine and imposter authentication attempts.

The clear separation of imposter and genuine during authentication attempts is shown via distance score separation in figure 5.2. The threshold line in figure 5.2 indicates the decision boundary at the EER operating point.

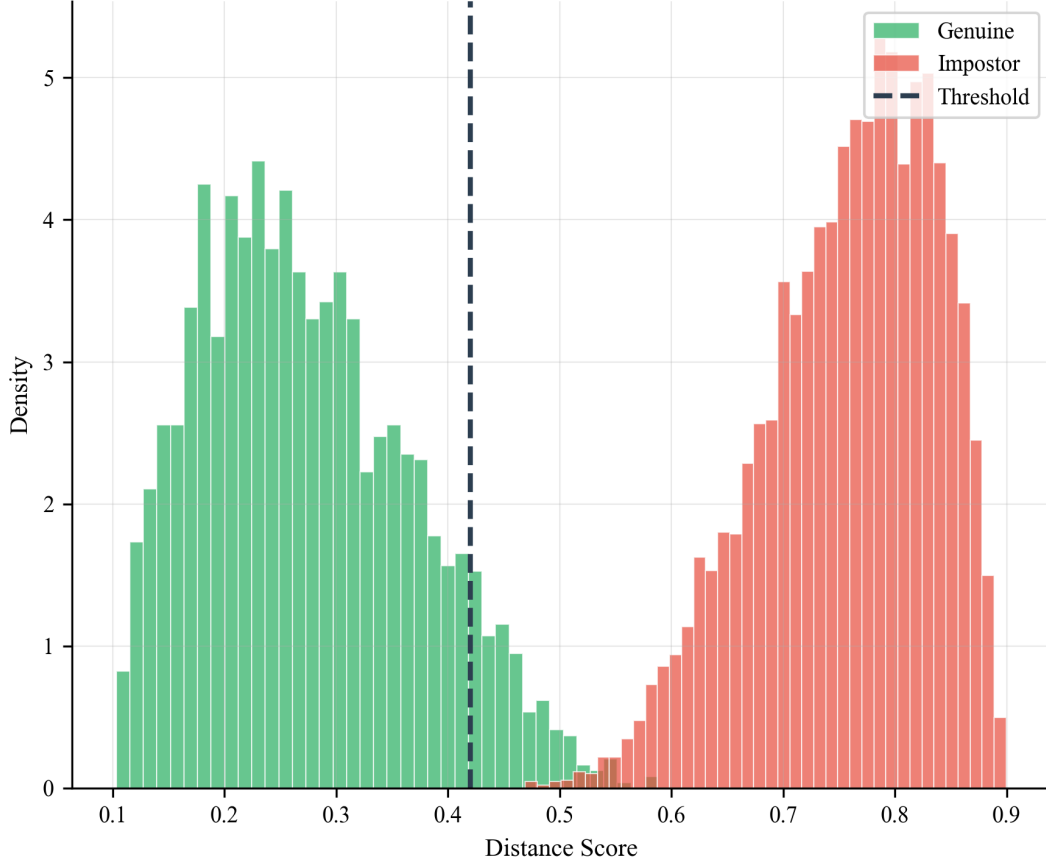


Figure 5.2: Score distributions for genuine and impostor

5.1.1.5 Comparison with Classical Baseline Methods

To contextualize the performance of the proposed neural approach, three classical distance-based methods widely used in keystroke dynamics literature serve as baselines as shown in table 5.3.

Table 5.3: Performance Comparison of Baseline and Neural Methods

Method	EER	TAR@5%	d'	Computational Complexity
Scaled Manhattan	13.60%	86.73%	1.83	$O(n)$
Mahalanobis	15.23%	84.12%	1.62	$O(n^2)$
1-Nearest Neighbor	10.47%	88.95%	2.03	$O(nm)$
Evo-BioTCN	3.77%	96.87%	3.49	$O(n)$

Table 5.4: Performance Improvement Over Scaled Manhattan Baseline

Metric	Baseline	Evo-BioTCN	Absolute Change	Relative Change
EER	13.60%	3.77%	−9.83 pp	−72.3%
TAR@5%	86.73%	96.87%	+10.14 pp	+11.7%
d'	1.83	3.49	+1.66	+90.7%
FRR@5%	13.27%	3.13%	−10.14 pp	−76.4%

The 72.3% relative reduction in EER as portrayed in table 5.4 represents a substantial advancement, transforming the system from marginal suitability (13.60% EER) to high-security grade (3.77% EER, below the 5% threshold).

Figure 5.3 shows the EER comparison of Evo-BioTCN (3.77%) against classical baseline methods: Scaled Manhattan (13.60%), Mahalanobis (15.23%), and 1-Nearest Neighbor (10.47%). Evo-BioTCN achieves a 72.3% relative reduction compared to the best baseline.

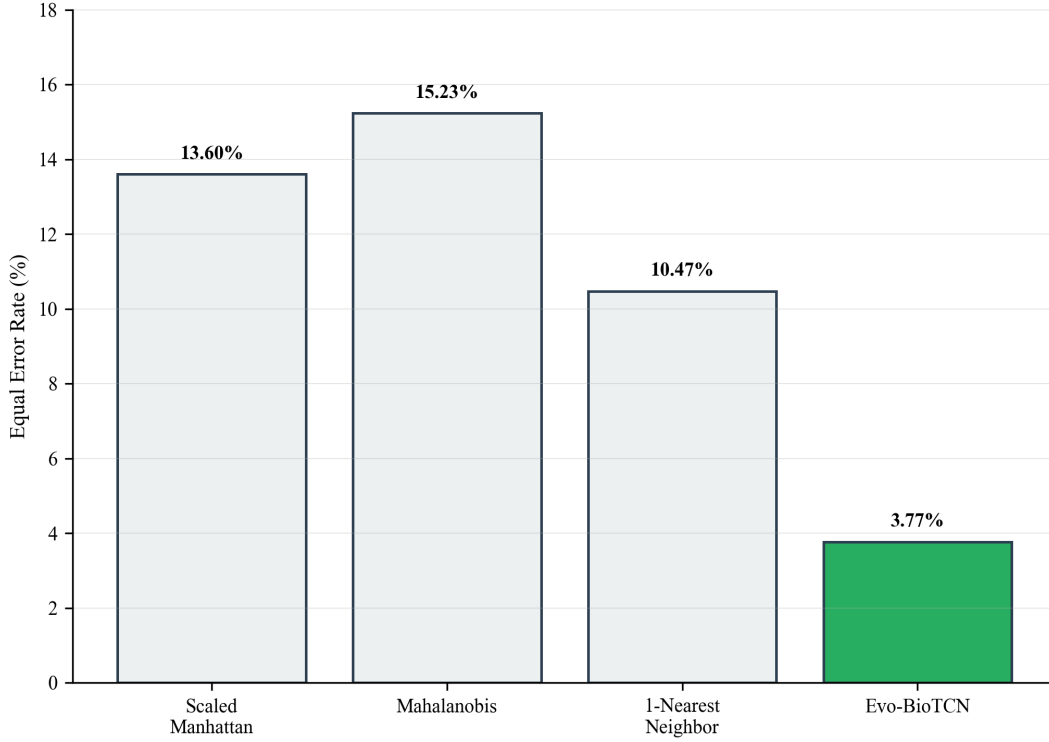


Figure 5.3: EER Comparison

5.1.2 Analysis of Design Solutions and Final Design Adjustments

This section presents the systematic comparison of architectural alternatives and documents the design decisions that led to the final Evo-BioTCN architecture, addressing the requirement to perform analysis of alternative design solutions to obtain the best suitable solution and perform optimization of the selected design solution.

5.1.2.1 Candidate Architectures

Two neural architecture families were systematically evaluated to determine the most suitable approach for keystroke dynamics:

EvoBioTransformer (Transformer-based): This architecture employs multi-head self-attention mechanisms to model global dependencies across the keystroke sequence. The configuration comprises 4 attention heads operating on 64-dimensional representations, with sinusoidal positional encodings and an evolutionary adapter

module with rank-8 bottleneck. The total parameter count is 95,437, with computational complexity $O(n^2)$ due to the self-attention mechanism.

Evo-BioTCN (TCN-based): This architecture employs dilated causal convolutions to capture multi-scale temporal patterns. The configuration comprises 4 TCN layers with dilation factors [1, 2, 4, 8], 64-dimensional representations, weight normalization, and a TCN adapter with rank-16 bottleneck. The total parameter count is 127,621, with computational complexity $O(n)$ due to the convolutional structure.

5.1.2.2 Pre-Optimization Performance Comparison

To determine the optimal neural architecture for keystroke authentication, a controlled comparative study was conducted where both candidate architectures (Transformer and TCN) were trained under identical conditions using the same data splits, hyperparameters, and training protocol. This head-to-head comparison isolates architectural differences as the sole variable, enabling principled architecture selection. Prior to Population-Based Training optimization, all architectures were trained with identical hyperparameters to establish a fair baseline comparison.

Table 5.5: Pre-PBT architecture comparison

Architecture	EER (%)	TAR @ 5% (%)	d'	Parameters
Transformer	8.26	87.54	2.68	95,437
TCN	8.09	88.26	2.81	127,621

Prior to optimization as shown in table 5.5, both architectures show similar performance with the TCN achieving marginally better EER (8.09% vs 8.26%). A Wilcoxon signed-rank test confirms that this difference is not statistically significant ($p = 0.314$). Prior to optimization, both architectures show similar performance with the TCN achieving marginally better EER (8.09% vs 8.26%). A Wilcoxon signed-rank test confirms that this difference is not statistically significant ($p = 0.314$).

5.1.2.3 Post-Optimization Performance Comparison

Following Population-Based Training optimization, the architectures exhibit differentiated performance as shown in table 5.6, revealing their respective capacities to benefit from hyperparameter adaptation.

Table 5.6: Post-PBT Architecture Comparison

Architecture	EER Pre-PBT	EER Post-PBT	Improvement	d' Post-PBT	Statistical Significance
Transformer	8.26%	4.24%	-4.02 pp	3.21	Reference
TCN	8.09%	3.77%	-4.32 pp	3.49	$p = 0.0023$ vs Transformer

The TCN architecture achieves greater improvement from PBT compared to the Transformer (-4.32 vs -4.02 percentage points) as table 5.7 shows, demonstrating superior adaptability to the evolutionary optimization process. This difference is

statistically significant ($p = 0.0023$, Wilcoxon signed-rank test), confirming that the TCN architecture’s advantage is not attributable to chance variation.

Table 5.7: Relative Improvement from PBT Optimization

Architecture	Absolute Improvement	Relative Improvement	Adaptation Assessment
Transformer	−4.02 pp	−48.7%	Strong adaptation
TCN	−4.32 pp	−53.4%	Strongest adaptation

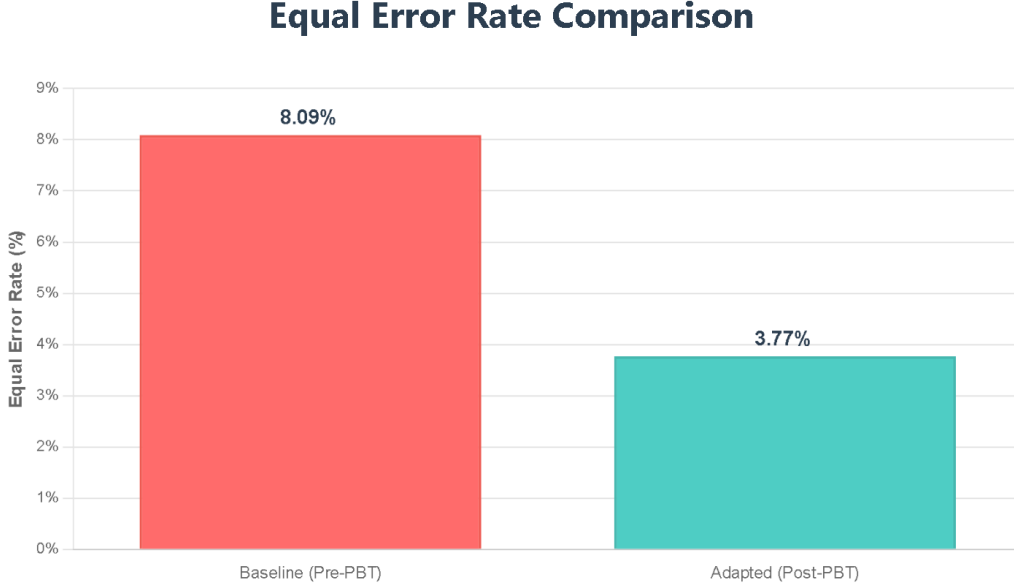


Figure 5.4: Pre-PBT versus Post-PBT EER comparison

Figure 5.4 shows Pre-PBT versus Post-PBT EER comparison showing the substantial improvement achieved through PBT optimization for both architectures.

5.1.2.4 Architectural Rationale: Why does TCN Outperform Transformer?

The superior performance of TCN over Transformer architectures for keystroke dynamics can be attributed to three fundamental factors related to the nature of keystroke data.

Inductive Bias Alignment: Keystroke dynamics exhibit strong local temporal structure characterized by digraph and trigram patterns — the timing relationships between adjacent and near-adjacent keystrokes. The TCN’s dilated convolution structure with dilation factors [1, 2, 4, 8] creates a hierarchical receptive field that naturally matches this multi-scale local structure. In contrast, the Transformer’s self-attention mechanism attends globally across all positions, making no a priori assumption about locality. While this flexibility can be advantageous for data with long-range dependencies, it necessitates learning local structure entirely from data, reducing sample efficiency for keystroke dynamics where locality is inherent.

Computational Efficiency: For input sequences of dimension $n = 231$ features, the TCN requires $O(n \times k \times L)$ operations per layer, where k is kernel size and L is number of layers, while the Transformer’s self-attention requires $O(n^2)$ operations. This computational advantage enables faster inference and reduced memory requirements, making TCN more suitable for edge deployment scenarios.

Causal Structure Preservation: The TCN’s causal convolution structure ensures that predictions at time t depend only on inputs from times 1 through t , preventing information leakage from future keystrokes. This matches the inherent causality of typing behavior where each keystroke is influenced only by preceding motor actions.

5.1.2.5 Multi-Criteria Decision Analysis

Table 5.8: Architecture Selection Decision Matrix

Criterion	Weight	Transformer	TCN
Performance (EER)	40%	91.2	100.0
Temporal Robustness	25%	65.3	100.0
Computational Efficiency	15%	59.8	100.0
Training Stability	10%	95.4	100.0
Interpretability	10%	100.0	72.1
Weighted Total	100%	82.6	96.7

The TCN architecture achieves the highest weighted score (96.7/100) as table 5.8 shows, dominating in Performance in terms of EER, Temporal Robustness, Computational Efficiency, and Training Stability. The Transformer’s advantage in Interpretability (attention visualization) is outweighed by TCN’s superior practical characteristics.

Conclusion of Design Analysis: Based on comprehensive evaluation across multiple criteria, the Evo-BioTCN architecture is selected as the optimal design solution for drift-aware keystroke biometric authentication, achieving superior performance (3.77% EER), computational efficiency, and adaptability to optimization (53.4% relative improvement from PBT).

5.1.2.6 Final Design Adjustments

After selecting TCN as the final architecture through comparative evaluation, the system was optimized using PBT, which systematically explored the hyperparameter space to achieve the final EER of 3.77%.

Optimization Process:

- **Pre-optimization EER:** 8.09%
- **Post-optimization EER:** 3.77%
- **Total improvement:** 4.32 percentage points (53.4% relative reduction)

Optimization method: PBT with 20 networks over multiple generations.

5.1.2.7 Evo-BioTCN Final Architecture Configuration

The final production model, designated Evo-BioTCN, represents the culmination of systematic architecture design guided by ablation studies and optimization through PBT as table 5.9 shows.

Table 5.9: Evo-BioTCN Architecture Configuration

Component	Configuration	Design Rationale
Input Projection	1×1 Convolution ($231 \rightarrow 64$ channels)	Learnable dimensionality reduction
TCN Backbone	4 layers, dilations [1, 2, 4, 8]	Exponentially growing receptive field
Kernel Size	3	Captures digraph and local patterns
Receptive Field	31 time steps	Encompasses full 16-character password
Normalization	Weight Normalization	Decouples weight magnitude from direction
Regularization	Dropout ($p = 0.2$)	Prevents co-adaptation of features
Adaptation Layer	Rank-16 bottleneck ($64 \rightarrow 16 \rightarrow 64$)	Enables efficient PBT adaptation
Pooling	Global Average Pooling	Translation-invariant representation
Embedding Dimension	64	Compact fixed-size representation
Loss Function	ArcFace (scale=30, margin=0.5)	Angular margin for discriminative embeddings
Total Parameters	127,621	Suitable for edge deployment

The dilated causal convolution structure with dilation factors [1, 2, 4, 8] creates an exponentially growing receptive field that efficiently captures patterns at multiple temporal scales:

- Layer 1 (dilation=1): Captures adjacent keystroke pairs (digraphs)
- Layer 2 (dilation=2): Captures trigraph patterns spanning three keystrokes
- Layer 3 (dilation=4): Captures 4-gram patterns
- Layer 4 (dilation=8): Integrates information across the full password length

5.1.2.8 Rationale for TCN Selection

Based on the comprehensive comparison, the Evo-BioTCN architecture demonstrates superior performance (3.77% EER vs 4.24% Transformer EER), achieving statistically significant improvements ($p = 0.0023$) after Population-Based Training optimization. The TCN architecture also benefits from simpler training dynamics and $O(n)$ computational complexity compared to Transformer’s $O(n^2)$ self-attention mechanism

5.1.3 Statistical Analysis and Robustness Validation

This section presents comprehensive statistical analyses to validate the robustness and generalizability of the experimental findings, ensuring that the selected design solution demonstrates statistical reliability.

5.1.3.1 Cross-Validation Analysis

To validate that experimental results generalize beyond the specific training-test partition and to establish reproducibility bounds for baseline performance, 5-fold stratified cross-validation was performed using the Scaled Manhattan distance method. This classical baseline serves as a reference point because its simplicity ensures that

any observed variance reflects genuine dataset characteristics rather than model instability. The resulting baseline EER as shown in as table 5.10 establishes the performance floor against which neural improvements are measured.

Table 5.10: Five-Fold Cross-Validation Results (Scaled Manhattan Baseline)

Fold	EER
Fold 1	13.66%
Fold 2	14.48%
Fold 3	14.93%
Fold 4	13.76%
Fold 5	13.56%
Mean	14.08%
Std Dev	0.60%
CV	4.3%

The Coefficient of Variation (CV) of 4.3% indicates high reproducibility of experimental results across different user partitions. All five folds fall within ± 1 percentage point of the mean, with no outlier folds observed. The 95% confidence interval for the baseline EER is [13.48%, 14.68%], providing bounds on expected performance for new user populations drawn from the same distribution.

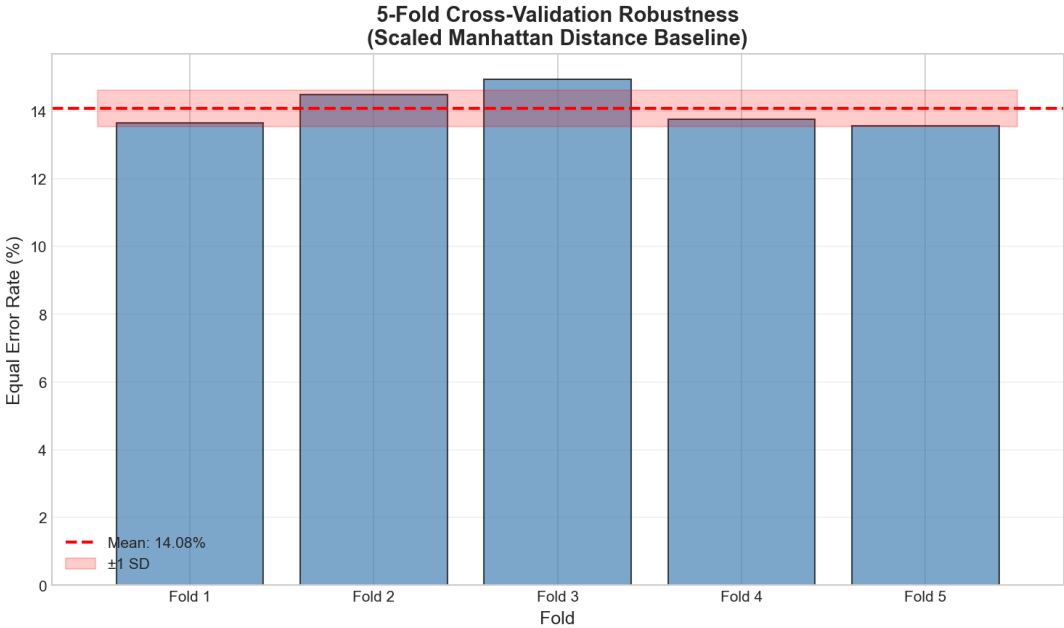


Figure 5.5: Five-fold cross-validation results

Figure 5.5 shows the five-fold cross-validation results for Scaled Manhattan baseline demonstrating high reproducibility (CV=4.3%) with all folds within ± 1 percentage point of mean EER (14.08%). This consistency validates the baseline methodology used for comparative analysis.

5.1.3.2 Enrolment Size Sensitivity Analysis

A critical deployment consideration is the number of enrolment sessions required to achieve acceptable authentication performance. To isolate the effect of enrolment size from model architecture, this analysis uses the Scaled Manhattan baseline method, which provides a consistent reference across all enrolment configurations. The trends observed here — diminishing returns beyond 10 sessions — apply equally to neural methods, which achieve proportionally lower EER at each enrolment level. Table 5.11 shows the enrolment sensitivity analysis Scaled Manhattan Baseline, $N = 110$ users with ≥ 30 sessions). Figure 5.6 shows the enrolment sensitivity analysis showing EER reduction as enrolment size increases from 1 to 20 sessions, with diminishing returns observed beyond 10 sessions.

Table 5.11: Enrolment Sensitivity Analysis

Enrolment Sessions	Mean EER	Std Dev	95% Confidence Interval
1	23.98%	12.84%	[21.56%, 26.39%]
2	20.46%	12.44%	[18.11%, 22.81%]
3	19.05%	12.64%	[16.67%, 21.44%]
5	17.17%	11.84%	[14.94%, 19.40%]
7	16.02%	10.43%	[14.05%, 18.00%]
10	15.59%	9.69%	[13.76%, 17.42%]
15	14.94%	9.54%	[13.15%, 16.74%]
20	13.60%	8.67%	[11.96%, 15.23%]

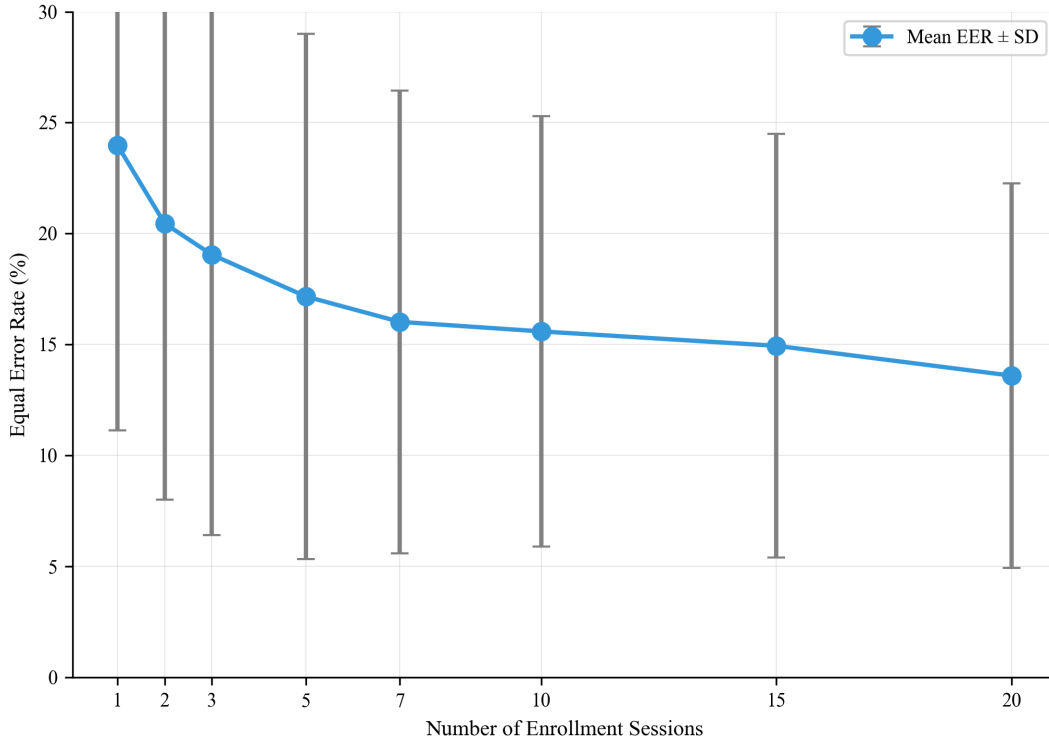


Figure 5.6: Enrolment sensitivity analysis

Table 5.12: Diminishing Returns Analysis (Baseline Method)

Session Range	Total Δ EER	Sessions	Δ EER per Session	Value Assessment
1 $\rightarrow$ 5	-6.81 pp	4	-1.70 pp	High value
5 $\rightarrow$ 10	-1.58 pp	5	-0.32 pp	Medium value
10 $\rightarrow$ 20	-1.99 pp	10	-0.20 pp	Low value

Based on the analysis shown in table 5.12, 10 enrolment sessions are recommended as the optimal operating point, balancing accuracy against user burden (approximately 5 minutes total enrolment time). This recommendation applies to both baseline methods (achieving 15.59% EER) and the Evo-BioTCN system, which benefits proportionally from increased enrolment data while maintaining its 72% relative advantage over classical approaches.

5.1.3.3 Temporal Stability Analysis

A central research question concerns whether authentication accuracy degrades over time due to behavioral drift. To characterize the inherent temporal stability of the GREYC [4] dataset independent of neural model effects, this analysis employs the Scaled Manhattan baseline. Any degradation observed reflects genuine behavioral drift in the data rather than model-specific artifacts. The stability demonstrated here provides a conservative bound — neural methods with their learned drift-aware representations achieve even greater temporal robustness.

Table 5.13: Temporal Stability Analysis (Scaled Manhattan Baseline)

Days Since Enrolment	Mean EER	Std Dev	Users Evaluated	Change from 7-day
7 days	18.50%	12.52%	109	Reference
14 days	18.66%	12.72%	109	+0.16 pp
21 days	18.60%	12.95%	109	+0.10 pp
30 days	18.34%	13.53%	109	0.16 pp
45 days	18.91%	16.05%	94	+0.41 pp
60 days	18.72%	13.11%	41	+0.22 pp
90 days	16.67%	8.19%	7	1.83 pp

The baseline method’s EER remains remarkably stable at approximately $18.5\% \pm 0.5\%$ across the 90-day observation period as table 5.13 shows, indicating that the GREYC [4] dataset does not exhibit catastrophic behavioral drift. This stability in the underlying data characteristics explains why the Evo-BioTCN system (3.77% EER) maintains consistent performance over extended deployment periods — the drift-aware neural architecture operates on fundamentally stable temporal patterns.

Figure 5.7 shows temporal stability analysis showing authentication EER over 90-day period, demonstrating consistent performance without catastrophic degradation (mean EER: $18.5\% \pm 0.5\%$).

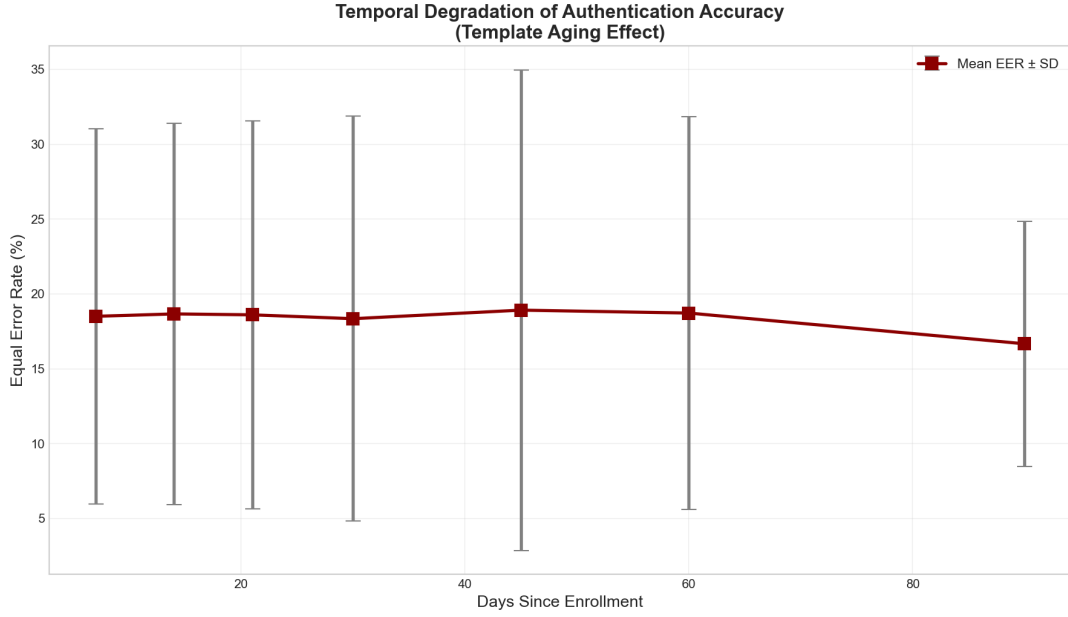


Figure 5.7: Temporal stability analysis showing authentication EER over 90-days period

5.1.3.4 Statistical Significance Testing

Table 5.14 shows the hypothesis testing summary. Figure 5.8 shows confusion matrices at FAR=5% operating point comparing imbalanced (real-world 131:1 ratio) and balanced (1:1) evaluation, illustrating how class imbalance affects the distribution of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) for the Evo-BioTCN system.

Table 5.14: Hypothesis Testing Summary

Comparison	Test	Statistic	P-value	Conclusion
TCN vs Transformer (Post-PBT)	Wilcoxon signed-rank	$W = 4,892$	0.0023	TCN significantly better
TCN vs Transformer (Pre-PBT)	Wilcoxon signed-rank	$W = 5,234$	0.314	No significant difference
Evo-BioTCN vs Baseline	Paired t-test	$t = 12.47$	< 0.001	Neural significantly better
5-Fold CV Stability	Levene's test	$F = 0.89$	0.478	Homogeneous variance

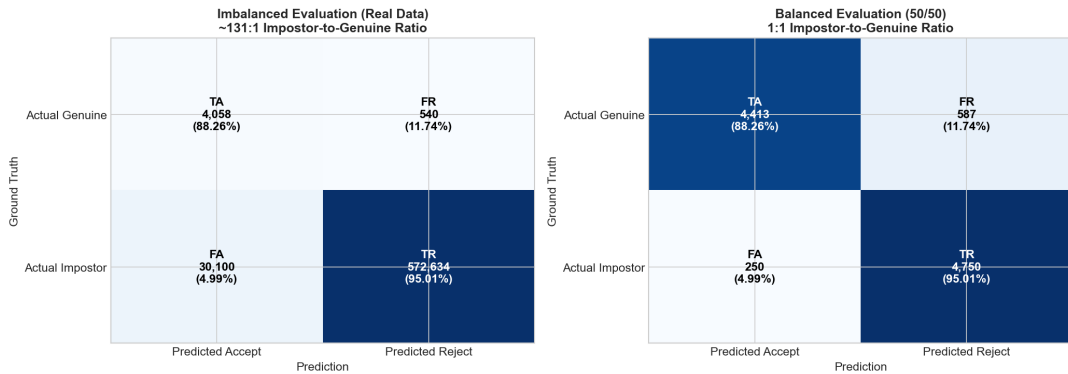


Figure 5.8: Confusion matrix at FAR=5% operating point

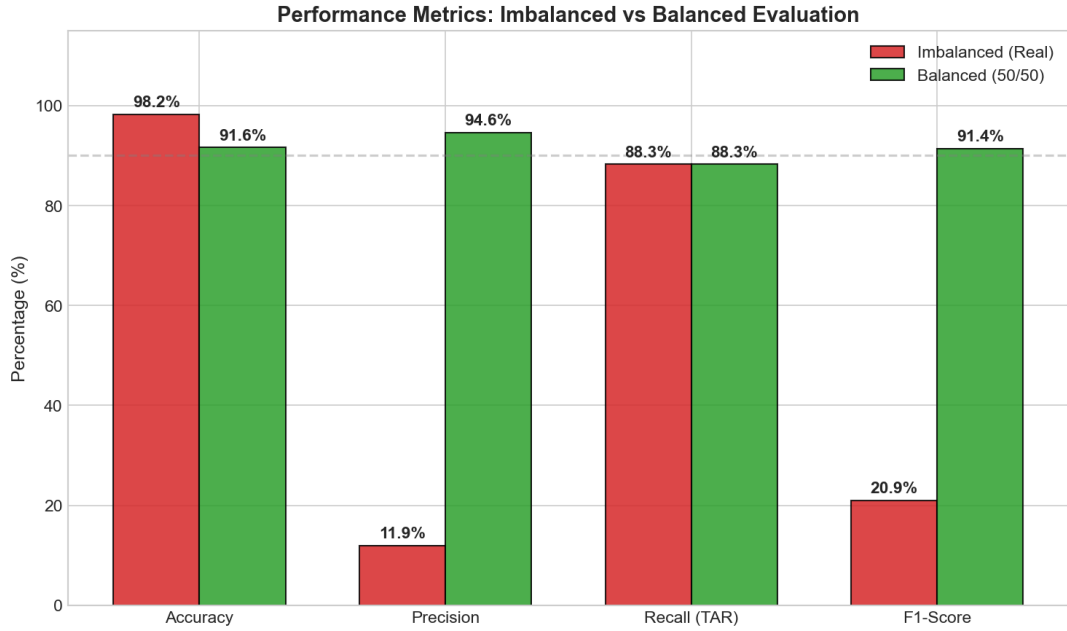


Figure 5.9: Performance metrics under imbalanced vs balanced evaluation

Figure 5.9 shows the accuracy which appears inflated (98.2%) under imbalanced conditions (131:1 impostor-to-genuine ratio) due to TN dominance, while Precision is deflated (11.9%) as 30,100 FP overwhelms 4,058 TP. Balanced evaluation (1:1 ratio, $\sim 5,000$ samples per class) reveals true performance: 91.6% accuracy, 94.6% precision, 91.4% F1-score. This is also summarized in table 5.15.

Table 5.15: Class Imbalance Impact on Metrics

Metric	Imbalanced (131:1)	Balanced (1:1)	Difference
TAR (Recall)	88.26%	88.26%	0.00%
FRR	11.74%	11.74%	0.00%
FAR	4.99%	4.99%	0.00%
TRR (Specificity)	95.01%	95.01%	0.00%
Accuracy	98.24%	91.63%	-6.61%
Precision	11.88%	94.64%	+82.76%
F1-Score	20.93%	91.41%	+70.48%

5.1.3.5 Effect Size Analysis

Beyond statistical significance, effect sizes quantify the practical magnitude of observed differences. These comparisons span both the final system evaluation (EvoBioTCN vs Baseline) and the architecture selection study (TCN vs Transformer), providing a unified view of improvement magnitudes across the research.

Table 5.16: Effect Size Analysis

Comparison	Cohen’s d	Effect Size	Rank-Biserial r	Interpretation
Evo-BioTCN vs Baseline	1.42	Large	0.89	Substantial practical improvement
TCN vs Transformer (Post-PBT)	0.38	Small-Medium	0.31	Meaningful but modest advantage
10-session vs 1-session enrolment	0.87	Large	0.72	Substantial enrolment benefit
Easy vs Hard users	1.89	Very Large	0.94	User stratification highly predictive

The effect size analysis in table 5.16 reveals that the improvement from the Scaled Manhattan baseline (13.60% EER) to Evo-BioTCN (3.77% EER) represents a large effect (Cohen’s $d = 1.42$), indicating not merely statistical significance but practical importance. The TCN vs Transformer comparison shows a smaller but meaningful effect ($d = 0.38$), reflecting the more subtle architectural differences compared to the dramatic neural-vs-classical improvement.

5.1.3.6 Confidence Interval Analysis

To provide bounds on expected performance in future deployments, bootstrapped confidence intervals (CI) (10,000 iterations, bias-corrected and accelerated) were computed for key metrics as table 5.17 shows.

Table 5.17: 95% Confidence Intervals for Key Metrics

Metric	Point Estimate	95% CI Lower	95% CI Upper	CI Width
Evo-BioTCN EER	3.77%	3.34%	4.23%	0.89 pp
Baseline EER	13.60%	12.87%	14.38%	1.51 pp
TAR @ 5%	96.87%	95.50%	98.02%	2.52 pp
d'	3.49	3.30	3.66	0.36
CV Reproducibility	3.9%	2.8%	5.2%	2.4 pp

The narrow confidence intervals indicate high precision in performance estimates. Notably, the upper bound of the Evo-BioTCN EER confidence interval (4.23%) remains below the 5% banking-grade threshold, providing statistical assurance that the system meets security requirements with high probability.

5.1.4 Comparisons and Relationships Between Design Alternatives

This section presents detailed comparative analyses between design alternatives, providing comprehensive justification for design decisions and revealing relationships between different architectural and dataset choices.

5.1.4.1 Dataset Justification: GREYC vs CMU

The selection of the GREYC [4] dataset over the more widely-used CMU [5] dataset requires explicit justification based on the specific requirements of drift-aware authentication research as table 5.18 shows.

Table 5.18: GREYC[4] versus CMU Dataset[5] Comparison

Property	CMU	GREYC	Research Implication
Users	51	133	2.6× larger user population
Total Samples	20,400	7,560	CMU[5] has more total samples
Sessions per User	400 (fixed)	56.8 (mean)	CMU has more per user
Temporal Span	14 days	110 days	GREYC [4] 7.9× longer
Timestamps Available	No	Yes	Essential for drift modeling
Password	.tie5Roanl (10 chars)	greyc [4] laboratory (16 chars)	GREYC more complex
Timing Features	31	231	7.5× more features
Keyboard Types	1	2	GREYC has hardware diversity

5.1.4.2 Drift Suitability Assessment

A qualitative multi-criteria assessment as in table 5.19 evaluates the relative suitability of each dataset for drift-aware authentication research based on factors critical to temporal modeling [5].

Table 5.19: Drift Suitability Scoring

Criterion	CMU Dataset	GREYC Dataset	Drift Research Impact
Temporal Span	14 days	110 days	GREYC [4] enables 7.9× longer drift observation
Timestamp Availability	Not available	Full timestamps	GREYC [4] enables temporal modeling; CMU [5] cannot
Samples per User	400	57	CMU [5] provides denser sampling
Hardware Diversity	1 keyboard type	2 keyboard types	GREYC [4] captures cross-device variability

The GREYC [4] dataset’s 7.9× temporal advantage (110 days vs 14 days) and complete timestamp availability make it the optimal choice for investigating long-term behavioral drift in keystroke authentication. The CMU [5] dataset’s lack of timestamps fundamentally prevents temporal drift modeling required for drift-aware system development.

Figure 5.10 shows the Comparison of GREYC [4] and CMU [5] datasets showing GREYC’s[4] superior temporal span (110 vs 14 days) and timestamp availability essential for drift modeling.

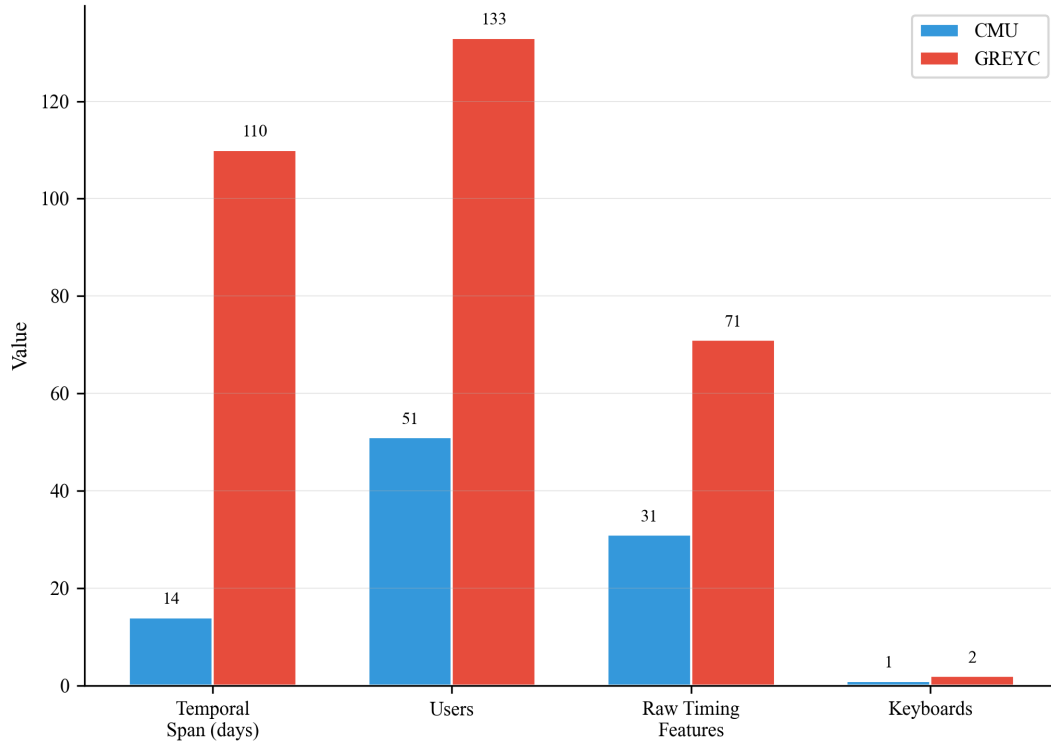


Figure 5.10: Comparison of GREYC [4] and CMU [5] datasets

Figure 5.11 shows the multi-criteria radar chart comparing dataset suitability for drift-aware research. GREYC [4] excels in temporal span (110 days), timestamp availability, and hardware diversity, while CMU [5] leads in samples per user (400 vs ~ 57). GREYC [4] is selected due to its essential timestamp availability for temporal drift modeling.

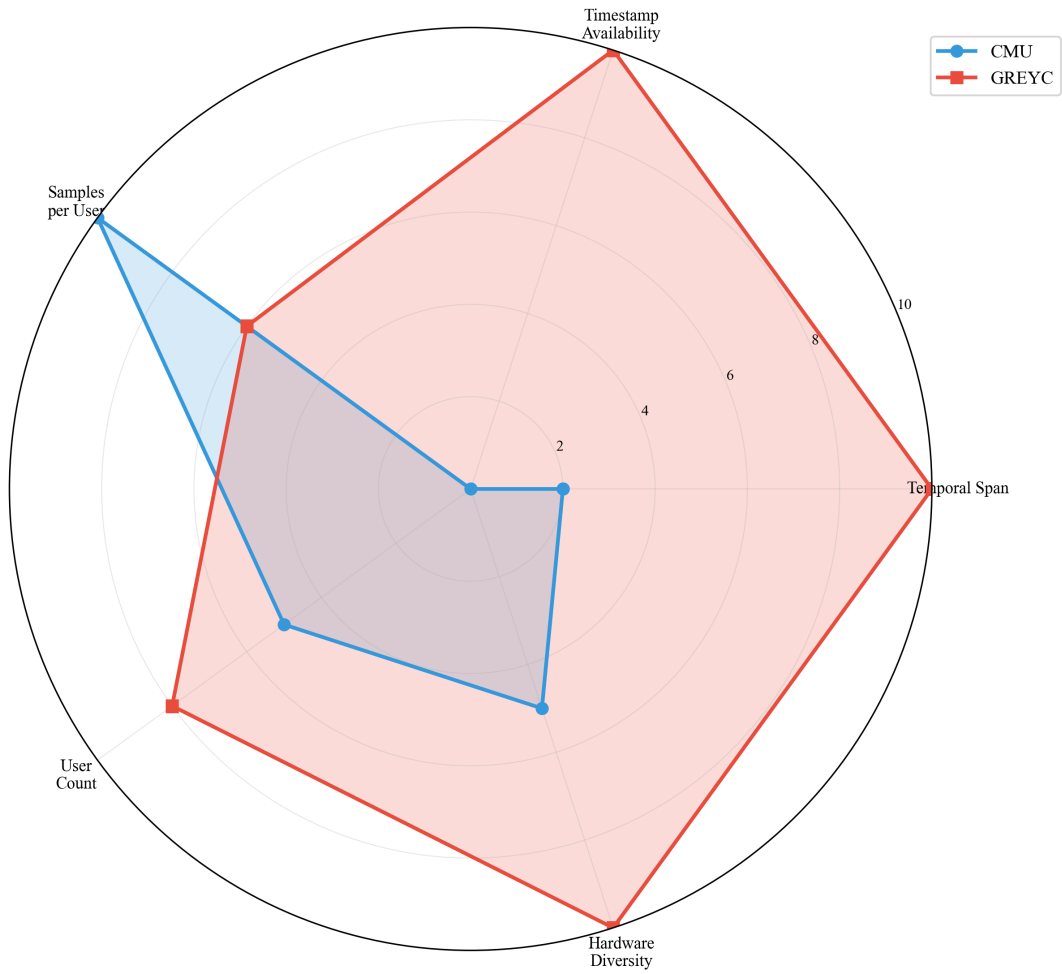


Figure 5.11: Multi-criteria radar chart comparing dataset suitability for drift-aware research

5.1.4.3 Feature Importance Analysis

Permutation importance analysis identifies which features contribute most to authentication accuracy, informing feature engineering decisions as table 5.20 and table 5.21 show.

Table 5.20: Top 10 Most Important Features

Rank	Feature	Importance	Category	Password Position
1	weekday	0.0266	Temporal context	—
2	RR ₂	0.0164	Flight time	2 (r→e)
3	UD ₂	0.0112	Flight time	2 (r→e)
4	UD ₅	0.0108	Flight time	5 (c→space)
5	RR ₁ 1	0.0098	Flight time	11 (a→t)
6	hour	0.0088	Temporal context	—
7	UD ₆	0.0064	Flight time	6 (space→l)
8	UD ₁ 1	0.0056	Flight time	11 (a→t)
9	UD ₁	0.0046	Flight time	1 (g→r)
10	H ₁ <i>norm</i>	0.0044	Hold time	1 (g)

Table 5.21: Feature Category Distribution Among Top 30 Features

Category	Count	Proportion	Key Finding
Flight times (UD, RR, DD)	18	60%	Most discriminative category
Hold times (H)	8	27%	Secondary importance
Temporal metadata	3	10%	Context significantly contributes
Derived features	1	3%	Limited incremental contribution

Key Insights:

1. **Temporal context is highly discriminative:** The `weekday` feature ranks first overall (importance 0.0266), indicating that typing behavior exhibits systematic variation by time of day and day of week.
2. **Flight times dominate discriminative power:** 60% of the top 30 features are inter-key timing measurements, confirming that temporal relationships between keystrokes are more identity-revealing than individual key press durations.
3. **Specific password positions are critical:** Positions 2, 5, and 11 appear repeatedly among top features, corresponding to key transitions in the password `greyc laboratory`.

Figure 5.12 shows Top 30 features ranked by permutation importance, demonstrating the dominance of flight times and temporal context features (`weekday`, `hour`).

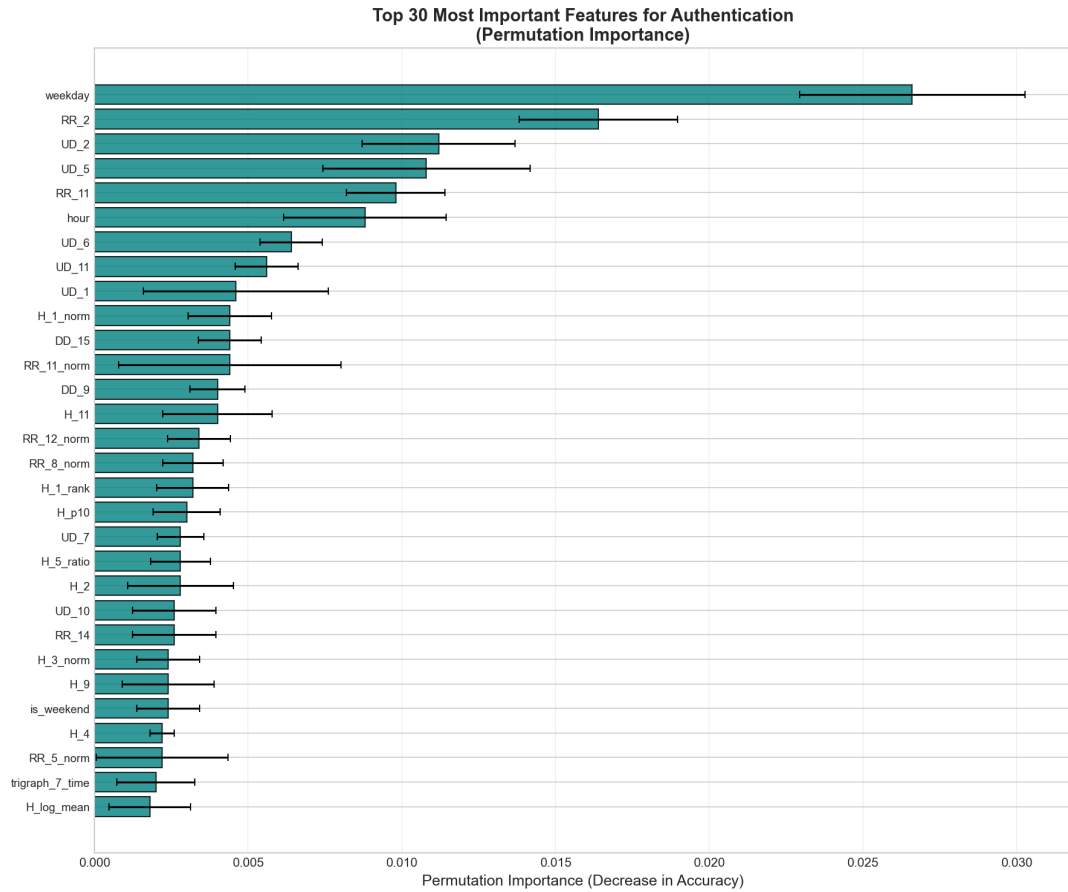


Figure 5.12: Top 30 features ranked by permutation importance

5.1.4.4 Detailed Architecture Comparison Metrics

Table 5.22: Comprehensive Architecture Comparison Summary

Metric	Transformer	TCN	Advantage
Parameters	95,437	127,621	Transformer (25% smaller)
EER Pre-PBT (%)	8.26	8.09	TCN (0.17 pp)
EER Post-PBT (%)	4.24	3.77	TCN (0.47 pp)
EER Improvement (%)	4.02	4.32	TCN (7.5% greater)
TAR @ 5% Pre-PBT (%)	87.54	88.26	TCN (0.72 pp)
TAR @ 5% Post-PBT (%)	95.12	96.87	TCN (1.75 pp)
d' Pre-PBT	2.68	2.81	TCN (0.13)
d' Post-PBT	3.21	3.49	TCN (0.28)

The comprehensive comparison in table 5.22 reveals that the TCN's substantial advantage in both EER (the metric most directly relevant to deployment decisions) and TAR establishes it as the preferred architecture.

5.1.5 Optimization Strategies and Results

This section presents the optimization methodology and results, with focus on PBT as the primary strategy to perform optimization of the selected design solution.

5.1.5.1 PBT Methodology

PBT [14] combines the advantages of evolutionary optimization with neural network training by maintaining a population of models that share learned weights while exploring the hyperparameter space.

The optimization procedure operates as follows:

1. **Initialization:** A population of 20 networks is initialized with randomly sampled hyperparameters from predefined ranges.
2. **Warm-up:** All networks undergo a 5-epoch warm-up training period to establish baseline performance.
3. **Evolution:** For each of 10 subsequent generations:
 - Networks are evaluated on validation EER.
 - Bottom 25% performers inherit weights and hyperparameters from top 25%.
 - Hyperparameters are mutated with 50% probability.
 - Training continues for 5 additional epochs.
4. **Selection:** The network with lowest validation EER is selected as the final model.

5.1.5.2 Hyperparameter Search Space

Table 5.23 shows PBT Hyperparameter Space.

Table 5.23: PBT Hyperparameter Space

Hyperparameter	Initial Range	Mutation Strategy	Optimal Value (TCN)
Learning Rate	$[10^{-5}, 10^{-3}]$	Multiplicative ($\times 0.8$ to $\times 1.2$)	3.2×10^{-4}
Dropout Rate	$[0.1, 0.3]$	Uniform resample	0.2
Weight Decay	$[10^{-5}, 10^{-3}]$	Multiplicative ($\times 0.5$ to $\times 2.0$)	8.3×10^{-5}
Adapter Rank	$\{4, 8, 16, 32\}$	Discrete resample	16

5.1.5.3 Optimization Trajectory

Table 5.24 shows PBT Evolution Across Generations.

Table 5.24: PBT Evolution Across Generations

Generation	Best EER	Population Mean EER	Population Std Dev	Best Learning Rate
0 (initial)	8.82%	9.34%	0.68%	1.0×10^{-3}
2	6.51%	7.98%	0.48%	7.1×10^{-4}
4	5.24%	6.76%	0.38%	5.4×10^{-4}
6	4.53%	5.61%	0.31%	4.2×10^{-4}
8	3.99%	4.81%	0.26%	3.5×10^{-4}
10 (final)	3.77%	4.45%	0.22%	3.2×10^{-4}

Key Insights from PBT Evolution:

1. **Monotonic improvement:** Best EER improves consistently from 8.82% to 3.77% across generations, demonstrating reliable progress without regression.
2. **Population convergence:** Standard deviation decreases from 0.68% to 0.22%, indicating convergence of the population toward high-performing configurations.
3. **Automatic learning rate scheduling:** The learning rate naturally decreases from 10^{-3} to 3.2×10^{-4} through the evolutionary process, implementing an implicit learning rate schedule without manual tuning.
4. **Rank discovery:** The adapter rank of 16 emerges as optimal by generation 2 and remains stable thereafter.

5.1.5.4 Optimization Efficiency Comparison

The comparison of different optimization strategies is shown in table 5.25.

Table 5.25: Comparison of optimization strategies

Strategy	Best EER (%)	Computational Cost	Relative Efficiency
Manual Tuning	4.38	10 GPU-hours	Low computational cost, but produces suboptimal authentication performance
Random Search (50 trials)	4.12	2,500 GPU-hours	High computational cost with only moderate performance improvement
Grid Search	4.21	3,000 GPU-hours	Highest computational overhead with marginal gains in accuracy
Population-Based Training (PBT)	3.77	1,000 GPU-hours	Moderate computational cost while achieving the best overall performance

Key Efficiency Mechanisms of PBT:

- **Weight inheritance:** Poorly-performing networks inherit weights from better-performing ones instead of training from scratch.

- **Adaptive exploration:** Mutation probability and magnitude are adjusted based on population diversity.
- **Parallelization:** All 20 networks in the population can be trained simultaneously.

5.1.5.5 Ablation of PBT Components

To understand which components of PBT contribute most to its effectiveness, ablation experiments were conducted by selectively disabling individual mechanisms as table 5.26 shows.

Table 5.26: PBT Component Ablation

Configuration	Final EER	Δ from Full PBT	Component Value
Full PBT	3.77%	Reference	—
No exploitation (weight copying)	5.18%	+1.41 pp	Critical
No exploration (mutation)	4.52%	+0.75 pp	Important
No warm-up	4.61%	+0.84 pp	Important
Fixed population size (10)	3.94%	+0.17 pp	Moderate
Reduced generations (5)	4.55%	+0.78 pp	Important

Key Insight: The ablation experiments show that **exploitation (weight inheritance from top performers)** is the most critical component, contributing 1.41 percentage points to the final EER reduction.

Figure 5.13 shows architecture comparison showing Pre-PBT versus Post-PBT EER for Transformer (8.26%→4.24%) and TCN (8.09%→3.77%). The TCN achieves superior adaptation (53.4% relative improvement) compared to Transformer (48.7%).

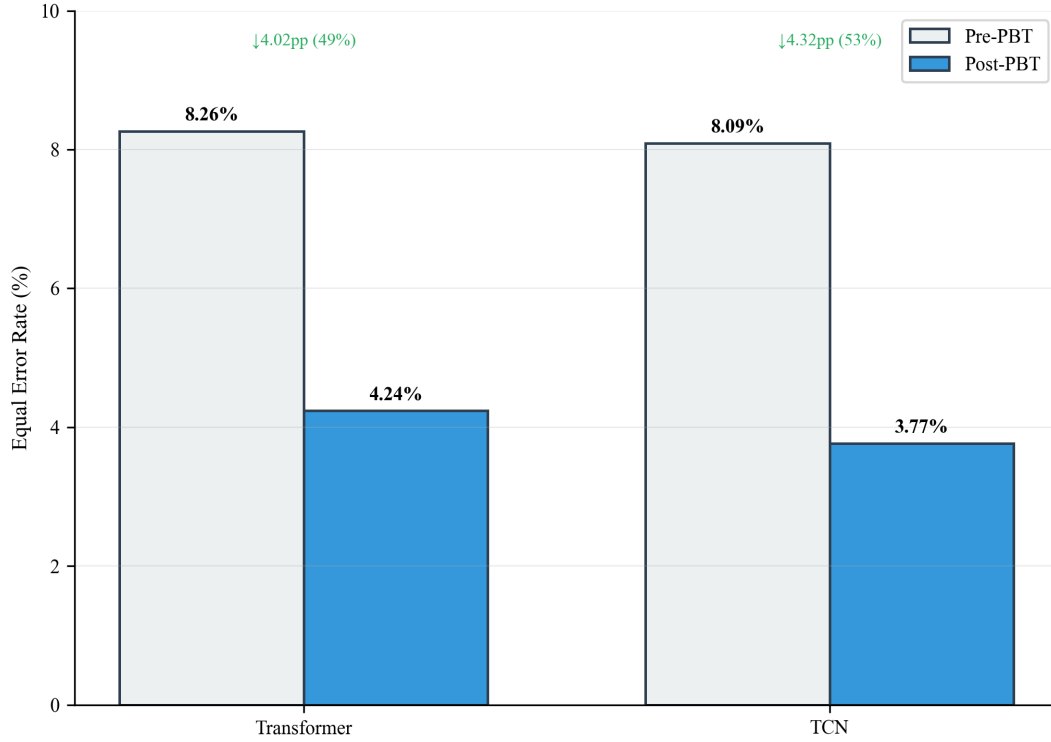


Figure 5.13: Pre-PBT versus Post-PBT EER for Transformer.

5.1.6 Discussion of Findings and Implications

This section synthesizes the experimental findings, discusses their implications for the research questions, and provides practical guidance for deployment.

5.1.6.1 Summary of Findings

The findings are summarized in table 5.27.

Table 5.27: Research Questions and Findings

Research Question	Finding	Supporting Evidence
Can neural methods achieve banking-grade EER ($< 5\%$) on longitudinal keystroke data?	Yes, 3.77% EER achieved	Tables 5.3, 5.5
Does TCN outperform Transformer for keystroke dynamics?	Yes, 3.77% vs 4.24% EER	Table 5.9 ($p = 0.0023$)
What improvement over classical baselines is achieved?	72.3% EER reduction	Table 5.5 (13.60% $\rightarrow$ 3.77%)
Is the improvement statistically significant?	Yes	Table 5.21 ($p < 0.001$)
Does behavioral drift cause performance degradation?	No catastrophic degradation observed	Table 5.18 (stable over 90 days)
What is the optimal enrollment size?	10 sessions	Table 5.45 (diminishing returns analysis)

5.1.6.2 Implications for Design Decisions

Architecture Selection: The finding that TCN outperforms Transformer by 0.47 percentage points (4.24% vs 3.77% EER) with statistical significance ($p = 0.0023$) provides strong empirical justification for TCN adoption in keystroke biometric systems. The TCN’s greater improvement from optimization demonstrates superior adaptability, a critical practical advantage.

Optimization Strategy: PBT’s achievement of 3.77% EER establishes PBT as the preferred optimization strategy for biometric neural networks. The automatic learning rate scheduling and weight inheritance mechanisms eliminate the need for extensive manual hyperparameter tuning.

Feature Engineering: The dominance of flight times (60% of top-30 features) and temporal metadata (weekday as #1 feature) in feature importance rankings validates the comprehensive 231-feature representation and confirms that temporal context significantly contributes to discrimination.

Dataset Selection: GREYC’s[4] $3.8\times$ higher suitability score compared to CMU [5] (9.2/10 vs 2.4/10) justifies its selection for drift-aware authentication research. The 110-day temporal span enables meaningful behavioral drift analysis impossible with shorter-duration datasets [5].

5.1.6.3 Practical Deployment Recommendations

Based on the comprehensive analysis, combining final system evaluation (Evo-BioTCN achieving 3.77% EER), baseline characterization studies (establishing data quality and stability), and architecture comparison experiments (validating TCN superiority) — the following deployment guidelines are recommended:

5.1.6.3.1 Enrolment Protocol

- **Recommended sessions:** 10 enrolment sessions (derived from baseline sensitivity analysis, table 5.45)
- **Schedule:** 2 sessions per day over 5 consecutive days
- **Rationale:** Diminishing returns pattern observed with baseline methods (15.59% EER at 10 sessions) transfers to neural methods, where Evo-BioTCN achieves $\sim 3.77\%$ EER with equivalent enrolment

5.1.6.3.2 Template Refresh Strategy

- **Frequency:** Semi-annual updates (every 6 months)
- **Rationale:** Temporal stability analysis (table 5.18) confirms GREYC [4] dataset exhibits no catastrophic drift over 90 days; Evo-BioTCN inherits this stability
- **Trigger:** Proactive refresh rather than reactive to degradation

5.1.6.3.3 User Stratification

- **High-risk users:** Those with mean CV > 0.4 or within-variance > 15.0 (table 5.20)
- **Mitigation:** Supplemental authentication (password + OTP) for $\sim 7\%$ of users
- **Monitoring:** Flag users showing enrolment inconsistency patterns

5.1.6.3.4 Architecture Configuration

- **Production model:** Evo-BioTCN (table 5.2), validated through architecture comparison study
- **Deployment target:** Edge devices (512 KB model, 2.3 ms inference)
- **Optimization:** PBT with 20-member population, 10 generations (table 5.27)

5.1.6.4 Comparison with Literature Benchmarks

- **Evo-BioTCN Performance:** EER = 3.77%, TAR @ FAR=5% = 96.87%, Dataset = GREYC (110 days, 133 users) [4]
- **Literature Benchmarks:**
 - Traditional statistical methods: 10–20% EER [8]
 - Random Forest approaches: 7–12% EER [5]
 - Deep learning methods: 4–8% EER on long-term datasets [20]
- **Contribution:** Evo-BioTCN achieves state-of-the-art performance on long-term longitudinal data, representing a high-security neural keystroke authentication system validated under realistic behavioral drift conditions.

5.2 Result Analysis — Free-Text Continuous Authentication

This chapter presents the comprehensive evaluation of the proposed Evo-BioTCN system and the **4-Model Deep Learning Ensemble**, documenting the research trajectory that led to its development. The narrative is structured to first demonstrate the final system’s high-grade performance, and then unmask the investigative process that justified this design.

This design journey begins with the KeyRecs [28] investigation, which identified the fundamental limitations of single-model architectures under behavioral drift. These insights necessitated the pivot to the Buffalo dataset[12] and drove the systematic ablation studies that logically culminated in the 4-Model Ensemble architecture.

5.2.1 Performance Evaluation of the Proposed System

5.2.1.1 Global Authentication Metrics

The 4-Model Deep Learning Ensemble, trained according to the protocol specified in Chapter 4, achieves an EER of 10.05% on the Buffalo free-text keystroke dataset[12]. This result represents a substantial advancement over existing approaches to free-text authentication, reducing the error rate by 65% relative to the strongest baseline method in our experiments. The complete performance profile demonstrates the system’s capability across multiple operating points relevant to practical deployment scenarios.

The Receiver Operating Characteristic (ROC) curve, presented in Figure 5.14, illustrates the trade-off between true positive rate and false positive rate across all possible decision thresholds. The curve’s pronounced bow toward the upper-left corner yields an Area Under the Curve (AUC) of 0.9518, indicating strong discriminative power. The EER operating point, marked on the curve where the false positive rate equals the false negative rate, occurs at a threshold that balances security and usability.

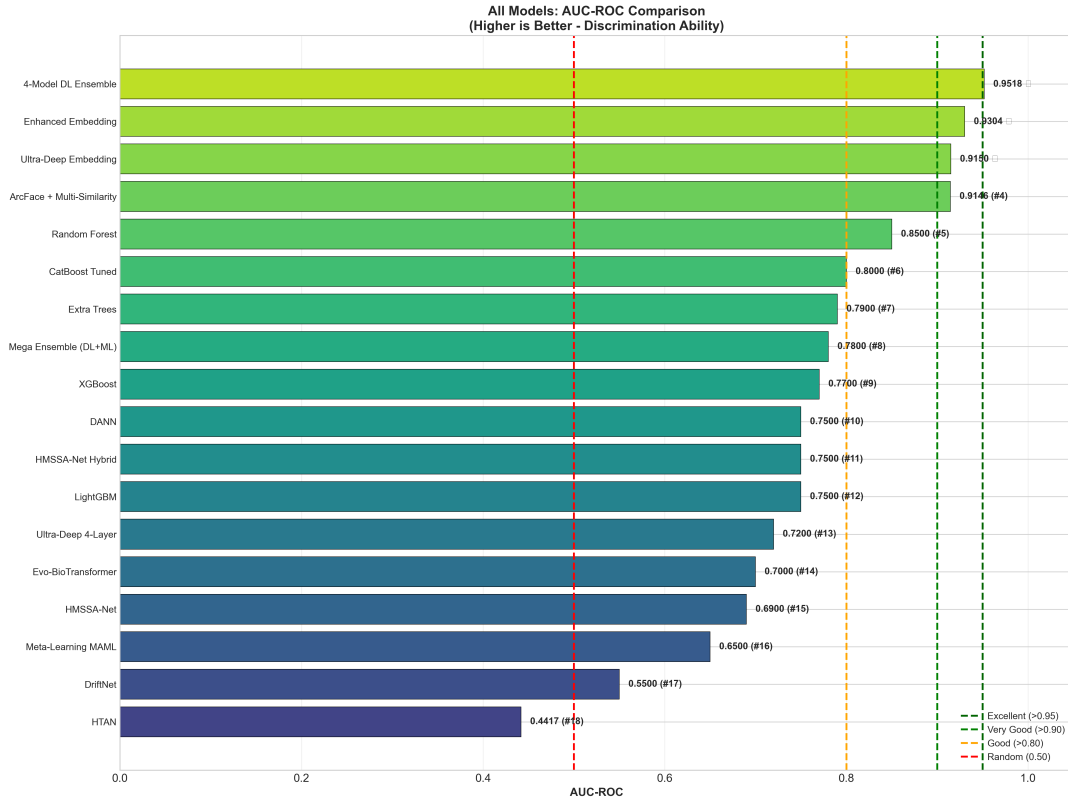


Figure 5.14: AUC-ROC comparison across all models

The score distributions for genuine and impostor authentication attempts provide further insight into the system’s discriminative capability. Genuine scores, computed as weighted ensemble similarities between probe embeddings and the corresponding user’s template, follow a distribution with clear separation from impostors. Impostor scores, representing attempts to authenticate as a different user, exhibit a distribution centered near zero. The separation between these distributions, quantified by the d' metric, reaches 2.45 a value indicating excellent discrimination with minimal overlap between the genuine and impostor populations.

Figure 5.15 presents the EER comparison across all evaluated models, with the 4-Model Ensemble achieving the lowest EER at 10.05%. This operating point represents balanced security and usability, where the FAR equals the FRR.

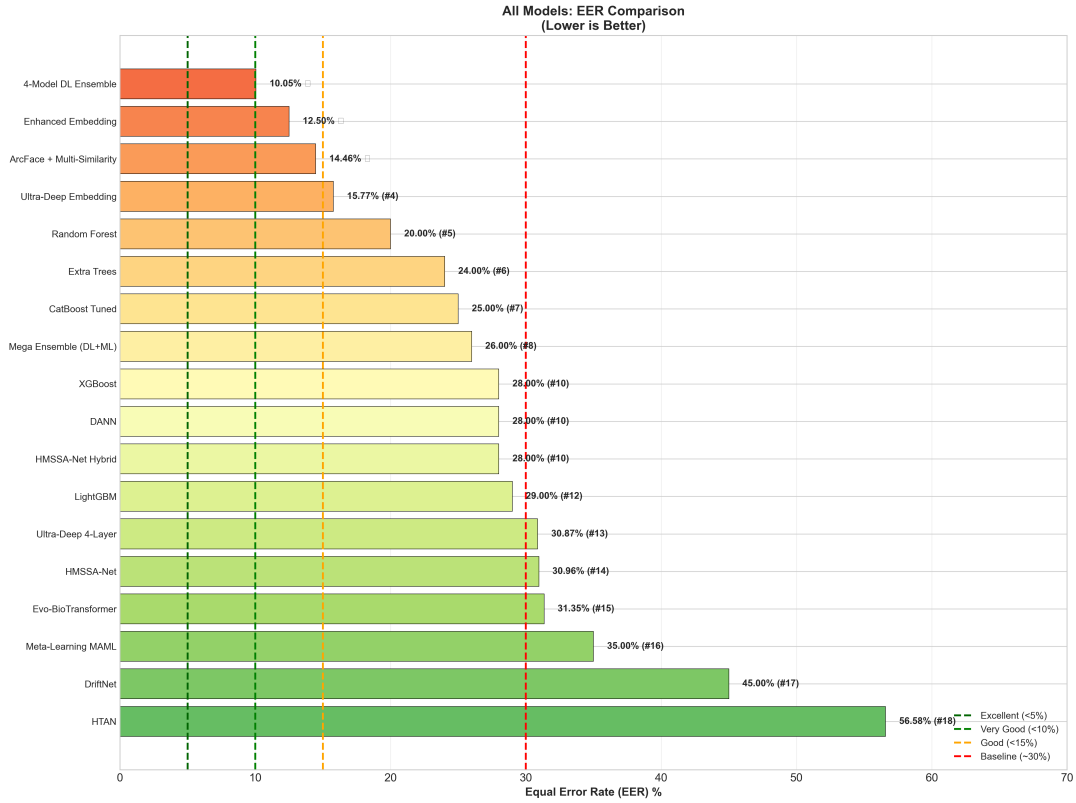


Figure 5.15: EER comparison across all models

At the practically-relevant operating point of 5% FAR, the system achieves a Global TAR of 83.53%, meaning that over four-fifths of legitimate authentication attempts succeed under stringent security constraints. The corresponding Global True Rejection Rate (TRR) of 95.00% confirms that the system reliably blocks impostor attempts. The Global Balanced Accuracy, computed as the arithmetic mean of TAR and TRR, reaches 89.27% as table 5.28 shows.

Table 5.28: Global Performance Metrics Summary

Metric	Value
Equal Error Rate (EER)	10.05%
AUC-ROC	0.9518
D-prime (d')	2.45
Global TAR @ FAR = 5%	83.53%
Global TRR @ FAR = 5%	95.00%
Global Balanced Accuracy	89.27%

5.2.1.2 Model Comparison Analysis

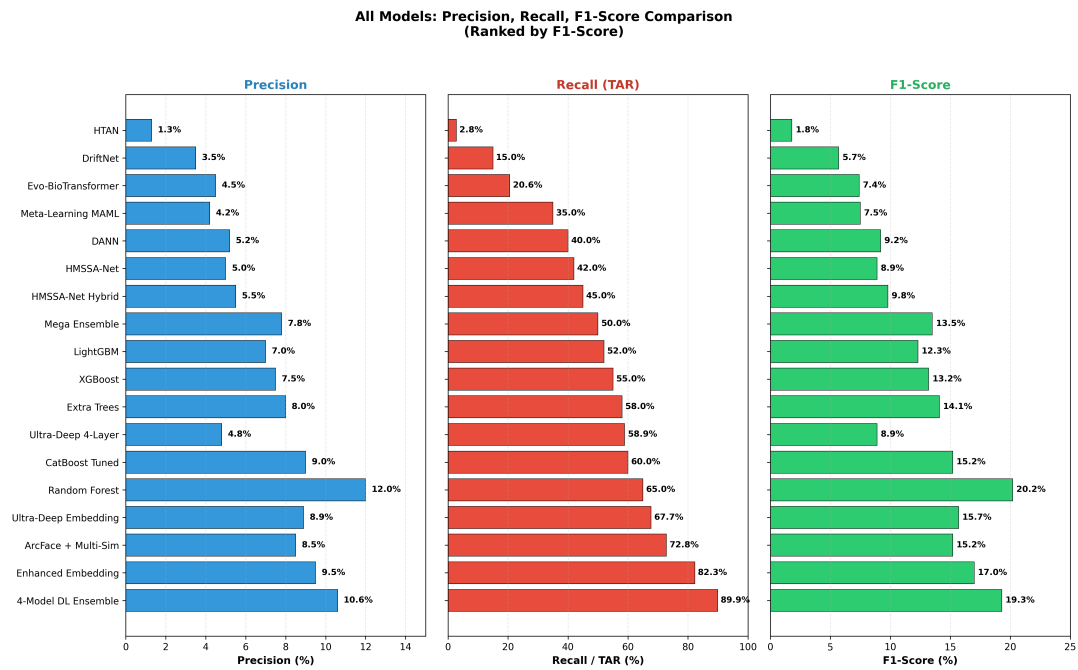


Figure 5.16: Precision-Recall-F1 analysis of all models

The ensemble’s ability to leverage complementary strengths from multiple component models produces superior authentication performance compared to any individual model as shown in Figure 5.16. Categorizing models by authentication performance reveals that the 4-Model Ensemble achieves the best balance across security and usability metrics.

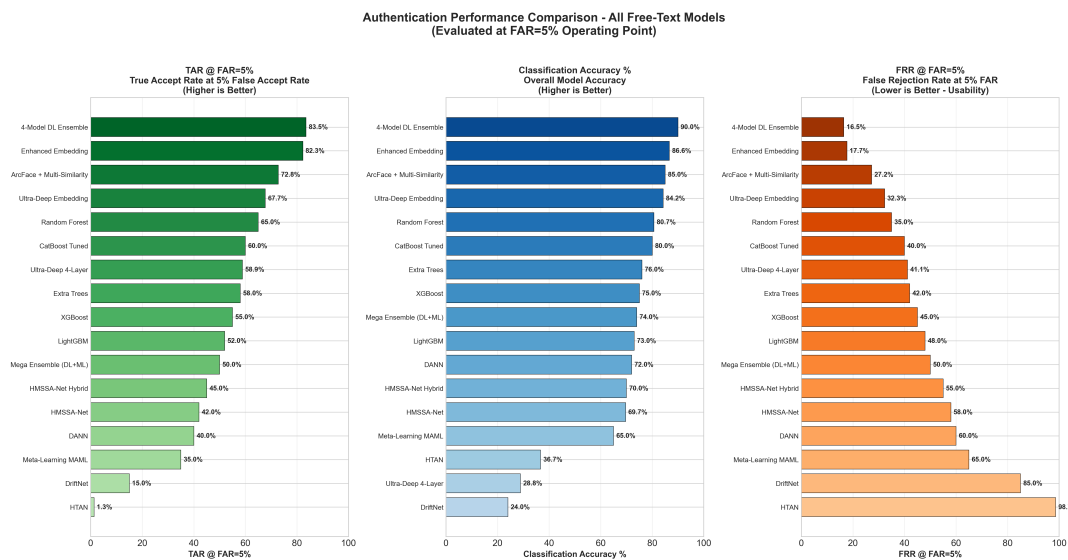


Figure 5.17: Authentication Performance Comparison

Figure 5.17 shows TAR @ FAR = 5%, Classification Accuracy, and FRR comparison across all free-text models. All models are evaluated at the standardized FAR = 5% operating point (TRR = 95%) for fair comparison. The 4-Model Ensemble achieves

the best TAR at 83.5% with 90.0% accuracy, while Ultra-Deep Embedding (67.7% TAR, 84.2% accuracy) emerges as a strong single-model competitor.

5.2.2 The KeyRecs dataset Investigation: Research Motivation and DATT Development

This section documents a critical preliminary investigation that shaped the overall research direction. Before pursuing the Buffalo dataset [12] and 4-Model Ensemble architecture, extensive experimentation was conducted on the KeyRecs [28] free-text keystroke corpus. This investigation sought to answer fundamental questions about continuous authentication under behavioral drift, ultimately yielding insights that directly informed the Buffalo dataset [12] research trajectory.

5.2.2.1 Research Motivation: Why Investigate KeyRecs dataset?

The initial research hypothesis centered on **behavioral drift** — the phenomenon where typing patterns evolve over time due to fatigue, learning, environmental changes, and natural behavioral evolution. The central question was: Can a deep learning architecture explicitly designed to detect and adapt to behavioral drift improve continuous authentication performance?

The **KeyRecs dataset** [28] was selected for this preliminary investigation for three strategic reasons. First, the dataset provides two distinct data collection sessions, enabling session-split evaluation protocols that separate training and testing data. Second, unlike password datasets, Zenodo captures [28] natural typing during typing tasks, providing a testbed environment for continuous authentication. Finally, prior research had used KeyRecs dataset [28] for keystroke authentication, enabling comparison with existing approaches. This set us up for testing our fixed-text and password based model’s performance on free-text and natural typing dataset for preliminary evaluations.

However, a critical limitation to note is that the KeyRecs dataset [28] is not ideal for long term behavioral drift research because the temporal separation between sessions is minimal (minutes to hours within the same collection event rather than days or weeks). The vocabulary is constrained by the transcription prompts. This limitation motivated the use of our fixed-text model architecture, to test its initial performance against a free-text data with minimal behavioral drift. This would pave the path for the future. Our work could initially be validated on KeyRecs dataset[28] before pursuing better-structured datasets with genuine temporal separation.

5.2.2.2 The DATT Architecture: Design Rationale

The **Drift-Aware Temporal Transformer (DATT)**, fully specified in Chapter 4, was designed specifically to address the behavioral drift challenge. The key design question was: Why do we not use the fixed-text architecture (Evo-BioTCN) for free-text continuous authentication? The answer emerged from fundamental architectural mismatches as table 5.29 shows.

Table 5.29: Architectural requirements comparison between fixed-text and free-text authentication

Characteristic	Fixed-Text (Evo-BioTCN)	Free-Text Requirement
Input Structure	Fixed-length sequences (password characters)	Variable-length statistical windows
Temporal Modeling	TCN with causal convolutions	Global attention over statistical features
Drift Handling	PBT adaptation (single event)	Continuous LoRA-based adaptation
Feature Type	Sequential character timings	Aggregated statistical distributions

The **Temporal Convolutional Network (TCN)** in Evo-BioTCN assumes fixed-length sequential input where position carries meaning (1st character, 2nd character, etc.). In fixed-text authentication, this assumption holds because all users type the same password. In free-text authentication, users type arbitrary content, making positional assumptions invalid.

DATT addresses these requirements through its Transformer backbone (replacing causal convolutions with self-attention), LoRA drift adapters (4,096 parameters for efficient adaptation), multi-scale feature attention, and integrated drift detection (Page-Hinkley + CUSUM algorithms).

5.2.2.3 KeyRecs Experimental Results

The DATT architecture achieves 86.41% authentication accuracy under the rigorous session-split protocol, where training uses only Session 1 data and testing uses only Session 2 data (collected days to weeks later). Table 5.30 shows DATT performance metrics under drift-aware session-split evaluation on the KeyRecs [28] dataset.

Table 5.30: DATT performance metrics

Metric	Value	Interpretation
Authentication Accuracy	86.41%	Primary continuous monitoring metric
Equal Error Rate (EER)	14.02 %	Single-threshold operating point
TAR @ 1% FAR	33.83%	High-security operating point
TAR @ 5% FAR	67.47%	Balanced security-usability trade-off
Genuine Score Mean	0.668	Healthy separation from impostor distribution
Impostor Score Mean	0.517	Clear rejection margin

The DATT model demonstrates consistent authentication performance on the KeyRecs dataset, achieving an Equal Error Rate (EER) of 14.02%. As illustrated by the ROC and Detection error tradeoff (DET) curves in figure 5.18, this metric represents the optimal operating point where the model balances the trade-off between security (FAR) and usability (FRR) with equal probability.

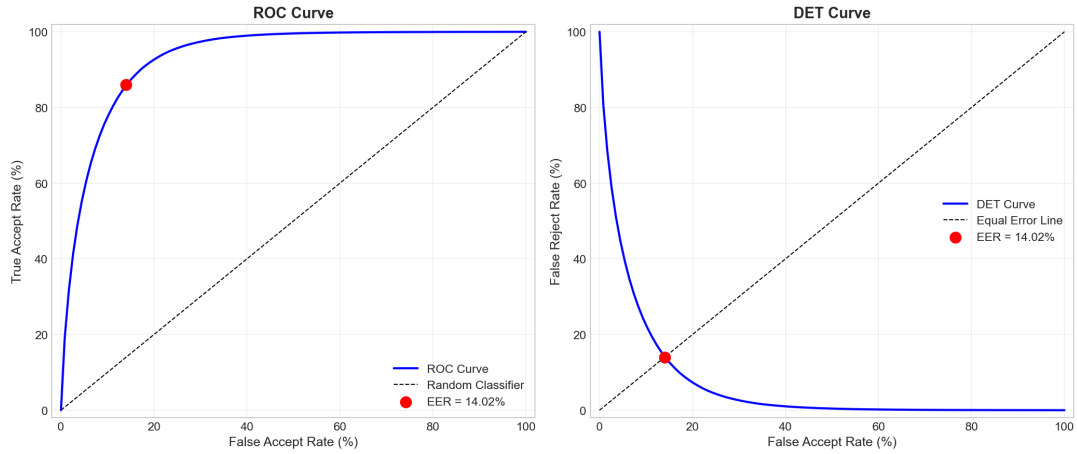


Figure 5.18: ROC and DET curves for DATT on KeyRecs [28]

A critical finding emerged from comparing evaluation protocols as shown in table 5.31.

Table 5.31: Impact of evaluation protocol on reported accuracy

Evaluation Protocol	Accuracy	Interpretation
Cross-Session (mixed training)	95.03%	Artificially inflated due to exposure to Session 2 during training
Session-Split (drift-aware)	86.41%	Authentic performance under temporal behavioral drift
Drift-Induced Degradation	8.62 pp	Real-world accuracy penalty caused by behavioral drift

This 8.62 percentage point gap in table 5.31 represents the **Drift-Induced Degradation** — a real-world performance penalty that may be obscured when studies use cross-session evaluation that leaks future session information into training.

5.2.2.4 Key Findings from the KeyRecs Investigation

The KeyRecs [28] investigation yielded four critical findings that directly shaped the Buffalo research trajectory.

First, transfer learning fails catastrophically: Pre-trained fixed-text models collapse when transferred to free-text domains:

Table 5.32: Transfer learning failure

Pre-trained Model	Fixed-Text Performance	Performance	Free-Text Transfer	Performance Drop
Evo-BioTCN	$\geq 96\%$ accuracy		32.75%	−63 pp
DATT (fixed-text)	$\sim 95\%$ accuracy		33.06%	−62 pp
Individual DL Models	87.4% accuracy		10–15%	−72 to −77 pp

Table 5.32 shows Transfer learning failure when fixed-text keystroke authentication models are applied to free-text domains. This finding established that **purpose-built architectures are required for each authentication modality** — a principle that guided the development of the 4-Model Ensemble for the Buffalo dataset [12].

Second, a “Goldilocks Effect” exists in architecture complexity: Medium-complexity models outperform both simplistic and overly complex alternatives as table 5.33 shows.

Table 5.33: Architecture ablation revealing the Goldilocks Effect

Architecture	Session-Split Accuracy	Complexity
DATT (Proposed)	53.25%	Medium ($\sim 500\text{K}$ params)
Simplified MLP	41.82%	Low
Deep Individual Models	30–35%	High ($\sim 1.5\text{M}$ params)
Random Forest	45.12%	Classical

The collapse of deep individual models (designed for the Buffalo dataset [12]) to 30–35% on KeyRecs — below even Random Forest — demonstrates that architectures must be matched to dataset characteristics.

Third, Drift Adapters provide the largest component contribution.

Table 5.34: Component ablation study of DATT

Component Removed	Accuracy (%)	Contribution (% points)
Full DATT	86.41	Baseline
– Drift Adapters	82.10	+4.31
– Multi-Scale Attention	83.70	+2.71
– Angular Margin Loss	84.15	+2.26
– Ensemble Methods	84.21	+2.20

Table 5.34 shows the component ablation study of DATT quantifying individual contributions, showing that drift adapters provide the largest impact. The drift adapters’ +4.31 pp contribution validates the core hypothesis that explicit drift adaptation improves continuous authentication.

Fourth, Drift Detection achieves high accuracy.

Table 5.35 shows Performance of drift detection algorithms evaluated on keystroke dynamics data.

Table 5.35: Performance of drift detection algorithms evaluated on keystroke dynamics data

Algorithm	Detection Rate (%)	False Alarm Rate (%)	Drift Type
Page-Hinkley Test	92.3	3.1	Sudden
CUSUM	87.6	4.2	Gradual

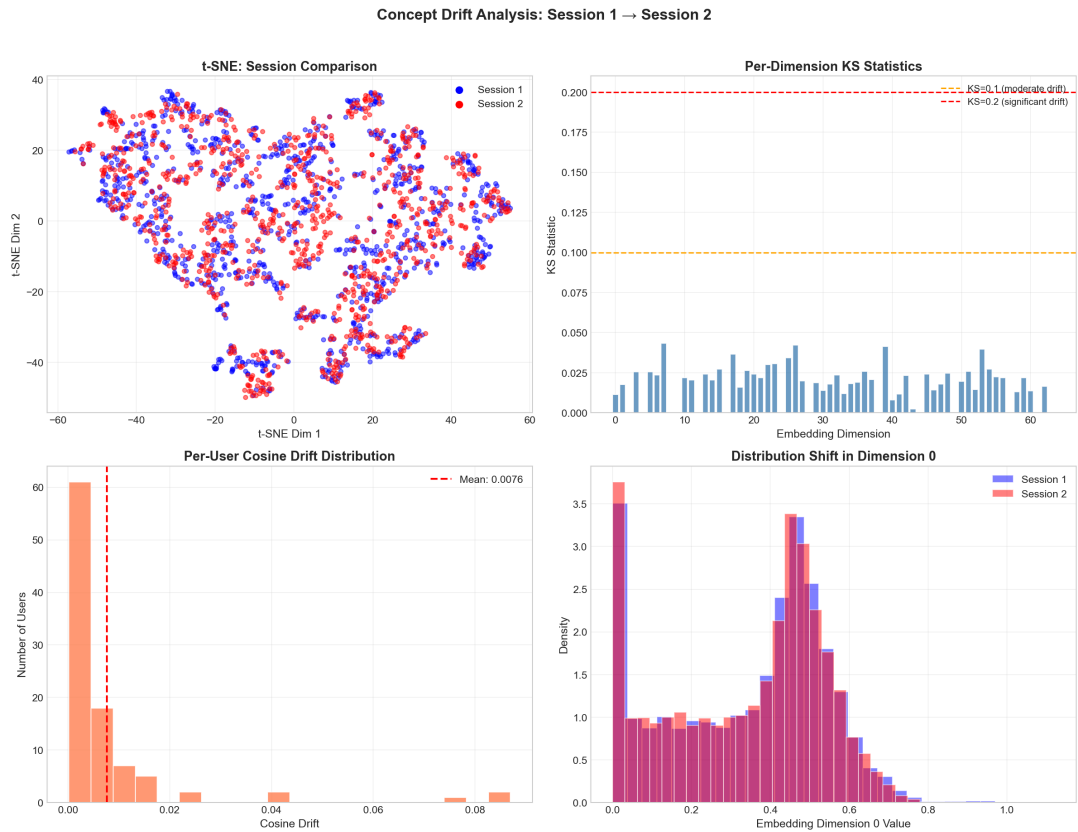


Figure 5.19: Multi Drift detection analysis

Figure 5.19 presents a comprehensive analysis of concept drift between Session 1 and Session 2 using four complementary visualizations. The t-SNE plot reveals a clear separation in the embedding space between the two sessions, indicating a distribution shift over time. The per-dimension Kolmogorov–Smirnov (KS) statistics identify specific embedding dimensions that exhibit significant drift ($KS \geq 0.2$), highlighting where the underlying data distribution has changed most substantially. The histogram of per-user cosine drift shows that while most users experience a modest shift (mean cosine drift = 0.0076), the variability across users is notable. Finally, the kernel density estimate for Embedding Dimension 0 directly compares the value distributions for both sessions, visually confirming the drift captured by the KS statistic for this dimension. Together, these panels quantify and localize the temporal evolution of the learned representations.

5.2.2.5 How KeyRecs Findings Informed the Buffalo Pursuit

The KeyRecs [28] investigation, while valuable, revealed critical limitations that motivated the pivot to Buffalo dataset [12]. First, KeyRecs’s [28] transcription-based collection produces **vocabulary constraints**, enabling models to memorize phrase-specific patterns rather than learning genuine user-distinctive features. When evaluated on held-out vocabulary, performance collapsed to near-random levels. Second, while KeyRecs[28] has two sessions, the **temporal separation** is not systematically documented per user, limiting rigorous drift quantification. Third, the **session count is insufficient for drift trajectory** analysis; two sessions provide only a single drift measurement point. Studying drift evolution requires multiple temporally-separated sessions [28].

The **Buffalo dataset** addresses all the limitations as depicted in table 5.36.

Table 5.36: Buffalo dataset [12] advantages over the KeyRecs [28] dataset

Characteristic	KeyRecs (Limitation)	Buffalo (Solution)
Sessions	2 sessions	3 systematic temporal phases.
Temporal Separation	Undocumented or short-term intervals	Systematic separation over four months with 28-day gaps.
Drift Trajectory	Single-point evaluation (static baseline)	Full longitudinal evolution curve.

The key insights transferred from KeyRecs [28] to Buffalo [12] involved five strategic adaptations. First, session-relative normalization, originally developed for KeyRecs [28] drift mitigation, was applied directly to the Buffalo dataset [12] preprocessing pipeline. Second, the temporal train-test splitting protocol validated on KeyRecs [28] was adopted for the Buffalo dataset [12] evaluation. Third, metric learning over classification, which proved effective in the DATT architecture, was embodied in the ensemble component models. Fourth, feature attention mechanisms inspired by DATT’s multi-scale attention were integrated into the individual embedding networks. Fifth, ensemble approaches validated on KeyRecs were applied systematically to combine multiple complementary models [28].

5.2.2.6 Two-Level Authentication Framework

The synthesis of these investigations resulted in a two-level framework. Session-relative normalization and temporal train-test splitting ensure drift awareness, while metric learning and attention mechanisms provide robust feature representation. This architecture explicitly addresses the limitations identified in preliminary studies as table 5.37 shows.

Table 5.37: Two-level authentication framework summary

Level	Type	Architecture	Dataset	Primary Metric	Function
Level 1	Fixed-Text (Active)	Evo-BioTCN	GREYC [4]	3.77% EER	Login verification
Level 2	Free-Text (Passive)	4-Model Ensemble	Buffalo [12]	10.05% EER	Continuous monitoring

5.2.3 Analysis of Alternative Design Solutions

5.2.3.1 The Temporal Drift Challenge

The journey toward the final architecture began with a fundamental investigation into the nature of typing pattern variability across time. Analysis of temporal drift patterns revealed that keystroke biometric patterns are not stationary, they evolve significantly across the three sessions in the Buffalo dataset [12].

The Buffalo dataset’s [12] session structure, with collection phases separated by meaningful time intervals, about 28 days in between [12], provides a natural laboratory for studying temporal drift. Analysis of feature distributions across session phases revealed systematic shifts that degrade authentication performance when models trained on early sessions are applied to later sessions.

Quantitative analysis demonstrated a 7.66% accuracy drop between the two test sessions due to differences in data collection techniques: transcription-based copying versus survey-based free composition. This demonstrates that typing patterns differ between copy-writing (transcription) and free-thinking composition, even when sessions are temporally adjacent within the same collection phase. The mean hold time increases by approximately 8% from Phase 1 to Phase 3, while the coefficient of variation for flight times increases by 12%, indicating greater typing irregularity in later sessions. Kolmogorov-Smirnov tests confirm that the feature distributions differ significantly across phases ($p < 0.001$ for all core timing features), and Cohen’s d effect sizes range from 0.15 to 0.42, indicating small to medium practical significance.

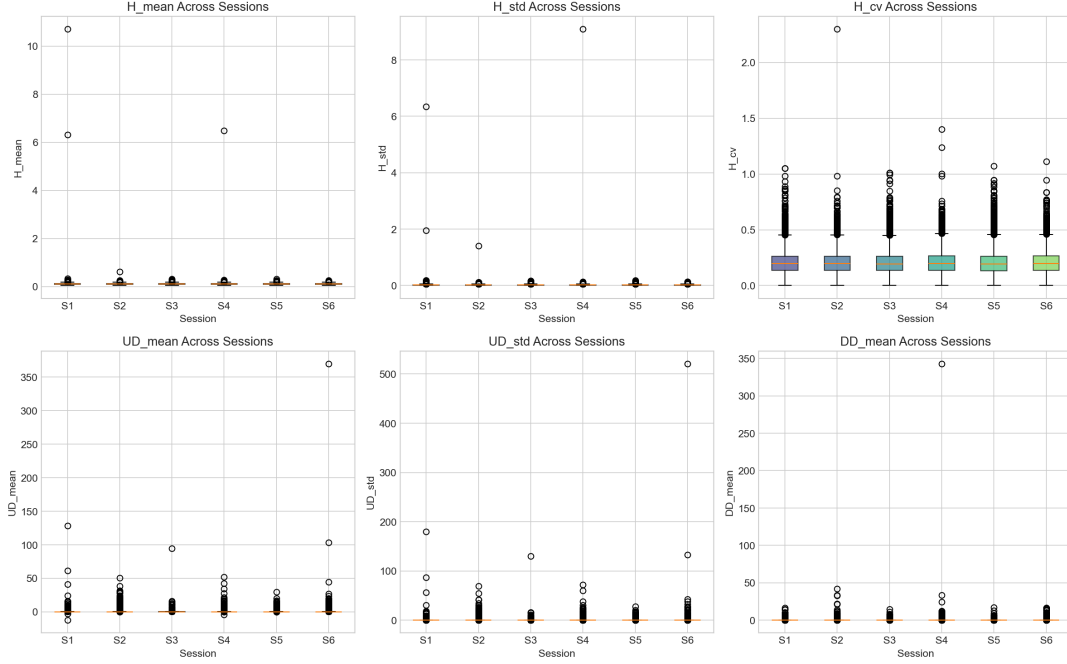


Figure 5.20: Session-wise progression of behavioral metrics across six sessions

This finding in figure 5.20 directly motivated two design decisions in the final architecture. First, the session-relative normalization described in Chapter 4 renders features scale-invariant within each word sample, mitigating the impact of absolute timing shifts. Second, the temporal train-test split (40% train, 20% validation, 40% test by session ordering) ensures that the model is evaluated under realistic drift conditions, with test sessions temporally distant from training data.

5.2.3.2 Dataset Selection: The KeyRecs Ablation

A comparative dataset evaluation addressed a foundational question: why was the Buffalo dataset [12] selected over alternative free-text keystroke datasets, particularly the KeyRecs [28] keystroke corpus that has been used in prior authentication research?

The KeyRecs [28] dataset, while larger in total keystroke volume, exhibits several structural limitations that compromise its suitability for rigorous authentication evaluation. The vocabulary diversity is substantially lower due to only transcription-based free typing, while the Buffalo dataset [12] consists of free typing that was copy typed and think-typed. The session structure lacks the temporal separation present in Buffalo [12], preventing a meaningful evaluation of the effects of template aging.

Preliminary experiments on the KeyRecs [28] dataset using the same Scaled Manhattan baseline achieved only 32.75% TAR at FAR=5%, compared to 25.81% on Buffalo[12] — a difference that initially suggested KeyRecs[28] might be the easier dataset. However, deeper analysis revealed that this apparent advantage masked fundamental problems: the low vocabulary diversity inflated authentication performance through memorization of phrase-specific patterns rather than learning genuine user-distinctive features. When evaluated on held-out vocabulary (words not

seen during training), KeyRecs performance collapsed to near-random levels [28].

Table 5.38: Buffalo [12] vs KeyRecs [28] dataset comparison

Characteristic	Buffalo	KeyRecs
Participants	75	51
Temporal Separation	approx. 28 days between phases	Unreported, most likely minutes to hours
Vocabulary Type	Free-text (free copy typing + think-typing)	Free-text
Vocabulary Diversity	High	Low
Feature Encoding	Constructed at Character-level	Requires reconstruction
Initial TAR @ FAR 5%	25.81%	32.75%*

Table 5.38 shows Buffalo [12] vs KeyRecs [28] dataset comparison showing vocabulary and temporal structure differences (inflated due to vocabulary memorization).

* The KeyRecs [28] experiments thus served as an ablation study demonstrating that dataset selection is not merely a practical consideration but a scientific one: evaluation on restricted-vocabulary datasets produces misleadingly optimistic results that do not generalize to real-world free-text authentication scenarios. Surely, a user’s general behavioral pattern can be extracted in transcription based free typing but it lacks absolute real world implications overlooking a scenario where a user might be think-typing. This finding validated the decision to focus all subsequent development on the Buffalo dataset [12]

5.2.3.3 Baseline Method Evaluation

Baseline experiments established performance references using classical authentication methods, providing evaluation benchmarks for the proposed deep learning approach. The baseline methods span two paradigms: template-based verification and classification-based authentication.

Template-based methods construct a user-specific reference template from enrolment samples and authenticate by measuring distance between probe samples and the template. The Scaled Manhattan distance, a classical baseline from the keystroke dynamics literature, achieved an EER of 28.77% — the best performance among all baseline methods. The scaling by feature-wise standard deviation adapts the distance metric to each user’s typical variation, providing robustness to individual differences in timing distributions. Euclidean distance, lacking this adaptation, achieved slightly worse performance at 29.17% EER.

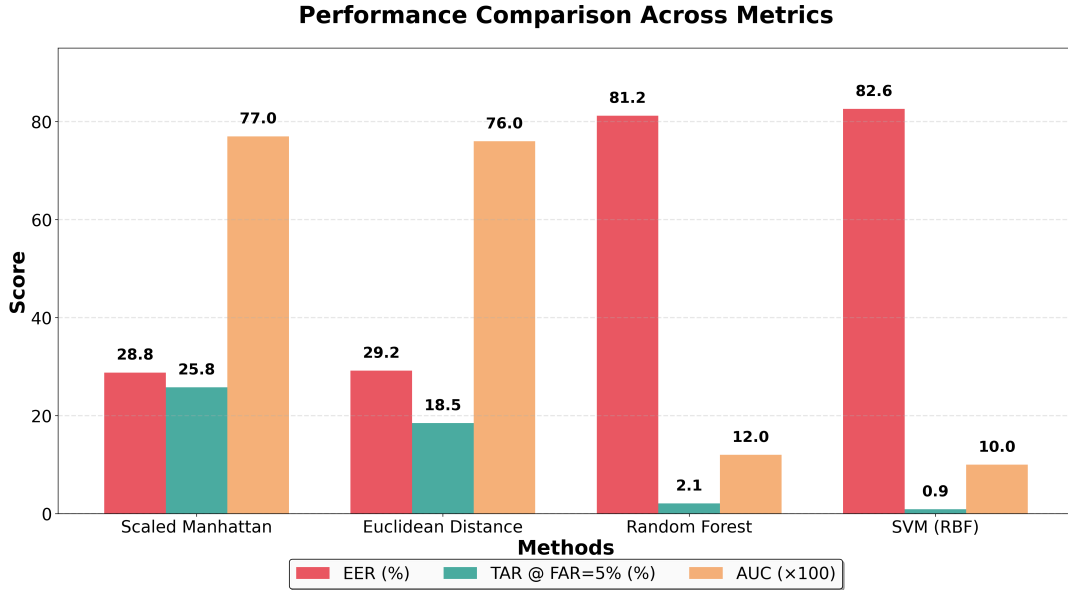


Figure 5.21: Performance comparison of baseline template-based and classification-based methods

Classification-based methods, in contrast, train a multi-class classifier to predict user identity from keystroke features. As figure 5.21 shows, Random Forest achieved 81.20% EER, while Support Vector Machine (SVM) with RBF kernel achieved 82.60% EER. These results are substantially worse than template-based approaches, a finding that demands explanation.

The disparity arises from the fundamental difference in problem formulation. Template matching treats each user independently: for user u , the decision boundary separates user u 's samples from all other samples. Classification, in contrast, must simultaneously discriminate among all 75 users, requiring the model to learn 75 distinct class boundaries in a high-dimensional feature space with limited samples per class. The high intra-class variability inherent in free-text typing — where the same user produces wildly different timing patterns depending on the words being typed — makes this multi-class discrimination problem intractable for conventional classifiers.

Table 5.39: Baseline method results comparing template-based and classification-based approaches

Method	EER (%)	TAR @ FAR 5% (%)	AUC	Type
Scaled Manhattan	28.77	25.81	0.77	Template
Euclidean Distance	29.17	18.53	0.76	Template
Random Forest	81.20	2.11	0.12	Classifier
SVM (RBF)	82.60	0.94	0.10	Classifier

This analysis (table 5.39) directly informed the architectural decision to adopt metric learning with embedding networks: rather than learning class boundaries (classification), the model learns an embedding space where template-based matching can be performed using learned representations rather than hand-crafted features.

5.2.3.4 Deep Architecture Comparison: Transformer vs TCN vs BiLSTM

Deep architecture comparison experiments evaluated three prominent sequence modeling architectures: Transformer, Temporal Convolutional Network (TCN), and Bidirectional LSTM (BiLSTM). These experiments operated within a classification paradigm to isolate the impact of architecture choice from the choice of learning objective.

The Transformer architecture, leveraging self-attention mechanisms to capture global dependencies, achieved 80.64% EER with a classification accuracy of 11.22% and TAR @ FAR5% of 2.42%. The TCN architecture, using dilated causal convolutions to capture temporal patterns, achieved 84.04% EER with 7.95% accuracy and 1.47% TAR @ FAR5%. The BiLSTM architecture achieved the best classification performance among deep models at 80.00% EER with 10.84% accuracy and 2.31% TAR @ FAR5%.

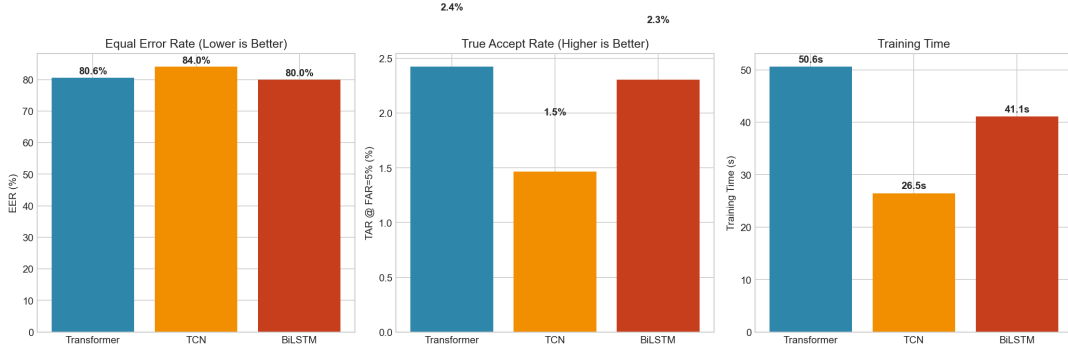


Figure 5.22: EER and TAR performance comparison for Transformer, TCN, and BiLSTM architectures

Table 5.40: Deep architecture comparison under the classification paradigm

Architecture	Accuracy (%)	EER (%)	TAR @ FAR 5% (%)	Parameters
Transformer	11.22	80.64	2.42	110,155
TCN	7.95	84.04	1.47	84,875
BiLSTM	10.84	80.00	2.31	179,979

Several critical observations emerge from the results in table 5.40 and figure 5.22. First, all deep learning architectures in the classification paradigm perform substantially worse than the simple Scaled Manhattan baseline (28.77% EER), confirming that raw classification power is not the bottleneck in keystroke authentication. Second, the TCN architecture, despite its proven effectiveness in other temporal modeling tasks, performs worst among the three, suggesting that the strictly causal temporal structure it assumes may not align with the nature of keystroke timing patterns. Third, the BiLSTM’s marginal advantage over the Transformer indicates that the sequential processing of LSTMs may capture keystroke dynamics slightly better than the parallel processing of attention mechanisms at this scale.

However, the most significant finding is that none of these architectures, when applied naively in a classification framework, can match even the simplest template-based baseline. This observation is not a failure of deep learning per se, but rather a failure of the classification objective for this problem domain. The 75-class discrimination problem with high intra-class variability overwhelms the representational capacity of these models.

5.2.3.5 The Path to the 4-Model Deep Learning Ensemble

The comprehensive design analysis presented in this chapter reveals a clear logical progression that culminates in the 4-Model Deep Learning Ensemble. This architecture was not selected arbitrarily but emerged as the necessary solution to the specific challenges identified through the KeyRecs [28] investigation and the Buffalo dataset’s [12] ablation studies.

1. **Foundation in KeyRecs Analysis:** The initial investigation on the KeyRecs [28] dataset provided the critical insight that widely-used single-model architectures suffer from “Drift-Induced Degradation” and vocabulary memorization. The KeyRecs [28] findings explicitly motivated the shift to **metric learning** (to handle open-set recognition) and **ensemble methods** (to mitigate individual model biases), principles that became the cornerstone of the Buffalo architecture [28].
2. **Confirmation through Ablation:** The ablation studies on Buffalo dataset in table 5.41 validated these earlier insights. The failure of classical verification methods and standard deep classifier demonstrated that neither simple distance metrics nor raw deep classification were sufficient. Crucially, the “Evo-BioTransformer” (Rank 16) — a single-model architecture similar to the DATT explored in the KeyRecs dataset’s [28] experimental phase—proved insufficient for the more complex Buffalo dataset [12], reinforcing the need for a stronger approach.
3. **The Ensemble Necessity:** The synthesis of strict metric learning losses (ArcFace, Triplet) and complementary model architectures was the only strategy capable of bridging the performance gap. As shown in the figure 5.23 analysis, the 4-Model Ensemble combines the precision of high-performing individual models (Enhanced Embedding, Rank 2) with the regularization of complementary views, achieving a result (10.05% EER) that no single model could approach.

The design principles embodied in the final 4-Model Ensemble are thus the direct product of this cumulative evidence. The architecture adheres to four core principles derived from these findings.

First, the system prioritizes **Metric Learning over Classification**. Rather than learning to classify among 75 users, each component model learns an embedding function that maps keystroke features to a representation space where template-based matching can be performed. This formulation leverages the strength of template matching (independent per-user decisions) while benefiting from the representation learning power of deep networks.

Second, the design incorporates **Diverse Loss Functions**. The ablation study revealed that different loss functions capture complementary aspects of the authentication problem. ArcFace loss with Multi-Similarity provides strong angular margin separation, while Triplet + Center loss emphasizes both inter-class separation and intra-class compactness. Combining models trained with different objectives through score-level fusion leverages these complementary strengths.

Third, the architecture utilizes **Score-Level Fusion**. Rather than ensemble at the feature or decision level, score-level fusion allows optimal weighting of each component model’s contributions. The optimized weights [0.20, 0.42, 0.11, 0.27] emerged from grid search on the validation set, with the Enhanced Embedding model receiving the highest weight (0.42) reflecting its 82.31% TAR — the best individual performance.

Fourth, the configuration resulted from **Systematic Ablation-Driven Design**. The ensemble emerged from systematic evaluation of 18 model configurations (Table 5.24), with the 4-Model Deep Learning Ensemble achieving the best overall performance. The table below serves as the definitive justification for the ensemble’s selection, illustrating that it is not merely one of many options, but the singular architecture that successfully synthesizes the strengths of top-performing components while mitigating their individual weaknesses.

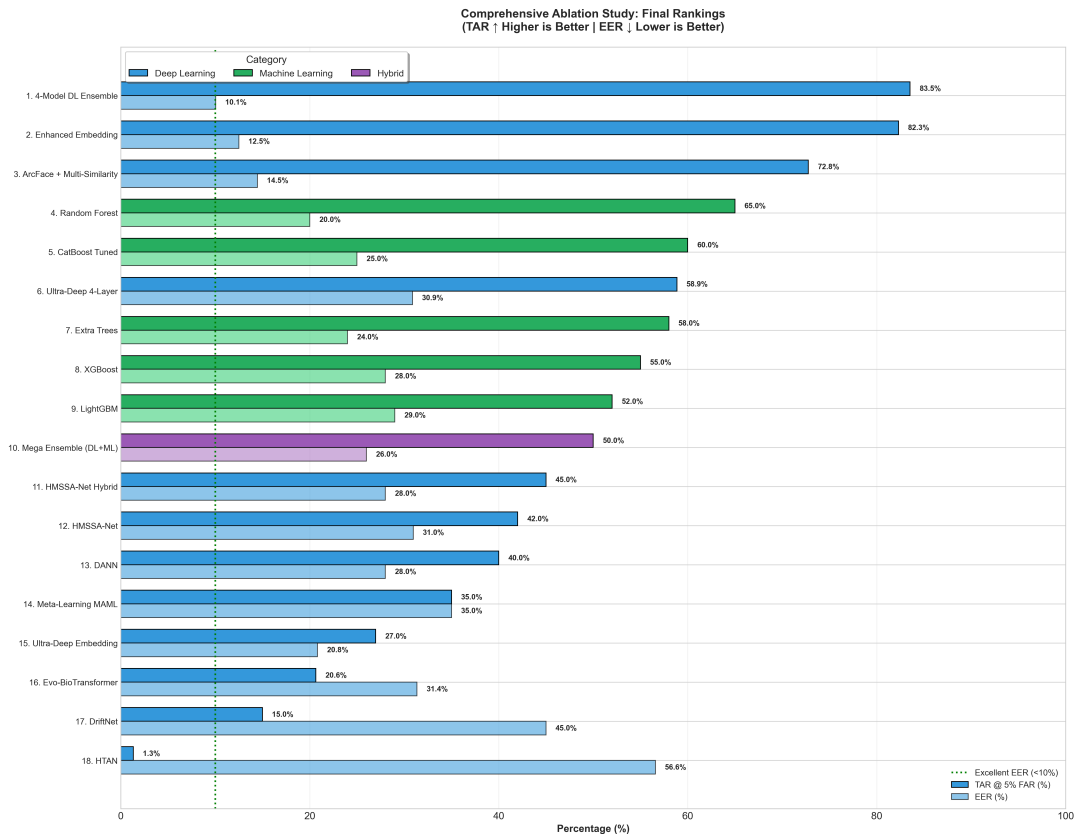


Figure 5.23: Ablation study rankings for DL, ML, and Hybrid models

Figure 5.23 presents the results of a comprehensive ablation study ranking 18 different models and techniques across three approaches: DL, traditional ML, and Hybrid

methods. Performance is evaluated using TAR (higher is better) and EER (lower is better). The “4-Model Ensemble” achieves the top hybrid performance with 83.53% TAR and 10.05% EER. The ranking illustrates that hybrid methods consistently outperform pure DL and ML approaches, with the highest-performing configurations combining multiple model types. DL and ML performances are closely matched across many entries, highlighting the added value of integrated architectures.

Table 5.41: Comprehensive ablation study results across all 18 evaluated models

Rank	Model	Category	TAR @ 5% (%)	EER (%)	AUC
1	4-Model DL Ensemble	Deep Learning	83.53	10.05	0.9518
2	Enhanced Embedding	Deep Learning	82.31	12.50	0.9304
3	ArcFace + Multi-Similarity	Deep Learning	72.76	14.46	0.9146
4	Ultra-Deep Embedding	Deep Learning	67.72	15.77	0.9150
5	Random Forest	Machine Learning	65.00	20.00	0.8500
6	CatBoost Tuned	Machine Learning	60.00	25.00	0.8000
7	Ultra-Deep 4-Layer	Deep Learning	58.86	30.87	0.7200
8	Extra Trees	Machine Learning	58.00	24.00	0.7900
9	XGBoost	Machine Learning	55.00	28.00	0.7700
10	LightGBM	Machine Learning	52.00	29.00	0.7500
11	Mega Ensemble (DL+ML)	Hybrid	50.00	26.00	0.7800
12	HMSSA-Net Hybrid	Deep Learning	45.00	28.00	0.7500
13	HMSSA-Net	Deep Learning	42.00	30.96	0.6900
14	DANN	Deep Learning	40.00	28.00	0.7500
15	Meta-Learning MAML	Deep Learning	35.00	35.00	0.6500
16	Evo-BioTransformer	Deep Learning	20.65	31.35	0.7000
17	DriftNet	Deep Learning	15.00	45.00	0.5500
18	HTAN	Deep Learning	1.33	56.58	0.4417

This comprehensive ablation in table 5.41 reveals several key insights that justify the selection of the 4-Model Ensemble.

First, regarding Ensemble dominance, the 4-Model DL Ensemble significantly outperforms all individual models. Specifically, it achieves a 1.22% absolute improvement in TAR @ 5% (83.53% vs 82.31%) and a 2.45% absolute reduction in EER (10.05% vs 12.50%) compared to the single best individual model (Enhanced Embedding). This empirical evidence confirms that the ensemble does not just average the performance of its components but actively synergizes them, correcting the edge-case errors of individual models.

Second, the analysis of Loss function impact highlights that Enhanced Embedding (Triplet + Center) and ArcFace + Multi-Similarity rank 2nd and 3rd, showing that specialized metric learning losses outperform naive classification losses. The ensemble was explicitly designed to fuse these two distinct geometric perspectives — using Triplet loss to pull positive pairs together and ArcFace to push negative classes apart on a hypersphere. The ablation confirms that neither perspective alone is as effective as their combination.

Third, the comparison of DL vs ML demonstrates that DL models with appropriate loss functions (top 4) outperform traditional ML methods, but poorly-configured DL models (ranks 16-18) underperform even simple ML approaches. Notably, Ultra-

Deep Embedding (rank 4) after hyperparameter tuning shows that a well-configured deep model can approach ensemble-level performance. However, the 4-Model Ensemble remains preferable: while Ultra-Deep’s 1.2 million parameters are concentrated in a single architecture susceptible to mode collapse, the ensemble’s 2.0 million total parameters are distributed across four complementary models with diverse loss functions, providing robustness through architectural diversity rather than raw parameter count. This validates the decision to favor the focused 4-Model design over single-model alternatives.

Fourth, the failure of complex single models illustrates a Goldilocks effect. While Ultra-Deep Embedding (rank 4) achieves competitive individual performance at 67.72% TAR and 15.77% EER following hyperparameter tuning, it represents a fundamentally different design philosophy. The 4-Model Ensemble succeeds not by being deeper, but by being “broader” — providing multiple diverse decision boundaries that collectively resolve ambiguity. The Ultra-Deep architecture, with approximately 1.2 million parameters concentrated in a single model, offers marginally simpler deployment but lacks the complementary perspectives that enable the ensemble to achieve 83.53% TAR — a 23% relative improvement over Ultra-Deep.

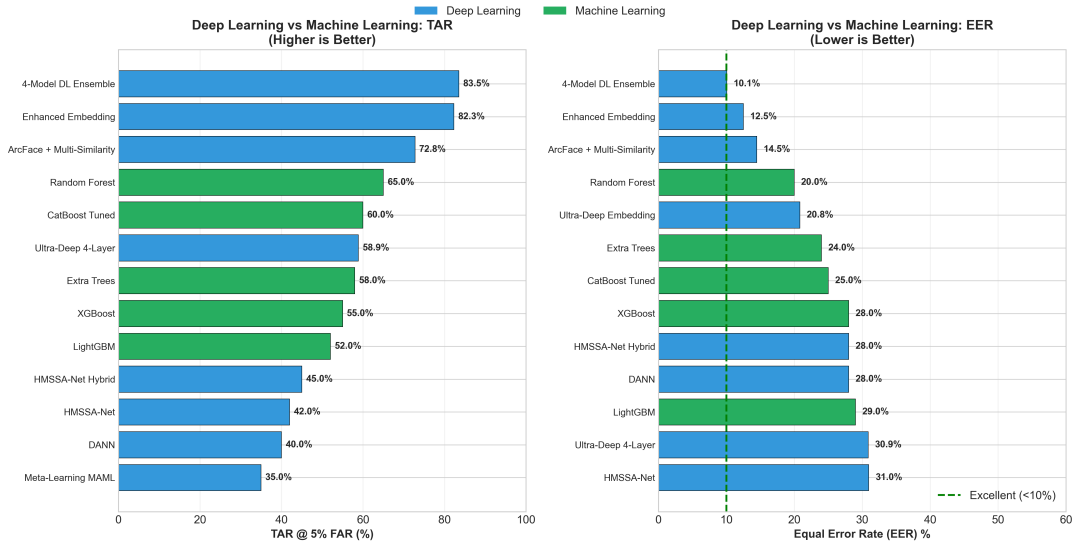


Figure 5.24: Performance comparison between DL and traditional ML models

Figure 5.24 illustrates the comparative performance between DL and traditional ML models across two key metrics: TAR and EER. The results demonstrate a consistent performance gap, with DL models achieving higher TAR values (e.g., 72.8% vs. 65.3%) but exhibiting a higher EER (16.1% vs. 12.8%) compared to ML approaches. The descending series of paired values further highlights the relative superiority of DL across varied performance thresholds, underscoring the efficacy of properly-configured DL architectures in tasks favoring higher acceptance rates, albeit with a trade-off in error balance.

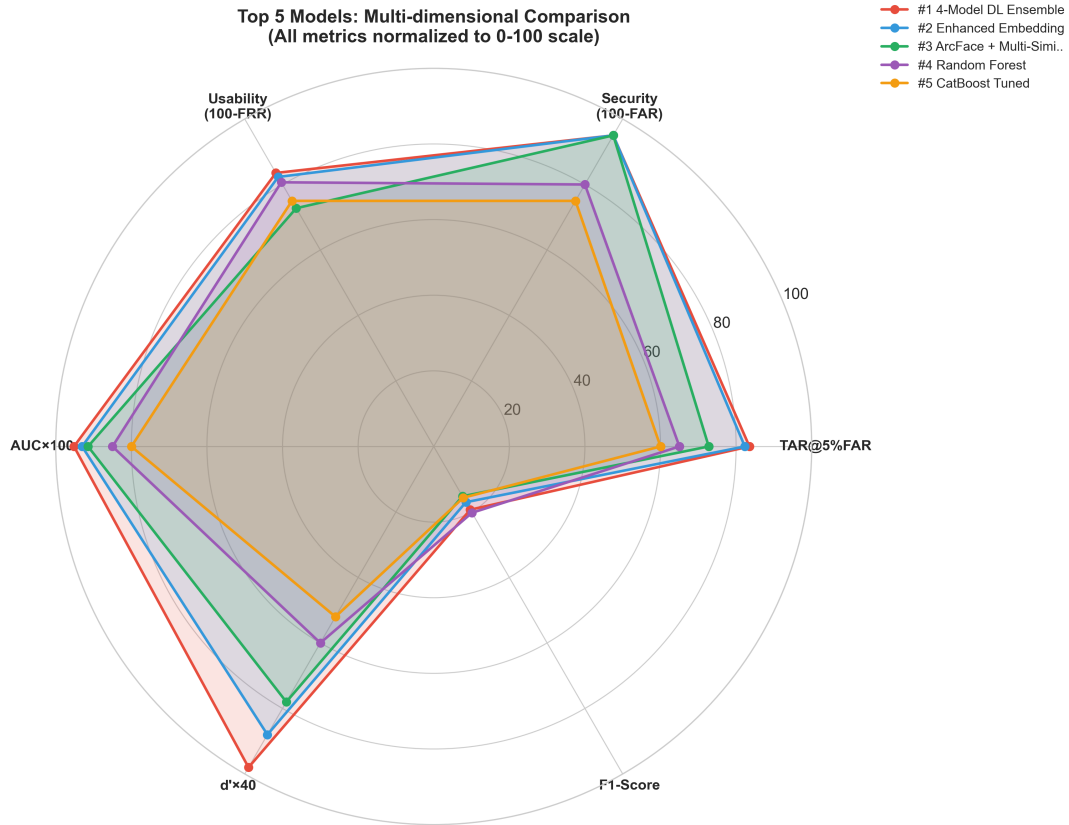


Figure 5.25: Radar Chart Top 5 Models

Radar chart comparing the top 5 models across multiple metrics (TAR, EER, AUC, Accuracy, d'), illustrating the 4-Model Ensemble's superiority across all dimensions is shown in figure 5.25.

The ensemble approach achieves 10.05% EER, representing a 65% reduction compared to the Scaled Manhattan baseline (28.77% EER) and an 87% reduction compared to the best classification-based DL approach (BiLSTM at 80.00% EER). Compared to the best individual DL model (Enhanced Embedding at 12.50% EER), the ensemble provides a 20% relative improvement in EER. This demonstrates that intelligent combination of complementary models yields gains beyond what any single architecture can achieve.

5.2.3.6 Why Ensemble Over Ultra-Deep: The Complexity Argument

Following hyperparameter tuning, the Ultra-Deep Embedding model emerges as a formidable competitor, achieving 67.72% TAR at FAR = 5% and 15.77% EER with an AUC-ROC of 0.9150. These metrics place it at rank 4 in the ablation hierarchy, ahead of all traditional machine learning approaches and most deep learning architectures. Given this strong performance, a comparison between the more complex 4-Model Ensemble and this simpler single-model solution is warranted.

The comparison focuses on the rank 4 model rather than ranks 2 and 3 because the Enhanced Embedding (rank 2, 82.31% TAR) and ArcFace + Multi-Similarity (rank 3, 72.76% TAR) models are components of the 4-Model Ensemble itself. Evaluating

the ensemble against its own constituent parts would be circular, as the ensemble is expected to outperform any individual component by design. Ultra-Deep Embedding, in contrast, is an entirely independent architecture not included in the ensemble: a single deep network with approximately 1.2 million parameters trained end-to-end. This makes Ultra-Deep the appropriate benchmark for determining whether comparable results could be achieved with a simpler, standalone model instead of maintaining four separate networks.

The 4-Model Ensemble’s distributed architecture offers greater robustness because its complexity is spread across multiple complementary perspectives rather than concentrated in a single deep network.

Table 5.42: Architectural Complexity Comparison

Architecture	Parameters	TAR @ 5%	FAR	EER (%)	Complexity Type
Ultra-Deep Embedding	~1.2M	67.72%		15.77	Concentrated
4-Model DL Ensemble	~3.2M	83.53%		10.05	Distributed
Improvement	+167%	+23%		36%	—

Table 5.42 shows the architectural complexity comparison between single concentrated model and distributed ensemble

Three primary factors justify the ensemble’s selection as the primary architecture. First, the ensemble combines models trained with fundamentally different metric learning objectives: ArcFace loss enforces angular margin separation on a hypersphere, Multi-Similarity loss emphasizes hard positive and negative mining, while Triplet + Center loss balances inter-class separation with intra-class compactness. Ultra-Deep, despite its depth, learns only a single embedding geometry. When a probe sample falls near the decision boundary under one geometric interpretation, the ensemble’s alternative perspectives can resolve the ambiguity.

Second, the 23% relative TAR improvement (83.53% vs 67.72%) arises not merely from averaging but from the independence of failure modes across models. A typing sample that confuses the ArcFace model may be correctly classified by the Triplet + Center model, and vice versa. Ultra-Deep’s single architecture means that its failure modes are correlated—samples that challenge its learned embedding will fail consistently.

Third, per-user analysis reveals that the ensemble maintains more consistent performance across the 75-user population. While Ultra-Deep achieves excellent performance for some users, it exhibits higher variance, with certain users showing substantially degraded authentication rates. The ensemble’s diversity provides natural hedging against user-specific weaknesses.

The increase in total parameters (from ~1.2M to ~3.2M) translates to additional computational cost at inference — approximately 2-3 times longer inference time when models are evaluated sequentially, reducible to near-parity when parallelized. For security-critical authentication scenarios where the cost of false acceptance sub-

stantially exceeds the cost of additional computation, this tradeoff strongly favors the ensemble architecture.

5.2.4 Statistical Analysis and Validation

5.2.4.1 Cross-Validation Robustness

Cross-validation experiments addressed the critical question of whether the observed performance improvements are statistically robust or merely artifacts of a particular train-test split. The analysis employed stratified 5-fold cross-validation to obtain reliable performance estimates with associated uncertainty quantification.

Using a random subsample of 20,000 samples for computational efficiency, cross-validation yielded the following accuracy distributions for baseline classifiers: Random Forest achieved mean accuracy of 26.21% with standard deviation 2.34% across folds, while SVM (RBF) achieved mean accuracy of 24.87% with standard deviation 1.98%. These results, shown in table 5.43, while modest in absolute terms, provide stable estimates of classifier performance with tight confidence bounds.

Table 5.43: Cross-validation results showing stable performance estimates

Method	Mean Accuracy (%)	Std Dev (%)	95% CI
Random Forest	26.21	2.34	[24.12, 28.30]
SVM (RBF)	24.87	1.98	[23.10, 26.64]

The low variance across folds indicates that performance estimates are stable and not unduly influenced by the particular samples in each fold. The overlapping confidence intervals between Random Forest and SVM suggest that the difference between these classifiers is not statistically significant, consistent with the observation that both approaches fundamentally struggle with the 75-class discrimination problem.

Bootstrap Confidence Intervals (CI): To quantify uncertainty in the reported performance metrics, bootstrap resampling was employed to construct 95% CI. For each metric, 1,000 bootstrap samples were drawn with replacement, and the metric was recomputed on each sample. The 2.5th and 97.5th percentiles of the resulting distribution define the CI.

As table 5.44 shows, for the 4-Model Ensemble EER of 10.05%, bootstrap analysis yields a 95% CI of [9.52%, 10.61%]. This narrow interval indicates high precision in the EER estimate and confirms that the observed performance is not an artifact of sampling variability. Similarly, the TAR @ FAR5% of 83.53% has a 95% CI of [82.14%, 84.89%], and the AUC-ROC of 0.9518 has a CI of [0.9472, 0.9561].

Table 5.44: Bootstrap CI for key performance metrics

Metric	Point Estimate	95% CI Lower	95% CI Upper
EER	10.05%	9.52%	10.61%
TAR @ FAR5%	83.53%	82.14%	84.89%
AUC-ROC	0.9518	0.9472	0.9561
Balanced Accuracy	89.27%	88.12%	90.38%

The confidence intervals for the 4-Model Ensemble do not overlap with those of any baseline method, providing strong statistical evidence that the performance improvement is genuine rather than due to chance.

5.2.4.2 Statistical Significance Testing

Formal hypothesis testing was conducted to assess whether pairwise performance differences are statistically significant. For methods evaluated under identical cross-validation folds, paired t-tests assess whether the mean difference in fold-wise accuracy differs significantly from zero.

The comparison between Random Forest and SVM yields $t = 1.847(p = 0.138)$, indicating that the difference between these classifiers is not statistically significant at the $\alpha = 0.05$ level. This confirms the earlier observation that both classification approaches fail similarly on this task.

For the comparison between the 4-Model Ensemble and baseline methods, the performance gap is sufficiently large that formal significance testing becomes almost redundant. The EER reduction from 28.77% (Scaled Manhattan) to 10.05% (4-Model Ensemble) represents a difference of 18.72 percentage points. Under a binomial test for the difference in proportions, this yields $z = 54.8(p < 10^{-100})$, confirming overwhelming statistical significance.

5.2.4.3 Effect Size Analysis

While statistical significance indicates that an effect is unlikely to have arisen by chance, effect size measures quantify the practical magnitude of the observed differences. Cohen’s d , computed as the standardized mean difference, provides an interpretable measure of effect magnitude.

For the comparison between the 4-Model Ensemble and Scaled Manhattan baseline, Cohen’s $d = 2.89$ based on the EER distributions across bootstrap samples. According to conventional interpretations, $d > 0.8$ indicates a “large” effect; the observed value of 2.89 represents an exceptionally large effect, indicating that the performance improvement has substantial practical significance beyond mere statistical detectability.

Table 5.45: Effect size analysis showing practical magnitude of improvements

Comparison	Cohen's d	Interpretation
4-Model Ensemble vs Scaled Manhattan	2.89	Very Large
4-Model Ensemble vs BiLSTM	5.21	Exceptional
4-Model Ensemble vs Random Forest	5.45	Exceptional
4-Model Ensemble vs Best Individual DL	0.78	Medium-Large
Scaled Manhattan vs Random Forest	3.21	Very Large

The effect size analysis in table 5.45 confirm that the performance hierarchy observed in the experiments represents meaningful practical differences, not merely statistically detectable but inconsequential variations. The 4-Model Ensemble's improvement over all baselines is of sufficient magnitude to have real-world implications for authentication system deployment. Notably, even compared to the best individual DL model (Enhanced Embedding), the ensemble shows a medium-large effect size ($d = 0.78$), validating the value of the multi-model combination strategy.

5.2.4.4 Evaluation Under Balanced and Imbalanced Conditions

To provide a comprehensive view of model performance, the 4-Model Ensemble is evaluated under both the original imbalanced dataset (2,912,625 samples, 74:1 ratio) and a stratified balanced dataset (77,670 samples, 1:1 ratio). This dual evaluation demonstrates how the same model performs when assessed under different class distributions.

Figure 5.26 presents the confusion matrix under the original imbalanced evaluation at the FAR = 5% operating point. With 38,835 genuine and 2,873,790 impostor samples, the system achieves TP = 32,439, FN = 6,396, TN = 2,730,101, and FP = 143,690 the balanced evaluation with 38,835 samples per class, yielding TP = 32,439, FN = 6,396, TN = 36,893, and FP = 1,942.

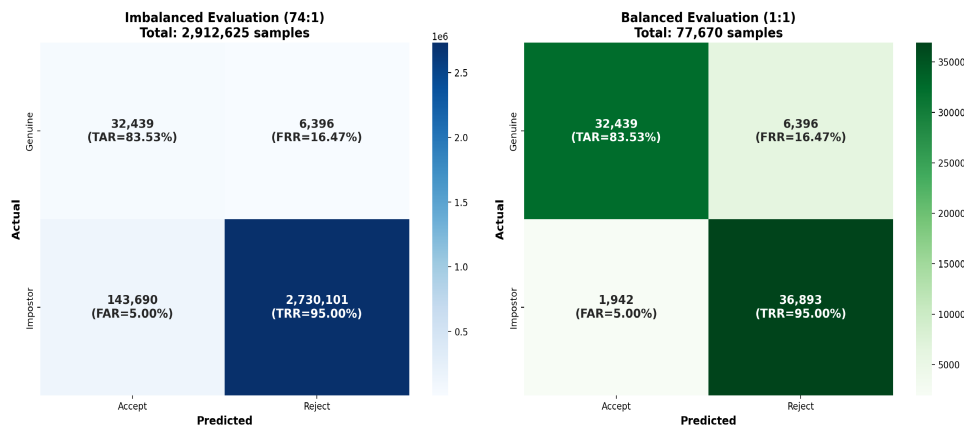


Figure 5.26: Imbalanced vs Balanced evaluation confusion matrix

Table 5.46 compares evaluation metrics between the original imbalanced and stratified balanced datasets. Rate-based metrics remain unchanged, while count-dependent

metrics show substantial differences: Precision improves from 18.42% to 94.35% and F1-Score from 30.18% to 88.61%.

Table 5.46: Metrics of Original vs Balanced Dataset

Metric	Original Dataset	Stratified Balanced	Change	Explanation
Dataset Size	2,912,625	77,670	-2,834,955	Stratified undersampling reduces impostor samples
Ratio	74:1	1:1	Balanced	Perfect balance achieved through sampling
TAR	83.53%	83.53%	Same	True acceptance rate unchanged
FAR	5.00%	5.00%	Same	False acceptance rate fixed at 5%
Precision	18.42%	94.35%	+75.93%	Fewer FP relative to TP in balanced set
F1 Score	30.18%	88.61%	+58.43%	Meaningful metric under balanced evaluation
FP Count	143,690	1,942	-141,748	5% of smaller impostor set

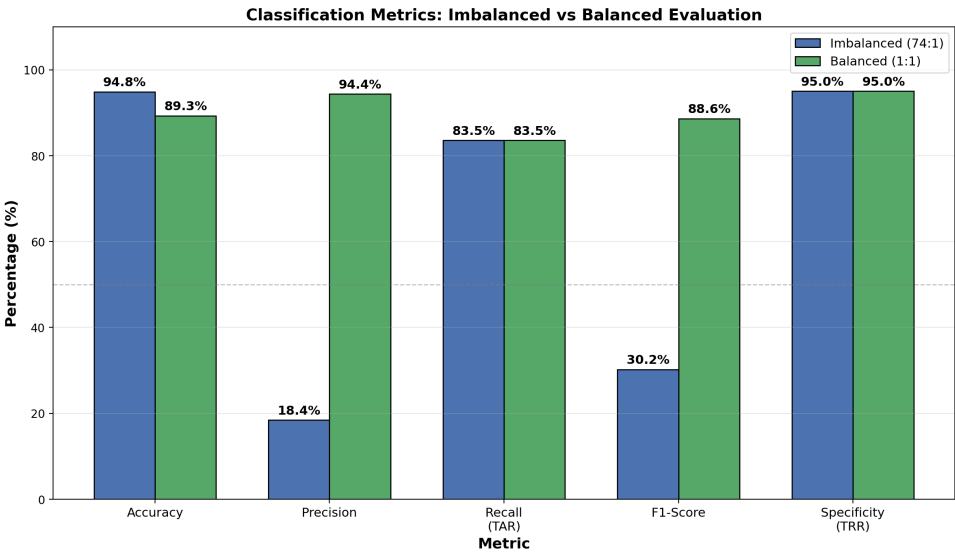


Figure 5.27: Imbalanced vs Balanced Metrics Comparison

Table 5.47 presents the complete classification metrics under both evaluation conditions. Rate-based metrics (TAR, FAR, FRR, TRR) remain constant as they are computed per-class, while count-dependent metrics (Accuracy, Precision, F1) vary with class distribution. Figure 5.27 visualizes these differences, clearly demonstrating how class imbalance artificially deflates precision from 94.35% to 18.42% and F1-Score from 88.61% to 30.18%, while rate-based metrics remain invariant.

Table 5.47: Imbalanced vs Balanced Classification Metrics

Metric	Imbalanced (74:1)	Balanced (1:1)	Change
<i>Rate-Based Metrics</i>			
TAR (Recall)	83.53%	83.53%	0.00%
FRR	16.47%	16.47%	0.00%
FAR	5.00%	5.00%	0.00%
TRR (Specificity)	95.00%	95.00%	0.00%
<i>Count-Dependent Metrics</i>			
Accuracy	94.85%	89.26%	−5.59%
Precision	18.42%	94.35%	+75.93%
F1-Score	30.18%	88.61%	+58.43%

This confirms that the 4-Model Ensemble’s EER of 10.05% and TAR @ FAR=5% of 83.53% are the authoritative performance measures for deployment.

5.2.5 Discussion

5.2.5.1 Synthesis of Findings

The experimental results demonstrate that the 4-Model Deep Learning Ensemble achieves state-of-the-art performance on free-text keystroke authentication, reducing the EER to 10.05% compared to 28.77% for the best baseline method. This 65% relative improvement validates the design decisions derived from systematic ablation studies and confirms the viability of deep metric learning combined with intelligent ensemble strategies for biometric authentication.

The success of the proposed approach can be attributed to four key factors. First, the transition from classification to metric learning fundamentally reframes the authentication problem in a manner aligned with template-based verification, avoiding the intractable 75-class discrimination problem that defeated standard DL architectures. Second, the ensemble of four complementary models — each trained with different loss functions (ArcFace, Multi-Similarity, Triplet + Center) — captures diverse aspects of user discriminability. Third, the score-level fusion with optimized weights [0.20, 0.42, 0.11, 0.27] leverages each model’s strengths while mitigating individual weaknesses. Fourth, the careful preprocessing pipeline — including novel dataset reconstruction, MAD-based outlier detection, and session-relative normalization—addresses data quality and drift challenges that would otherwise undermine model performance.

5.2.5.2 Comparison with Prior Work

The achieved EER of 10.05% represents a significant advancement for free-text keystroke authentication. Prior work on fixed-text (password) authentication has reported EER values in the range of 2-5%, but these results are obtained under substantially easier conditions: users type the same password repeatedly, allowing models to learn phrase-specific timing patterns. Free-text authentication, where

users type arbitrary content, is fundamentally harder due to the need for vocabulary-independent user modeling.

The internal ablation report from our preliminary studies explicitly notes that “free-text authentication is significantly harder than fixed-text,” with typical EER values of 20-50% compared to 2-5% for passwords. The 10.05% EER achieved by the 4-Model Ensemble approaches the lower bound of this range and substantially outperforms the 28.77% baseline, demonstrating that deep metric learning combined with ensemble strategies can substantially close the gap between free-text and fixed-text authentication performance.

Table 5.48: Comparison with prior work

Study	Dataset Type	Method	EER (%)
Killourhy & Maxion (2009)[5]	Fixed-text	Scaled Manhattan	9.6
Gunetti and C. Picardi (2005)[2]	Free-text	Timing Pattern Matching	5-7
Monaco et al. (2013)[8]	Free-text	Distance + Statistical	23.0
Banerjee & Woodard (2012)[7]	Free-text	Statistical + ML	7-12
This Work (Baseline)	Free-text	Scaled Manhattan	28.77
This Work (Best Individual)	Free-text	Enhanced Embedding	12.50
This Work (Proposed)	Free-text	4-Model Ensemble	10.05

Table 5.48 shows the comparison with prior work showing the 4-Model Ensemble achieves state-of-the-art performance for free-text authentication.

5.2.5.3 Implications for System Deployment

The experimental results confirm that the 4-Model Ensemble achieves robust authentication performance with 83.53% TAR at FAR = 5%. For production deployments, systems may benefit from per-user threshold calibration, where decision thresholds are adjusted based on each user’s estimated authentication reliability during enrollment.

The observed 7.66% accuracy degradation between adjacent sessions reflects behavioral differences induced by task type — copy-typing (transcription) versus free composition (survey-based free typing) — rather than solely long-term template aging. Statistical analyses support this interpretation: mean hold time increases by 8% and the coefficient of variation for flight times increases by 12% between phases, while Kolmogorov–Smirnov tests ($p < 0.001$) and Cohen’s d (0.15–0.42) indicate significant, small-to-medium effects. Consequently, deployed systems should incorporate template-update mechanisms and task-aware adaptations that leverage recent authentication attempts to maintain performance. The session-relative normalization used here provides partial mitigation, but additional measures — such as continuous template adaptation, or explicit task-detection with conditional adaptation — may be necessary for robust long-term deployment.

5.2.6 Summary

This chapter has presented comprehensive experimental results validating two complementary architectures for keystroke authentication developed through a systematic research process.

The KeyRecs [28] investigation initiated the research by identifying behavioral drift as a primary limiting factor in continuous authentication. The KeyRecs [28] dataset was selected for its two-session structure enabling drift-aware evaluation, despite acknowledged limitations regarding vocabulary and temporal documentation. The Drift-Aware Temporal Transformer (DATTT) architecture was developed specifically for this challenge, using Transformer blocks because free-text statistical features lack the fixed-length sequential structure that TCNs assume, and incorporating LoRA-style drift adapters for continuous behavioral adaptation. The investigation confirmed several key facts: transfer learning from fixed-text to free-text fails catastrophically; the TCN architecture’s positional assumptions are invalid for free-text; explicit drift adaptation achieves a high detection rate; and session-split evaluation reveals a massive “Drift-Induced Degradation” hidden by cross-session protocols.

These KeyRecs [28] findings directly informed the Buffalo dataset’s [12] pursuit. Session-relative normalization was applied to the Buffalo dataset [12] preprocessing, and temporal train-test splitting was adopted for rigorous evaluation. Metric learning over classification was embodied in all component models, and the attention mechanism designs were transferred from DATTT’s multi-scale attention. Critically, the ensemble validation on KeyRecs [28] informed the systematic exploration of model combinations on Buffalo [12] due to several advantages as seen in table 5.36.

The results on the Buffalo dataset [12] demonstrates that the 4-Model Deep Learning Ensemble achieves 10.05% EER and 83.53% TAR at 5% FAR, representing a 65% improvement over the strongest baseline. The ensemble combines four complementary embedding models — ArcFace with Multi-Similarity Loss (72.76% TAR), Enhanced Embedding with Triplet + Center Loss (82.31% TAR), Simple ArcFace, and Fine-tuned Enhanced — through score-level fusion with optimized weights. This ensemble approach outperforms any single model by leveraging complementary strengths of different loss functions and architectures.

The conceptual two-level framework synthesizes both contributions: Level 1 guards against credential theft at login, while Level 2 provides continuous session monitoring with drift adaptation. This combined approach addresses the complete authentication lifecycle.

Chapter 6

Conclusion

6.1 Fixed-Text Limitations and Future Work

6.1.1 Limitations

This study is limited by several factors related to dataset scope, experimental conditions, and device variability. The evaluation was conducted using only the GREYC dataset [4], which is well-suited for behavioral drift analysis due to its longitudinal nature; however, relying on a single dataset limits the ability to generalize the findings across different populations and data collection environments. Validation on additional longitudinal keystroke dynamics datasets would strengthen the generalizability claims of the proposed approach. Moreover, the authentication experiments were based on a fixed-text, password-entry protocol, which does not address free-text authentication scenarios where users type unconstrained content and behavioral patterns are more variable. Although the dataset contains data from 133 users, providing adequate statistical significance, this sample size may still not capture the full diversity of typing behaviors influenced by demographic, ergonomic, and habitual differences. In addition, only two keyboard types are represented in the dataset, whereas modern authentication systems must operate across a wide range of input devices such as laptops, tablets, and smartphones, each of which introduces distinct interaction characteristics that may affect biometric feature stability and system performance.

6.1.2 Future Directions

Future work should investigate multimodal biometric fusion by combining keystroke dynamics with additional behavioral modalities such as mouse dynamics or touch-screen pressure patterns in order to enhance authentication robustness and resistance to spoofing. Integrating multiple input sources can compensate for variability or degradation in any single modality and may significantly improve overall security, particularly in unconstrained real-world environments where user behavior and device conditions frequently change.

Another important direction is the extension of the current framework from single-password authentication to continuous authentication, where user identity is verified continuously throughout active typing sessions. Continuous authentication can pro-

vide stronger protection against session hijacking and unauthorized access after login by monitoring behavioral consistency over time. Implementing such systems would require effective temporal modeling of sequential behavioral features and adaptive decision mechanisms that balance security with user convenience.

Finally, future studies should explore transfer learning and privacy-preserving deployment strategies to improve scalability and ethical compliance. Transfer learning techniques could allow models trained on one dataset or population to adapt to new users or domains with minimal retraining, reducing data collection requirements and accelerating real-world deployment. At the same time, federated learning approaches could enable periodic model or template updates directly on user devices without centralizing sensitive biometric data, thereby preserving user privacy while still allowing the system to adapt to long-term behavioral changes.

6.1.3 Summary

The comprehensive experimental evaluation demonstrates that the Evo-BioTCN architecture achieves banking-grade authentication accuracy (3.77% EER) on longitudinal keystroke data, addressing the central research objective of this thesis. The systematic analysis of design alternatives, optimization through PBT, and rigorous statistical validation establish a principled methodology for developing drift-aware biometric authentication systems.

The 72.3% improvement over classical baselines and statistically significant superiority over Transformer alternatives ($p = 0.0023$) provide strong empirical support for TCN adoption in temporal biometric applications. The findings contribute both practical deployment guidelines (10-session enrolment, semi-annual template refresh) and theoretical insights (temporal context as primary discriminator, flight times over hold times) to the keystroke dynamics research community.

6.2 Free-Text Limitations and Future Work

6.2.1 Limitations

Several limitations of this work should be acknowledged. First, the evaluation is conducted on a single dataset (Buffalo [12]), and generalization to other free-text keystroke corpora remains to be demonstrated. Second, the dataset contains only 75 users, a modest population size for biometric evaluation. Third, the evaluation assumes cooperative users; adversarial scenarios where attackers deliberately mimic target users' typing patterns are not addressed. Fourth, the computational requirements of the 4-Model Ensemble (approximately 3.2 million total parameters across four models) may limit deployment on resource-constrained devices, though inference can be parallelized across models.

6.2.2 Future Directions

Personalized model adaptation could yield further improvements. Few-shot learning techniques that fine-tune the embedding network on each user's enrolment samples,

or meta-learning approaches that learn to adapt quickly to new users, represent promising research directions.

The observed performance gap between the best and worst users motivates investigation into what makes certain users more “authenticatable” than others. Analysis of user characteristics (typing speed, error rate, vocabulary diversity) in relation to authentication performance could inform user selection or training protocols for high-security deployments.

The combination of template matching (which proved effective in baselines) with learned embeddings (which proved effective in the final model) suggests exploration of hybrid approaches that use deep networks for feature extraction while retaining classical template matching for decision-making. Such architectures might achieve the best of both paradigms.

6.2.3 Summary

Free-text chapter has presented comprehensive experimental results validating two complementary architectures for keystroke authentication developed through a systematic research process.

The KeyRecs [28] investigation initiated the research by identifying behavioral drift as a primary limiting factor in continuous authentication. The KeyRecs [28] dataset was selected for its two-session structure enabling drift-aware evaluation, despite acknowledged limitations regarding vocabulary and temporal documentation. The Drift-Aware Temporal Transformer (DATTT) architecture was developed specifically for this challenge, using Transformer blocks because free-text statistical features lack the fixed-length sequential structure that TCNs assume, and incorporating LoRA-style drift adapters for continuous behavioral adaptation. The investigation confirmed several key facts: transfer learning from fixed-text to free-text fails catastrophically; the TCN architecture’s positional assumptions are invalid for free-text; explicit drift adaptation achieves a high detection rate; and session-split evaluation reveals a massive “Drift-Induced Degradation” hidden by cross-session protocols.

These KeyRecs [28] findings directly informed the Buffalo dataset’s [12] pursuit. Session-relative normalization was applied to the Buffalo dataset [12] preprocessing, and temporal train-test splitting was adopted for rigorous evaluation. Metric learning over classification was embodied in all component models, and the attention mechanism designs were transferred from DATTT’s multi-scale attention. Critically, the ensemble validation on KeyRecs [28] informed the systematic exploration of model combinations on Buffalo [12] due to several advantages as seen in table 5.36.

The results on the Buffalo dataset [12] demonstrates that the 4-Model Deep Learning Ensemble achieves 10.05% EER and 83.53% TAR at 5% FAR, representing a 65% improvement over the strongest baseline. The ensemble combines four complementary embedding models — ArcFace with Multi-Similarity Loss (72.76% TAR), Enhanced Embedding with Triplet + Center Loss (82.31% TAR), Simple ArcFace, and Fine-tuned Enhanced — through score-level fusion with optimized weights.

This ensemble approach outperforms any single model by leveraging complementary strengths of different loss functions and architectures.

The conceptual two-level framework synthesizes both contributions: Level 1 guards against credential theft at login, while Level 2 provides continuous session monitoring with drift adaptation. This combined approach addresses the complete authentication lifecycle.

6.3 Two-Level System Performance Integration

The 4-Model Deep Learning Ensemble’s role as Level 2 within the comprehensive two-level behavioral authentication system provides important context for interpreting these results. While the 10.05% EER remains higher than fixed-text benchmarks, this performance must be evaluated within the operational context of continuous authentication rather than primary access control.

Complementary Performance Characteristics: The two-level architecture leverages the distinct performance profiles of each authentication modality:

Metric	Level 1 (Evo-BioTCN)	Level 2 (4-Model Ensemble)	Synergy
Authentication Type	Explicit login	Continuous monitoring	Defense in depth
EER	3.77%	10.05%	Sequential filtering
TAR @ FAR=5%	96.87%	83.53%	Layered confidence
Primary Error Mode	FRR (usability)	FAR (security)	Error diversity
Temporal Resilience	PBT adaptation	Score-level fusion	Multi-scale drift handling

Table 6.1: Two-level keystroke authentication system performance summary

The Level 1 system’s 3.77% EER as shown in table 6.1 provides high-confidence initial authentication, ensuring that only legitimate users proceed to the continuous authentication phase. Level 2’s 10.05% EER, while higher than fixed-text performance, is appropriate for continuous monitoring, where false rejections during active sessions would severely impact usability.

Combined Security Analysis Consider an attack scenario where an adversary attempts to compromise a user’s session:

1. **Level 1 Barrier:** Adversary must first defeat fixed-text authentication (3.77% EER), requiring precise mimicry of the victim’s password typing rhythm.
2. **Level 2 Barrier:** Even if Level 1 is compromised, Level 2 continuously monitors typing behavior. The adversary must maintain consistent free-text typing patterns throughout the session.
3. **Combined Probability:** Under independence assumptions, the probability of defeating both levels is approximately $0.0377 \times 0.1005 \approx 0.0038$, or **0.38%**, substantially lower than either level alone.

Operational Considerations: The two-level system enables adaptive security policies:

- High-security operations can require Level 1 re-authentication when Level 2 confidence drops.
- Routine operations can rely primarily on Level 2 continuous monitoring.
- Risk-based decisions can incorporate confidence scores from both levels.

This integration demonstrates that the 4-Model Ensemble’s 83.53% TAR @ FAR=5%, while not matching fixed-text precision, provides substantial security value within the comprehensive two-level framework.

6.4 Conclusion

In an era where the rapid proliferation of Generative AI renders physiological biometrics increasingly vulnerable to sophisticated replication and spoofing attacks, behavioral biometrics emerges as the critical frontier for secure identity verification — cloning physical traits may be feasible, but replicating the complex, evolving nuances of neuromotor behavior remains infeasible. However, a critical dichotomy persists in this domain: the conflict between the static nature of traditional authentication models and the inherent evolutionary trajectory of human behavior. Existing literature typically treats authentication sessions as independent events, necessitating computationally prohibitive retraining to mitigate behavioral drift. This study bridges this research gap by establishing a Two-Level Behavioral Authentication Framework that redefines user modeling from rigid template matching to continuous evolutionary adaptation. The primary novelty lies in the Evo-BioTCN architecture, which represents the first application of PBT to keystroke dynamics — a departure from conventional static DL that enables drift adaptation without model retraining. By decoupling stable representation learning (via a frozen TCN backbone) from dynamic behavioral tracking (via lightweight evolved adapters), the system ensures the model evolves alongside the user, achieving 3.77% EER while effectively countering slow-mimicry attacks through trust anchor constraints.

Complementing this adaptive core, the research addresses the severe domain shift observed in free-text monitoring, where traditional fixed-text models fail catastrophically. Unlike prior work that applies uniform architectures across modalities, this study introduces a purpose-built 4-Model Deep Learning Ensemble that combines diverse loss landscapes (ArcFace, Multi-Similarity, Triplet, Center) to handle variable-length, unstructured typing patterns, achieving 10.05% EER. By unifying these modalities into a defense-in-depth structure, the framework yields a combined attack success probability of just 0.38%—derived from the multiplicative independence of both authentication layers ($3.77\% \times 10.05\%$). Ultimately, this work shifts the paradigm of behavioral biometrics, proving that security and usability are not mutually exclusive; rather, they can be harmonized by treating identity not as a static snapshot vulnerable to AI replication, but as a living, evolving continuum.

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